Unit 9

Thermodynamics

The gasoline engine in an automobile, the jet engine in an airplane, the rocket engine, the steam engine, all use the heat of combustion of their fuel to perform mechanical work that moves the vehicle or machine. What is more, the working principle of air conditioner, refrigerator, cooker as well as the answers to deep questions such as "how will the universe end?" all depend on the laws of thermodynamics.

Thermodynamics: This term is derived from the Greek words *therme*, meaning heat, and *dynamics*, meaning power. Hence,

Thermodynamics: Is the study of energy conversion between heat and other forms of energy such as mechanical energy.

9.1 Atoms and Moles

In thermodynamics, we often deal with a gas in a cylinder. The gas consists of atoms and molecules and their number is measured using a unit called *Mol*.

Mole: Is the measure of discrete particles in a substance (solid, liquid, gas).

$1 \text{ mol} = 6.02 \times 10^{23}$ particles

The number, 6.02×10^{23} is called *Avogadro's number* (N_A)

1 mol of helium gas contains 6.02×10^{23} particles of helium and, 1 mol of water contains 6.02×10^{23} molecules of water, this means 6.02×10^{23} atoms of oxygen and 1.20×10^{23} atoms of hydrogen.

The relationship between the number of moles, the Avogadro's number and the number of particles is given by

$$N = n N_A$$

Where, N = number of particles

n = number of moles

N_A = Avogadro's number

Molar mass: Is the mass of one mole of a substance.

For example, 1 mole of C – 12 has a molar mass of 12 g

1 mole of Hydrogen molecule has a molar mass of 2g

1 mole of oxygen molecule has a molar mass of 32 g

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- Mass of the given substance is given by:

$$\boxed{m = nM}$$

Where, m = mass of substance
n = number of moles
M = molar mass

- Mass of each particle (m_p) is calculated using

$$m_p = \frac{M}{N_A} \quad \text{or} \quad m_p = \frac{m}{N}$$

Conceptual Examples

1. Which, if either, contains a greater number of molecules, a mole of hydrogen or a mole of oxygen? Explain

Explanation: Both of them contain equal number of molecules which is 6.02×10^{23} molecules.

2. Suppose two different substances, A and B have same densities. In general, does one mole of substance A have the same mass as one mole of substance B? Justify your answer.

Explanation: From the relation, $m = nM$ and $\rho = \frac{m}{v}$

Here, $\rho_A = \rho_B$

$$\frac{m_A}{v_A} = \frac{m_B}{v_B} \Rightarrow \frac{n_A M_A}{v_B} = \frac{n_B M_B}{v_B}, \text{ for } n_A = n_B = 1$$

$$\frac{M_A}{v_A} = \frac{M_B}{v_B} \text{ if } V_A = V_B, \text{ then } M_A = M_B$$

Illustrative Examples

3. Find the mass of an oxygen molecule.

Solution: $M_{O_2} = \frac{M}{N_A} = \frac{32}{6.02 \times 10^{23}} = 5.3 \times 10^{-23}\,g$

4. A block of Carbon – 12 contains 5 mol particles. Calculate:
   a) The number of particles in the block.
   b) The mass of the block.
   c) The mass of each carbon – 12 atom.

Solution: a) $N = nN_A = (5)(6.02 \times 10^{23}) = 3 \times 10^{24}$ particles
b) $m = nM = (5)(12) = 60g = 0.06\,kg$

c) $m_P = \frac{M}{N_A} = \frac{12}{6.02\times10^{23}} = 2\times10^{-23}\,g = 2\times10^{-26}\,kg$

Or, $m_P = \frac{M}{N} = \frac{60}{3\times10^{24}} = 2\times10^{-23}\,g = 2\times10^{-26}\,kg$

5. A droplet of mercury has a radius of 5mm. How many mercury atoms are there in the droplet? (take M = 200g, $\rho = 13600\ kg/m^3$)

Solution: Assume the droplet is spherical

Droplet volume, $v = \frac{4}{3}\pi r^3 = \frac{4}{3}(3.14)(5\times10^{-3})^3 = 5.2\times10^{-7}\,m^3$

Droplet mass, $m = \rho v = (13600)(5.2\times10^{-7}) = 7.1\times10^{-3}kg$

Mass of mercury atom, $m_P = \frac{M}{N_A} = \frac{0.2}{6.02\times10^{23}} = 3.3\times10^{-25}\,kg$

Number of atoms in droplet, $N = \frac{m}{m_P} = \frac{7.1\times10^{-3}}{3.3\times10^{-25}} = 2.1\times10^{22}$

6. a) How many moles of gas are present in 200g of CO_2.
   b) How many molecules are present? M = 44g/mol

Solution: a) $n = \frac{m}{M} = \frac{200}{44} = 4.6$ moles

b) $n = \frac{N}{N_A} \Rightarrow N = nN_A$

$N = (4.6)(6.02\times10^{23}) = 2.8\times10^{24}$ parties

9.2 Temperature, Heat and Zeroth Law of Thermodynamics

Temperature is something that we all have experience of. In solids, particles vibrate. In liquids and gases, the particles move freely. In all cases, the particles have kinetic energy. If the particles in the substance are moving faster, we say their kinetic energy is larger, the substance is hotter and its temperature is higher.

Temperature: Is a measure of the average random kinetic energy of particles of a body and it determines the direction of heat flow.

Temperature is a scalar quantity and its SI unit is the Kelvin (k). If we place two bodies of different temperature in contact, then the particles at the boundary will collide and the kinetic energy of particles is

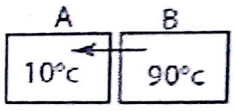
transferred backwards and forwards between them. So there is a net transfer of thermal energy from the hotter body to the colder body.

Heat: Is a net transfer of thermal energy between bodies because of temperature difference.

- Heat flows from a hotter body to the colder body, that is from a high temperature region to a low temperature region. And, heat is a scalar quantity and its SI unit is the Joule (J).
- **Important point to remember**
  When there is heat transfer between two bodies, the bodies are said to be in *thermal contact* and the heat transfer from the hotter body to the colder body continues until thermal equilibrium is reached.

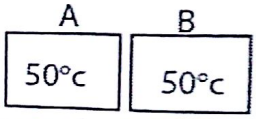
Thermal Equilibrium: Is a condition in which two bodies are at the same temperature and there is no net transfer of thermal energy between them.

Example,



Thermal contact

heat flows from body B to body A (from hotter to colder)



Thermal equilibrium

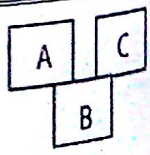
the bodies have the same temperature (no heat exchange)

Zeroth Law of Thermodynamics

It is based on the idea that thermal equilibrium is determined by temperature.

Zeroth Law of Thermodynamics states that; "Two bodies that are separately in thermal equilibrium with a third body must be in thermal equilibrium with each other."

That is, if A is in thermal equilibrium with B and C is in thermal equilibrium with B, then A is also in thermal equilibrium with C.
(If A = B and C = B then A = C).



Conceptual Examples

7. Is it valid to say that a hot object contains more heat than a cold object? Explain

Explanation: No! a body does not contain or possess heat but a hot body is a region of higher temperature.

8. In which case will you feel warmer, rubbing your palm on a metal surface or on a wooden surface? Why

Explanation: On the wooden surface. Because, wood is thermal insulator and hence heat will not leak into it easily.

9. A cold block of metal feels colder than a block of wood at the same temperature. Why?

Explanation: Because, metal is good conductor of heat but wood not it.

Illustrative Examples

10. Temperature of a body is a measure of (UEE 2002).
    A. Heat content of a body.
    B. Average molecular kinetic energy.
    C. Average pressure in a gas.
    D. Average in a gas.

Answer: B

11. Heat travel from one object to another when these objects differ in (UEE 1999)
    A. Specific heat
    B. Heat Capacity
    C. Temperature
    D. State

Answer: C

12. The zeroth law of thermodynamics states that (UEE 2003)
    A. Two bodies that are separately in thermal equilibrium with a third body must be in thermal equilibrium with each other.
    B. The increase in internal energy of a system equals the sum of energy entering through heating and work done on it.
    C. No process, is possible in which there is an overall decrease in the entropy of the universe.
    D. No process is perfectly reversible and no refrigerator or heat engine can be perfectly efficient.

Answer: A

Temperature Scale

The Celsius, Fahrenheit and Kelvin scales can be used as temperature measuring scales but the SI unit of temperature is the Kelvin. And, The three temperature scales have the following relationships.

$$T_k = T_c + 273.15 \quad \text{and} \quad T_F = \frac{9}{5}T_C + 32$$

Absolute zero (0k): Is a temperature at which all the random motion of particles in a substance ceases (stops).

☛ It is impossible to cool an object below the absolute zero moreover absolute zero cannot be attained.

Conceptual Example

13. At what temperature will
    a) $T_C = T_F$ b) $T_C = T_k$

Explanation: a) $T_F = \frac{9}{5}T_C + 32$, now $T_C = T_F = T$

$$T = \frac{9}{5}T + 32 \Rightarrow T - \frac{9}{5}T = 32$$

$$-\frac{4}{5}T = 32 \Rightarrow -4T = 160$$

$T = -40°$, Thus, $-40°F = -40°C$

b) $T_k = T_C + 273$, if $T_k = T_C = T$

$T = T + 273$

$O = 273 \rightarrow$ False, hence the Celsius and kelvin scales can never by equal.

Illustrative Example

14. Convert 10°C into
    a) Kelvin b) Fahrenheit

Solution: a) $T_k = T_C + 273 = 10 + 273 = 283k$

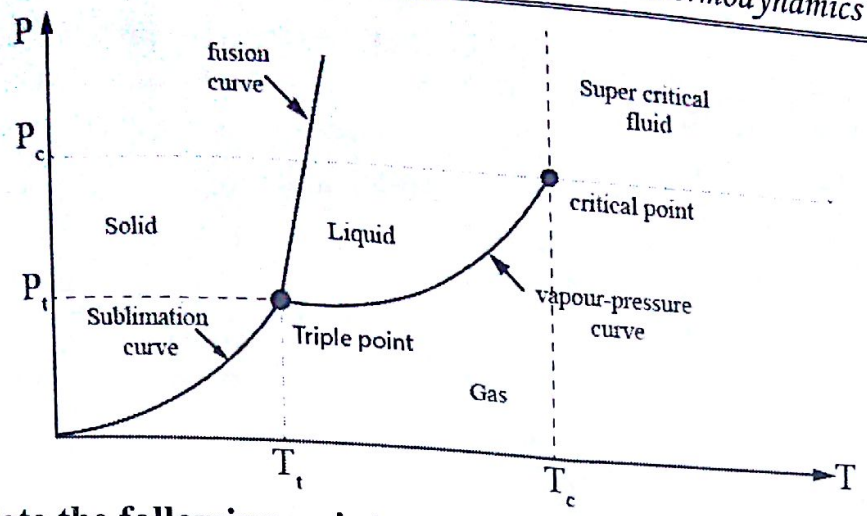
b) $T_F = \frac{9}{5}T_C + 32 = \frac{9}{5}(10) + 32 = 50°F$

Phase of Matter

The solid, liquid, gas states are often called *phases of matter.*

Phase (state of matter): Is a point at which all the physical properties such as shape and volume with in a material are uniform. At a given phase, a material will have the same density and refractive index.

For instance, water exists in the three states at different temperature and pressure. That is, solid → ice, liquid → water and gas → vapour. Changing the pressure and temperature of a substance will affect its phase. Plotting a graph of pressure V_s temperature is often called *phase diagram.*



☛ **Note the following points**

- **Sublimation curve:** along which solid and gas phases are in equilibrium.
- **Fusion curve**: indicates where the solid and liquid phase are in equilibrium.
- **Vapor – pressure curve:** which separates liquid and gas phases.

Triple point: Is a specific temperature and pressure at which all the three states of matter coexist, i.e., become in equilibrium.

→ For water, the triple point is at temperature of 273.16k and pressure of 611.2Pa.

Critical point: Is a point beyond which there is no distinction between the liquid and gas. The two phases merge together and become one and the same forming the fluids often called *super critical fluids*.

→ For water, the critical point is around 647k and 22Mpa.

Conceptual Example

15. Is there any difference between boiling and critical points explain.

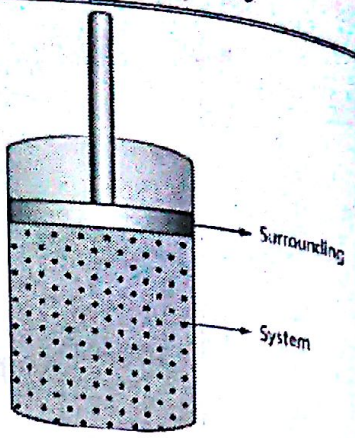
Explanation: Yes! A substance can have a number of boiling points depending on the pressure. But; the critical point only occur at one specific temperature and pressure.

Internal Energy and the First Law of Thermodynamics

In thermodynamics, the object for which attention is being focused is called the *system*, while everything else around the system is called *surrounding*.

System: Is a body (solid, liquid or gas) that is subject of study.
Surrounding: Is everything outside the system which has a direct effect on the system.

For example; consider a gas enclosed in a cylinder fitted with a piston.
The gas can be the thermodynamic system, while the piston, the wall of the cylinder and the outside air can be the surrounding.



The thermodynamic variables (parameters) that help us to study the behavior of a thermodynamic system may be:

a) Microscopic, at the atomic and molecular level, such as the speed of particles.
b) Macroscopic, at the large scale level, such as temperature, pressure and volume of a gas in a container. The three macroscopic variables, volume, pressure and temperature are all a sequence of the motion of the atoms of the gas.

☛ **Remember**: The volume is a result of the freedom the atoms have to spread throughout the container, the pressure is a result of the collision of the atoms with the wall of the container, and the temperature has to do with the kinetic energy of the atoms.

Remark: We can speak about the temperature, pressure or volume of a gas. But, it makes no sense to speak about the temperature, pressure or volume of an individual gas molecule.

Internal Energy of a System

Matter consists of atoms and molecules which in turn are made up of particles having kinetic and potential energies. The sum of all the kinetic and potential energies of particles of a system is called *internal energy*.

> **Internal Energy**: Is the total energy content of a system. It includes, kinetic, potential, thermal, chemical, electrical, nuclear and other forms of energies possessed by the atoms and molecules of the system.

In symbol, internal energy is written as u, while change in internal energy is written as Δu.

☛ **Remember**, for an ideal gas, there is no intermolecular interaction among particles other than collision, and hence there is no potential energy. As a result, the internal energy of the system is the sum of the kinetic energy of each of its molecules.

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First Law of Thermodynamics

A thermodynamic system may exchange energy with the surrounding in two ways.

a) Via work done by or on the system which involves displacement due to the application of external force or pressure.

b) Via heat transfer into or out of the system.

The exchange of energy between the system and the surrounding may lead the internal energy of the system to change, and this is well explained by the first law of thermodynamics.

> **First Law of Thermodynamics**, states that; "The change in the internal energy of a system is equal to the sum of the energy entering or leaving the system through heat and the work done on or by the system."

In symbols,

$$\Delta u = \Delta Q + \Delta W$$

Where; $\Delta u = u_f - u_i$ is the change in internal energy

ΔQ = heat added or removed from the system

ΔW = work done on or by the system

Sign Convention

While using the above equation, we should stick to the following sign conventions.

i) Heat flow into the system is taken as positive, while heat flow out of the system is taken as negative.

ii) Work done on the system is taken as positive, while work done by the system is taken as negative.

iii) Increase in the internal energy of a system is taken as positive, while decrease in the internal energy of the system is taken as negative.

The other option of writing the first law is the relation,

$$\Delta u = \Delta Q - \Delta W$$

In this case, the other sign conventions are the same as above except that of Δw. So here,

→ Work done on the system is taken as negative, while work done by the system is taken as positive.

The first law can also be written in the form,

$$\Delta u = \Delta Q - P\Delta v$$

Where; P = pressure exerted on or by the system

$\Delta v = v_f - v_i$ = change in volume of the system and,

$\Delta w = P\Delta v$, because $\Delta w = F\Delta s$, $P = \frac{\text{Force}}{\text{Area}}$

$\Rightarrow F = PA$, substitute for F

$\Delta w = (PA)(\Delta s) = P\Delta V$, where $\Delta v = (A)(\Delta s)$

☛ **Important points to remember**

Work: Is the energy transferred from one body to another due to the action of a force.

Heat: Is the transfer of energy between the system and its surrounding because of temperature difference.

Internal energy: Is the total energy associated with the particles of a system.

Isolated system: Is a system that does not interact with its surrounding so its internal energy remains constant. Isolated system do not exchange energy by heat and work. Thus, $\Delta Q = 0$ and $\Delta W = 0$. Hence for isolated system, $\boxed{\Delta u = 0}$.

Insulted System: Is a system that allows no transfer of energy by heating, $\Delta Q = 0$.

Conceptual Examples

16. Is it possible for the temperature of a substance to rise without heat flowing into it? Explain

Explanation: Yes. When you pump a tire, you may notice that the pump and the tire get warm. This temperature increase is due to the compressed air.

17. Is it possible to convert internal energy to mechanical energy? Explain

Explanation: Yes it is possible. An expanding gas in a cylinder pushes the piston upward and hence does mechanical work on the surrounding.

18. A gas expands, doing 100J of work. How much heat must be added to this system for its internal energy to increase by 200J.

Explanation: $\Delta u = \Delta Q + \Delta W \Rightarrow 200 = \Delta Q - 100$

$\Rightarrow \Delta Q = 300J$

Illustrative Examples

19. An ideal gas absorbs 5×10^3J of heat and does 2×10^3J of work on the environment. Calculate the change in the internal energy of the gas.

Solution: Method I Method II

$\Delta u = \Delta Q + \Delta W$
$= 5 \times 10^3 + (-2 \times 10^3)$
$= 3 \times 10^3 J = 3kJ$

$\Delta u = \Delta Q - \Delta W$
$= 5 \times 10^3 - (2 \times 10^3)$
$= 3 \times 10^3 J = 3kJ$

20. A thermodynamic system undergoes a process in which its internal energy decreases by 500J. If at the same time 220J of work is done on the system. Find the heat energy transferred into or form it.

Solution: $\Delta u = \Delta Q + \Delta W$
$-500 = \Delta Q + 220$
$\Delta Q = -720J \rightarrow$ 720J of heat flows out (lost)

21. A gas is compressed at a constant pressure of 0.8 atm from 9L to 2L. In the process, 400J of heat leaves the gas. What is the change in its internal energy?

Solution: $1 \text{ atm} = 1.01 \times 10^5$ Pa
$P = 0.8 \text{ atm} \Rightarrow 0.8 \times (1.01 \times 10^5) = 8.08 \times 10^4$ Pa
$1L = 10^{-3} m^3$
$v_1 = 9L = 9 \times 10^{-3} m^3$ and $v_2 = 2L = 2 \times 10^{-3} m^3$
$\Delta u = \Delta Q - P\Delta v$
$= -400 - (8.08 \times 10^4)(2 \times 10^{-3} - 9 \times 10^{-3})$
$= -400 - (-566) = 166J$

22. A 40W heat source is applied to a gas sample for 25 sec during which the gas expands and does 750J of work on its surrounding. By how much does the internal energy of the gas change?

Solution: $\Delta u = \Delta Q + \Delta W$, $\Delta Q = (P)(t)$
$= (40)(25) - 750 = 250J$

23. What is the change in internal energy of a system if it absorbs 400kJ of heat from the surrounding and does 500kJ of work on the surrounding?

A. 100kJ B. -100kJ C. 400kJ D. 500kJ

Solution: $\Delta u = \Delta Q + \Delta W = 400 - 500 = -100kJ$ **Answer: B**

24. The first law of Thermodynamics is a restatement of the law of conservation of

A. Energy C. Momentum
B. Mass D. Charge **Answer: A**

25. The internal energy of a gas is given by $u = 1.5pv$. It expands from 100 cm^3 to 200cm^3 against a constant pressure of 1×10^5Pa. Calculate the heat absorbed by the gas in the process.

A. 250J B. 25J C. 2.5J D. 50J

Solution: Work done at constant pressure.

$\Delta W = P\Delta v = P(V_2 - V_1)$
And internal energy, $u = 1.5pv$
$\Delta u = u_2 - u_1 = 1.5pv_2 - 1.5pv_1$
$\Delta u = 1.5P(v_2 - v_1)$
From first law of thermodynamics,
$\Delta u = \Delta Q + \Delta W$, here ΔW is negative (expansion)
$\Rightarrow \Delta u = \Delta Q - \Delta W$
$\Delta Q = \Delta u + \Delta W = 1.5P(v_2 - v_1) + P(v_2 - v_1)$
$\Delta Q = 2.5P(v_2 - v_1) = 2.5 \times 10^5 (2 \times 10^{-4} - 1 \times 10^{-4}) = 25J$

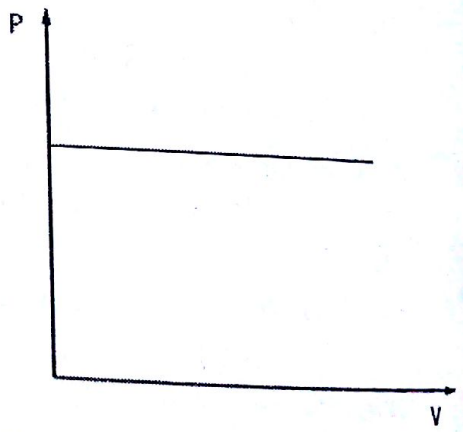
Answer: B

Thermodynamic Processes

A thermodynamic process is said to take place when the state of the system changes due to changes of variables such as temperature, pressure or volume with time. The following are common thermodynamic processes.

A) **Isobaric Process**: It takes place at constant pressure.

In a p – v graph an isobaric process is represented by a horizontal line.



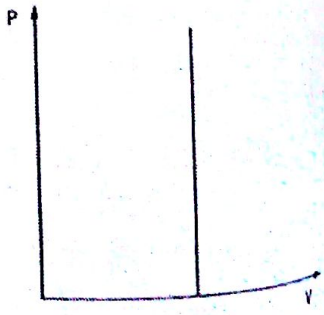
For isobaric process, the first law of thermodynamics takes the usual form.

$\Delta u = \Delta Q - P\Delta v$

☛ Isobaric process can be achieved by adding heat slowly at constant pressure.

B) **Isochoric (isovolumetric) Process:** It takes place at constant volume.

In a p – v graph, an isochoric process is represented by a vertical line



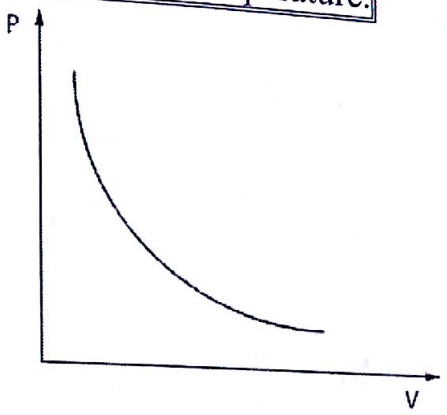
For isochoric process the first law of thermodynamics is modified to;
$\Delta u = \Delta Q - P\Delta v$, but $v \rightarrow$ constant $\Rightarrow \Delta v = 0$
here work done is zero because $\Delta w = P\Delta v = 0$

$\boxed{\Delta u = \Delta Q}$

- Isochoric process can occur when a gas is heated in a container of fixed (rigid) volume – a volume that does not change with time.

C) Isothermal Process: It takes place at constant temperature.

In a p – v graph, an isothermal process is represented by a curve called *isotherm* for an isothermal process, the internal energy of the system is constant. And in this case, the first law of thermodynamics is modified to;



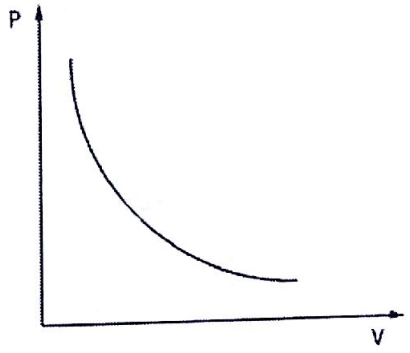
$\Delta u = \Delta Q + \Delta W$, but $u_i = u_f \Rightarrow \Delta u = 0$

$\boxed{\Delta Q = -\Delta W}$ $= -P\Delta V$

- A process has to be extremely slow to be isothermal.

D) **Adiabatic Process:** It takes place without transfer of heat into or out of the system.

In a p – v graph, an adiabatic process is represented by a curve that is steeper than the isothermal curve.



For an adiabatic process, the first law of thermodynamics becomes,

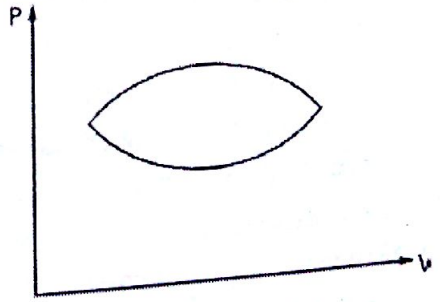
$\Delta u = \Delta Q + \Delta W$, but $\Delta Q = 0$

$\boxed{\Delta u = \Delta W}$

- A process has to be sudden and rapid to be adiabatic.

E) **Cyclic Process:** Is a process that returns to its initial state after a series of changes.

In a p – v graph, a cyclic process is represented by a closed loop.
In this case, the change in internal energy is zero, and hence the first law of thermodynamics is written as



$\Delta u = \Delta Q + \Delta W$, $\Delta u = 0$

$\boxed{\Delta Q = -\Delta W}$

D) **Quasi – Static Process**: It takes place extremely slowly such that at every instant of time, the temperature and pressure are the same in every part of the system.

Conceptual Examples

26. Is it possible to change the thermal energy of a system without adding to or removing heat from it? Explain

Explanation: Yes it is possible. When a gas is compressed its thermal energy increases because of collision of particles. And when the gas expands rapidly, it cools. In each case energy is transferred to or from the system as work.

27. a) Why is it a must for an isothermal process to be a very slow process.
    b) Why is it necessary for the change in the internal energy of the system to be zero? Explain
    c) Can heat flow into or out of a system undergoing isothermal process.

Explanation: a) For isothermal process the temperature must be constant. For this purpose, the exchange of heat between the system and its surrounding must be slowly so that thermal equilibrium is maintained.

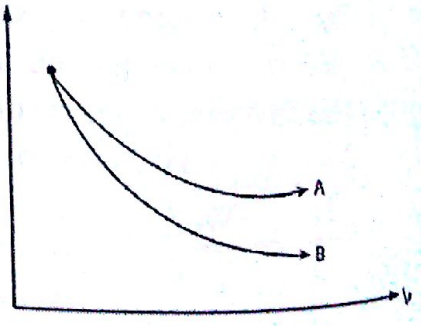
b) Temperature is a measure of kinetic energy or internal energy. Hence if the temperature is constant, the internal energy is also constant. Thus the change in internal energy is zero.

c) Yes. There is exchange of thermal energy between the system and the surrounding at constant temperature.

28. It is said that, adiabatic process can be achieved either by thermally insulting the system from its surrounding or by performing the process rapidly. How? Explain

Explanation: In both cases, there is no chance and there is no enough time to exchange heat.

29. From the p – v graph shown, identify the isothermal and the adiabatic process.



Explanation: An adiabatic curve is steeper than the isothermal curve. In our case, curve A represent isothermal while curve B represents

adiabatic process.

30. Fill +, −, or 0 for each of the following cases.

	ΔQ	ΔW	Δu
a) Rapidly pumping a bicycle tire	0	+	+
b) Water in a pan sitting on a hot stove	+	0	+
c) Air quickly leaving out of a ball	0	−	−

Explanation: a) Because the pump is rapid, it is adiabatic, $\Delta Q = 0$ and work is done on the system and hence the internal energy increases.

b) No work is done, it is isochoric, $\Delta W = 0$ and heat is supplied on the system and the internal energy increase.

c) It is quick process, so adiabatic, $\Delta Q = 0$ but the gas do work on the surrounding hence internal energy decreases.

Illustrative Example

31. 300J of energy is transferred to a system in the form of heat and the internal energy is decreased by 150J. How much work is done on or by the system?

Solution: $\Delta u = \Delta Q + \Delta W \Rightarrow -150 = 300 + \Delta W \Rightarrow \Delta W = 450J$

32. 10J of heat is removed from a gas and 20J of work is done on it by compression. What is the change in the internal energy of the gas. Does the temperature of the gas increase or decrease?

Solution: $\Delta u = \Delta Q + \Delta W \Rightarrow \Delta u = -10 + 20 = 10J$ the temperature increased because the internal energy is positive.

Molar Heat Capacity

Molar Heat Capacity: Is heat capacity per mole.

We have molar heat capacity at constant volume and molar heat capacity at constant pressure.

i) **Molar heat capacity at constant**

$$C_v = \frac{\Delta Q}{n\Delta T} \Rightarrow \boxed{\Delta Q_v = C_v n\Delta T} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

First law of thermodynamics at constant volume is,

$$\boxed{\Delta u = \Delta Q_v} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

ii) **Molar heat capacity at constant pressure**

$$C_p = \frac{\Delta Q_p}{n\Delta T} \Rightarrow \boxed{\Delta Q_p = C_p n\Delta T} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3)$$

First law of thermodynamics at constant pressure is,

$$\boxed{\Delta u = \Delta Q_p - P\Delta v} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

Now, from equation (2) $\Delta u = \Delta Q_v$ substituting it in to equation (4)

$$\Delta Q_v = \Delta Q_P - P\Delta v$$

And from equations (1) and (3) we have,

$$\boxed{C_v n\Delta T = C_P n\Delta T - P\Delta v} \quad \ldots\ldots\ldots\ldots\ldots\ldots (5)$$

From ideal gas equation, pv = nRT, at constant pressure, volume changes as temperature changes

$$\boxed{P\Delta v = nR\Delta T} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (6)$$

Substituting $nR\Delta T$ for $P\Delta v$ in equation (5) we get

$$C_v n\Delta T = C_P n\Delta T - nR\Delta T$$

Rearranging this gives

$$\boxed{C_P - C_v = R} \leftarrow \text{Mayer's equation}$$

Where R = 8.3 J/mol.k is the *universal gas constant.*

i) For a monatomic ideal gas, $C_v = \frac{3}{2}R$

From, $C_P - C_V = R$

$$C_P = R + C_v = R + \frac{3}{2}R$$

$$\Rightarrow C_P = \frac{5}{2}R$$

The molar heat capacity ratio for monoatomic gas is then

$$\frac{C_P}{C_v} = \gamma = \frac{\frac{5}{2}R}{\frac{3}{2}R}$$

$$\gamma = \frac{5}{3} = 1.67 \leftarrow \text{Monoatomic gases such as He, Ar, kr}$$

ii) For a diatomic ideal gas, $C_v = \frac{5}{2}R$

Fom, $C_P - C_v = R$

$$C_P = R + C_v = R + \frac{5}{2}R$$

$$C_P = \frac{7}{2}R$$

The molar heat capacity ratio for diatomic gas is then

$$\frac{C_P}{C_v} = \gamma = \frac{\frac{7}{2}R}{\frac{5}{2}R}$$

$$\gamma = \frac{7}{5} = 1.4 \leftarrow \text{Diatomic gases such as } N_2, O_2, H_2$$

Conceptual Example

33. From the equation $C_P - C_v = R$, molar heat capacity at constant pressure is greater than that at constant volume. Why?

Explanation: At constant volume, $\Delta Q_v = \Delta u$, the heat added goes entirely to increase the internal energy. While, at constant pressure, $\Delta Q_P = \Delta u + P\Delta v$, the heat supplied increase the internal energy and does mechanical work. Hence the case $C_P > C_v$.

In isochoric, volume is constant, in isobaric pressure is constant, in isothermal, temperature is constant but in adiabatic, all the variables, that is V, P and T may change. And we have the following;

- For a reversible, adiabatic process, it is true that,

• $PV^\gamma = \text{constant}$
$\rightarrow P_1V_2^\gamma = P_2V_2^\gamma$
• $TV^{\gamma-1} = \text{constant}$
$\rightarrow T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$
• $P^{1-\gamma}T^\gamma = \text{constant}$
$\rightarrow P_1^{1-\gamma}T_1^\gamma = P_2^{1-\gamma}T_2^\gamma$

Proof A) Adiabatic relation between P and v for ideal gas.

We know, $\Delta u = \Delta Q - P\Delta v$, 1st law of thermodynamics

$\Delta u = -P\Delta v$, $\Delta Q = 0$, adiabatic

$C_v n\Delta T = -P\Delta v$,

$C_v n dT + Pdv = 0$, ... (1)

From ideal gas equation, $pv = nRT$

differentiating it, $pdv + vdp = nRdT$

$\Rightarrow dT = \frac{pdv + vdp}{R}$, substituting in equation (1)

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$$C_v n\left(\frac{Pdv + vdp}{R}\right) + pdv = 0, \text{ for one mole } n = 1$$

$$C_v\, Pdv + C_v VdP + PRdv = 0$$

$$(C_v + R)\, Pdv + C_v Vdp = 0,\ C_P = C_v + R$$

$$C_P pdv + C_v vdp = 0$$

Dividing both sides by $C_v PV$, we have

$$\frac{C_P Pdv}{C_v pv} + \frac{C_v Vdp}{C_v Pv} = 0$$

$$\gamma\frac{dv}{v} + \frac{dp}{P} = 0, \frac{C_P}{C_v} = \gamma$$

Integrating $\int 0 = \text{constant}$

$$\int \gamma\frac{1}{v}dv + \int\frac{1}{p}dp = \text{constant} \qquad \int\frac{1}{x}dx = \ln x$$

$$\gamma \ln v + \ln P = \text{constant} \qquad k \ln y + \ln x = \ln xy^k$$

$$\ln pv^\gamma = \text{constant} \qquad \text{Antilog (constant)} = \text{constant}$$

$$\boxed{pv^\gamma = \text{constant}}$$

B) Adiabatic relation between volume and temperature

From ideal gas equation, $pv = nRT \Rightarrow P = \frac{nRT}{v}$ and $P_1v_1^\gamma = P_2v_2^\gamma$,

adiabatic $P_1 = \frac{nRT_1}{v_1}, P_2 = \frac{nRT_2}{v_2}$

$$\left(\frac{nRT_1}{V_1}\right)\left(V_1^\gamma\right) = \left(\frac{nRT_2}{V_2}\right)\left(V_2^\gamma\right) \qquad \frac{x^a}{x} = x^{a-1}$$

$$nRT_1\frac{V_1^\gamma}{V_1} = nRT_2\frac{V_2^\gamma}{V_2}$$

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\boxed{TV^{\gamma-1} = \text{constant}}$$

C) Adiabatic relation between pressure and temperature

From ideal gas equation, $PV = nRT \Rightarrow V = \frac{nRT}{P}$ and $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$$, V_1 = \frac{nRT}{P_1}, V_2 = \frac{nRT_2}{P_2}$$

$$P_1\left(\frac{nRT_1}{P_1}\right)^{\gamma} = P_2\left(\frac{nRT_2}{P_2}\right)^{\gamma}$$

$$\frac{P_1 n^{\gamma} R^{\gamma} T_1^{\gamma}}{P_1^{\gamma}} = \frac{P_2 n^{\gamma} R^{\gamma} T_2^{\gamma}}{P_2^{\gamma}}$$

$$\frac{P_1}{P_1^{\gamma}} T_1^{\gamma} = \frac{P_2}{P_2^{\gamma}} T_2^{\gamma}$$

$$P_1^{1-\gamma} T_1^{\gamma} = P_2^{2-\gamma} T_2^{\gamma}$$

$$\boxed{P^{1-\gamma} T^{\gamma} = \text{constant}}$$

Illustrative Examples

34. A 20 cm^3 monoatomic ideal gas at 12°C and 1×10^5 Pa is adiabatically compressed to 0.5cm^3. Calculate
    a) The new pressure
    b) The new temperature

Solution: a) $T_1 = 12 + 273 = 285k$

$$P_1v_1^{\gamma} = P_2v_2^{\gamma}, \text{ adiabatic}$$

$$P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma}, \gamma = \frac{5}{3} = 1.67, \text{monatomic}$$

$$= 1\times10^5\left(\frac{20}{0.5}\right)^{1.67} = 4.37\times10^7 \text{ Pa}$$

b) $$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$T_2 = T_1\left(\frac{v_1}{v_2}\right)^{\gamma-1} = 285\left(\frac{20}{0.5}\right)^{1.67-1} = 337k$$

35. A container with an initial volume of 0.06m^3 holds 2.5 mol of monatomic ideal gas at a temperature of 315 k. If the gas is compressed adiabatically to 0.03m^3. Find:
    a) The final pressure.

b) The final temperature.

Solution: a) From ideal gas equation, pv = nRT

$$P_1v_1 = nRT_1 \Rightarrow P_1 = \frac{nRT_1}{v_1} = \frac{(2.5)(8.31)(315)}{(0.06)} = 1.1\times10^5\,Pa$$

Then,

$$P_1v_1^{\gamma} = P_2v_2^{\gamma} \Rightarrow P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma} = 1.1\times10^5\left(\frac{0.06}{0.03}\right)^{1.67} = 3.5\times10^5\,Pa$$

b) $P_2v_2 = nRT_2$

$$T_2 = \frac{P_2v_2}{nR} = \frac{(3.5\times10^5)(0.03)}{(2.5)(8.31)} = 505k$$

36. A certain ideal gas $\left(\gamma = 3/2\right)$ with an initial pressure P and initial temperature T is adiabatically compressed to one – fourth of its original volume. Calculate

a) The final pressure.

b) The final temperature.

Solution: a) For the gas, $\gamma = \frac{3}{2}$

$$P_1v_1^{\gamma} = P_2v_2^{\gamma},\ P_1 = P \text{ and } v_2 = \frac{1}{4}v_1$$

$$P_2 = P\left(\frac{V_1}{\frac{1}{4}V_1}\right)^{\gamma} = P(4)^{3/2} = 8P$$

b) $T_1\,v_1^{\gamma-1} = T_2v_2^{\gamma-1}$

$$T_2 = T\left(\frac{v}{1/4\,v}\right)^{\gamma-1} = T(4)^{3/2-1} = 2T$$

37. A diatomic ideal gas at pressure P and temperature T is adiabatically compressed so that its density becomes twice that of initial value. Calculate

a) The final pressure

b) The final temperature

Solution: a) For diatomic ideal gas, $\gamma = 7/5$, and $P_1 v_1^{\gamma} = P_2 v_2^{\gamma}$

$$\rho = \frac{m}{v} \Rightarrow v = \frac{m}{\rho}$$

$$P_1\left(\frac{m_1}{\rho_1}\right)^{\gamma} = P_2\left(\frac{m_2}{\rho_2}\right)^{\gamma}, m_1 = m_2 = m$$

$$P_1\frac{m^{\gamma}}{\rho_1^{\gamma}} = P_2\frac{m^{\gamma}}{\rho_2^{\gamma}},$$

$$P_2 = P_1\left(\frac{\rho_2}{\rho_1}\right)^{\gamma}, \rho_2 = 2\rho_1$$

$$P_2 = P\left(\frac{2\rho_1}{\rho_1}\right)^{7/5} = 2.6P$$

b) $T_1 v_1^{\gamma-1} = T_2 v_2^{\gamma-1}$

$$T_1\left(\frac{m^{\gamma-1}}{\rho_1^{\gamma-1}}\right) = T_2\left(\frac{m^{\gamma-1}}{\rho_2^{\gamma-1}}\right)$$

$$T_2 = T\left(\frac{\rho_2}{\rho_1}\right)^{\gamma-1} = T\left(\frac{2\rho_1}{\rho_1}\right)^{1.4-1}$$

$$T_2 = 2^{0.4}T \simeq 1.3T$$

Work Done in Thermodynamic Process

In general, the work done by a thermodynamic system is the area under a $p-v$ graph.

> The work done by a system on its surrounding is equal to the area under the $p-v$ graph.

Specifically, the work done by the system on each thermodynamic process is given as follows.

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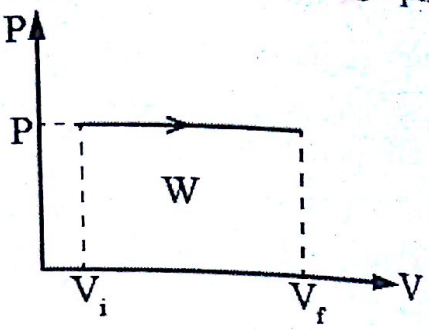
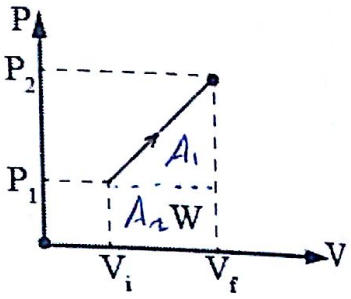
A) Work done during Isobaric process

In isobaric process, the pressure is constant and hence the p – v graph is horizontal line.



$$W = p\,\Delta v = P(v_f - v_i)$$

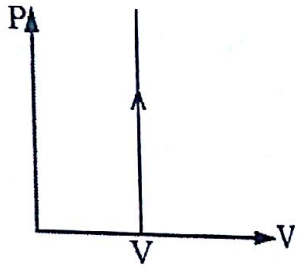
- If the pressure is changing linearly from P_1 to P_2 and the volume is changing from v_i to v_f, then we have.



$$W = (P_{av})(\Delta v) = \left(\frac{P_1 + P_2}{2}\right)(v_f - v_i)$$

B) Work done during Isochoric process

In isochoric process, the volume is constant and the p – v graph is a vertical line.

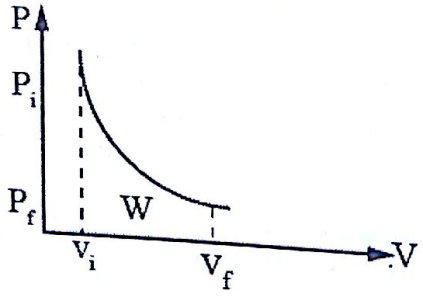


$$W = P\Delta v = P(v_i - v_f) = 0,\ v_i = v_f$$

Work done is zero during isochoric process.

C) Work done in an Isothermal process

In isothermal process, the temperature of the system is constant and the p – v graph is a hyperbolic curve.



$$W = \int_{v_i}^{v_f} p\,dv,\ \text{but } pv = nRT \Rightarrow p = \frac{nRT}{v}$$

$$W = \int_{v_i}^{v_f} nRT\frac{dv}{v} = nRT = \int_{v_i}^{v_f} \frac{dv}{v} = nRT[\ln]_{v_i}^{v_f}$$

$$W = nRT \ln\left(\frac{v_f}{v_i}\right)$$

D) Work done during adiabatic process

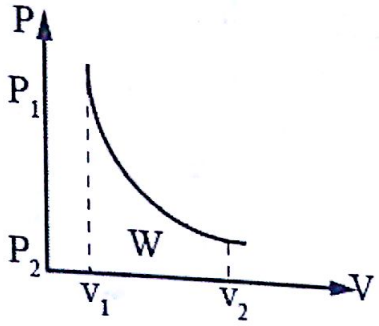
In adiabatic process, there is no thermal energy exchange between the system and the surrounding and the p – v graph is a hyperbolic curve that is steeper than that of the isothermal curve.

From 1st law of thermodynamics

$\Delta u = \Delta Q - \Delta W$, $\Delta Q = 0 \rightarrow$ adiabatic

$du = -dw$, but $du = C_v n dT$

$\Rightarrow dW = -C_v n dT$, integrating both side



$$W = \int_{T_1}^{T_2} -C_v n dT$$

$$W = -C_v n \int_{T_1}^{T_2} dT = -C_v n [T]_{T_1}^{T_2}$$

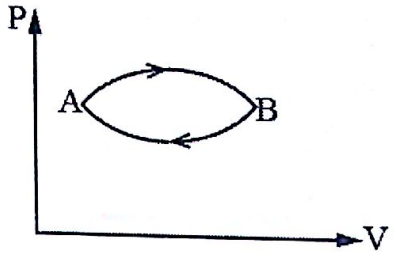
$$W = -C_v n [T_2 - T_1]$$

$$\boxed{W = C_v n (T_1 - T_2)} \quad \ldots\ldots\ldots\

$$W = \frac{P_1v_1 - P_2v_2}{\gamma - 1}$$

E) Work done in cyclic process

In a cyclic process, the internal energy of the system is constant and hence $\Delta u = 0$. In this case, the p – v graph is a closed loop.



Work done = area enclosed by the cycle, say ABA is this case.

Remark: Work done depends on initial and final states as well as on the path followed.

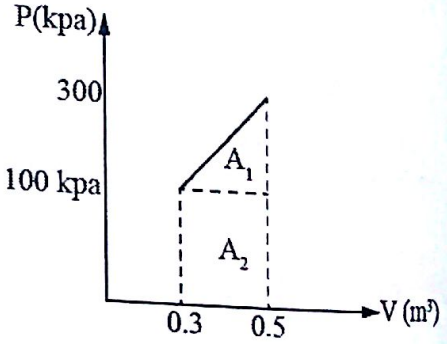
Illustrative Examples

38. A gas with a constant pressure of 1.5×10^5Pa expands from a volume of $0.76m^3$ to volume of $0.92m^3$. How much work does the gas do?

Solution: At constant pressure, $W = P\Delta v$

$W = P\Delta v = (1.5 \times 10^5)(0.92 - 0.76) = 24{,}000J = 24kJ$

39. A gas expands from an initial volume of $0.3m^3$ to a final volume of $0.5m^3$ as the pressure increases linearly from 100kPa to 300kPa. Find the work done by the gas.



Solution: **Method I** graphically, picturing the problem we have,

Work done = Area under the p – v graph

$W = A_1 + A_2$

$$= \frac{1}{2}(0.5 - 0.3)(3 \times 10^5 - 1 \times 10^5) + (0.5 - 0.3)(1 \times 10^5) = 4 \times 10^4 J$$

Method II: Analytically, taking average pressure.

$$W = (P_{av})(\Delta v) = \left(\frac{P_1 + P_2}{2}\right)(v_2 - v_1)$$

$$= \frac{(1 \times 10^5 + 3 \times 10^5)}{2}(0.5 - .3) = 4 \times 10^4 J$$

40. A 2 mol monatomic ideal gas at a temperature of 25°C expands isothermally from a volume of $0.036m^3$ to $0.072m^3$. Find:
a) The change in internal energy of the gas.
b) The heat flow into or out of the gas.

Solution: a) $T = 25 + 273 = 298k$

$$W = nRT \ln\left(\frac{v_f}{v_i}\right) = (2)(8.31)(298)\ln\left(\frac{0.072}{0.036}\right)$$

$$= 3400J = 3.4kJ$$

In isothermal process, $u = \frac{3}{2}nRT$ for monatomic gas and

$$\Delta u = \frac{3}{2}nR\Delta T \text{ , but } \Delta T = 0$$

$$\Delta u = 0$$

b) From first law, $\Delta u = \Delta Q + \Delta W$, $\Delta u = 0$
$\Delta Q = \Delta W = 3400J$

41. A balloon of volume $2m^3$ contains 3.5 moles of helium at a temperature of 300k. If the gas is compressed adiabatically so as to have a final temperature of 400k. What is the work done by the gas?

Solution: $\Delta W = \frac{3}{2}nR\Delta T$, monatomic

$$= \frac{3}{2}(3.5)(8.31)(400-300) = 4400 = 4.4kJ$$

42. A diatomic ideal gas expands adiabatically from initial pressure and volume of 2 atm and 2L to final pressure and volume of 0.758am and 4L. Find the work done by the gas.

Solution: $W = \frac{P_1v_1 - P_2v_2}{\gamma - 1}$, $\gamma = \frac{7}{5} = 1.4$ diatomic

$$W = \frac{(2)(2)-(0.758)(4)}{1.4}, \quad 1 \text{ atm} = 1.01 \times 10^5 Pa, \quad 1L = 10^{-3}m^3$$

$$= 2.42L \text{ atm} = (2.42)(10^{-3})(1.01 \times 10^5) = 245J$$

43. In a cylinder 2 moles of an ideal mono tomic gas initially at $1 \times 10^6 Pa$ and 300k expands until its volume is doubled. Compute the work done if the expansion is:

a) Isobaric b) Isothermal c) Adiabatic

Solution: a) $\Delta W = P\Delta v$, from $pv = nRT$

$$v_1 = \frac{nRT}{P} = \frac{(2)(8.31)(300)}{1\times10^6} \simeq 5\times10^{-3}\,m^3 \quad v_2 = 2v_1$$

$$\Delta W = P\Delta v = P(v_2 - v_1) = 10^6(10\times10^{-3} - 5\times10^{-3} = 4980J$$

b) $$\Delta W = nRT\ln\left(\frac{v_2}{v_1}\right) = (2)(8.31)(300)\ln(2) = 3452J$$

c) $\Delta W = \frac{3}{2}nRT$, from $P_1v_1^{\gamma} = P_2v_2^{\gamma} \Rightarrow P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma}$

$$P_2 = 10^6(0.5)^{5/3} \simeq 3\times10^5 Pa \text{ and } T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$\Rightarrow T_2 = T_1\left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

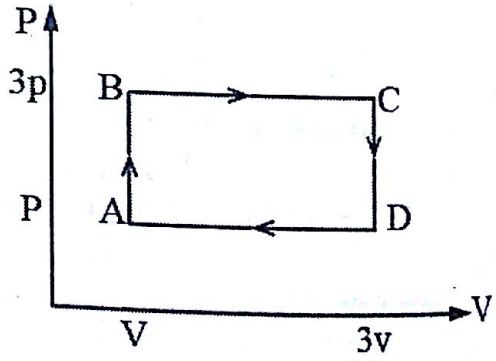
$$T_2 = 300\,(0.5)^{2/3} \simeq 189k$$

$$\Delta W = \frac{3}{2}(2)(8.31)(300-189) = 2764J$$

44. An ideal monatomic gas is taken round the cycle ABCDA as shown. What is the work done during the cycle?

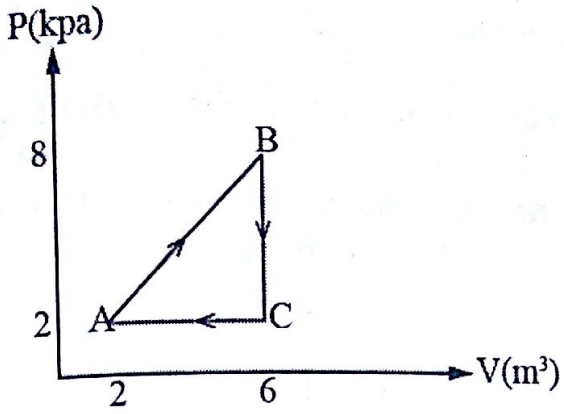
Solution: Work done = Area of ABCDA

$W = (3v - v)(3P - P)$
$W = 4Pv$



45. A gas is taken through a cyclic process shown. Find the work by the gas at the end of the cycle.

Solution: $$W = \frac{1}{2}(6-2)(8-2)\times10^3 = 24000J = 24kJ$$



46. A 2 mole helium gas at initial pressure and volume of 2.5×10^5Pa and 20L expands isobaricaly from a to b to a volume of 40L. Then it expands adiabatically from pressure and volume of 2.5×10^5 Pa and 40L to a final values of 0.5×10^5 Pa and 113L along b – c. Calculate the net work done in the process.

Solution: Part I (along a → b) for isobaric, $W = A_1$

$W_1 = (40 - 20)(2.5 \times 10^5) = 5000J = 5\ kJ$

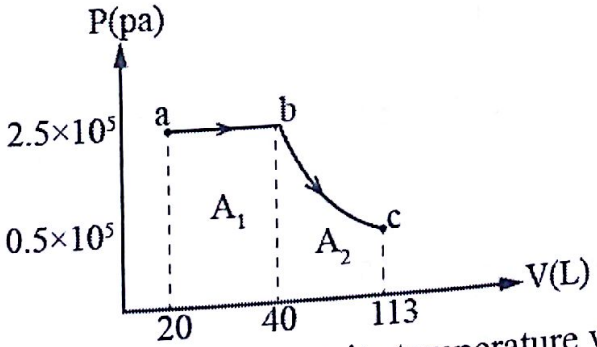
Part II: (along b → c) for adiabatic,

$$W_2 = \frac{P_1v_1 - P_2v_2}{\gamma - 1} = \frac{(2.5\times10^5)(20\times10^{-3}) - (0.5\times10^5)(40\times10^{-3})}{5/3 - 1}$$

$= 4500J = 4.5kJ$

$W = W_1 + W_2 = 5kJ + 4.5kJ$

$W = 9.5kJ$



47. Suppose you want to heat a gas so its temperature will be as high as possible. Would you heat it under conditions of constant pressure or constant volume? Why?

Explanation: At constant volume. Because, when the gas is heated at constant volume, the heat energy entirely goes to increase its internal energy. But when it is heated at constant pressure part of the thermal energy increase the internal energy while the rest does mechanical work.

That is, at constant volume, $\Delta Q_v = \Delta u$, at constant pressure

$\Delta Q_P = \Delta u + P\Delta v$

48. For an ideal gas, show that, the work done is $W = \frac{P_1v_1 - P_2v_2}{\gamma - 1}$.

Derivation: For an adiabatic process, $W = -\Delta v$ and

$\Delta u = nC_v\Delta T$, for any process.

$W = -nC_v\Delta T = nC_v(T_1 - T_2)$

From, $pv = nRT$, $T = \frac{pv}{nR}$

$$W = nC_v\frac{P_1v_1 - P_2v_2}{nR} = \frac{C_v}{R}(P_1v_1 - P_2v_2)$$

But, $R = C_P - C_v$, so

$$\frac{C_v}{R} = \frac{C_v}{C_P - C_v} = \frac{1}{\left(C_P/C_v\right) - 1} = \frac{1}{\gamma - 1}$$

Then we have, $\boxed{W = \frac{P_1v_1 - P_2v_2}{\gamma - 1}}$

49. A certain gas obey's the relation, $P = 45 + 3v$. Calculate the work done by the gas as it expands from $v_1 = 1m^3$ to $v_2 = 3m^3$.

Solution: Method I (analytical)

$$W = \int_{v_1}^{v_2} pdv = \int_{v_1}^{v_2} (45 + 3v)dv = 45v + \frac{3v^2}{2}\Bigg|_{v_1}^{v_2}$$

$$W = 45v_2 + 3\frac{v_2^2}{2} - \left[45v_1 + 3\frac{v_1^2}{2}\right]$$

$$= 45(3) + 3\left(\frac{3^2}{2}\right) - 45(1) - 3\frac{(1)^2}{2} = 102J$$

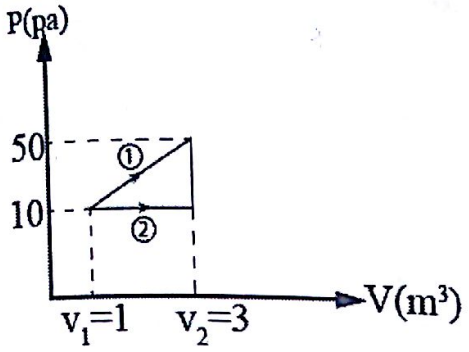
Method II (Graphical)

$P_1 = 45 + 3(1) = 48Pa$

$P_2 = 45 + 3(3) = 54Pa$

$$W = A_1 + A_2 = = \frac{1}{2}(2)(6) + (2)(48) = 6 + 96 = 102J$$

50. A gas expands from v_1 to v_2 via paths 1 and 2. In which case does the gas do more work?

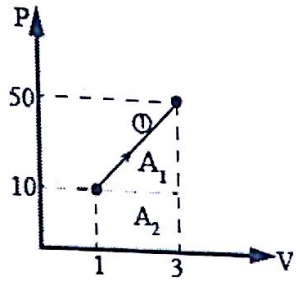


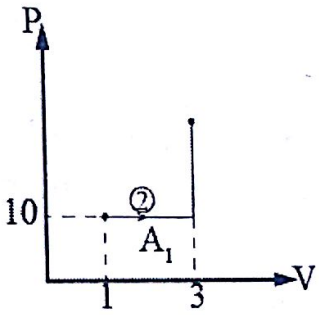
Solution: Work done along path 1 is

$$W = A_1 + A_2 = \frac{1}{2}(2)(40) + (10)(2) = 60J$$

Work done along path 2 is

$$W = A_1 = (10)(2) = 20J$$





The gas does more work while expanding along path 1.

the wor

☛ **Note that**: Work done by a gas is a path dependent.

51. A sample of gas is expanded to twice its original volume of $1m^3$ in a quasi – static process for which $p = \alpha v$, with $\alpha = 5 \times 10^5 Nm^5$. How much work was done by the expanding gas (UEE 2003)

A. 0.75MJ B. 1MJ C. 7.5MJ D. 1.5MJ

Solution: Method I (Analytical)

$$W = \int pdv = \int (\alpha v)dv = \alpha \frac{v^2}{2}\Big|_{v_1}^{v_2} = \alpha \frac{v_2^2}{2} - \alpha \frac{v_1^2}{2}$$

$$W = \frac{5\times10^5}{2}(2^2 - 1^2) = 7.5\times10^5 J = 0.75MJ$$

Method II (Analytical)

$$W = (P_{av})(\Delta v) = \left(\frac{P_1 + P_2}{2}\right)(\Delta v) = \left(\frac{\alpha v_1 + \alpha v_2}{2}\right)(v_2 - v_1)$$

$$= \alpha\left(\frac{1+2}{2}\right)(2-1) = 1.5\alpha = (1.5)(5\times10^5) = 7.5\times10^5 J = 0.75MJ$$

Answer: A

Gas Laws and Ideal gas Equation

The gas laws are empirical laws obtained by experiment.

Boyle's law states that; "At constant temperature, the volume of the gas is inversely proportional to its pressure."

pv = constant, T, constant

$$\Rightarrow \boxed{p_1v_1 = p_2v_2} \leftarrow \text{Boyle's law}$$

Charle's Law states that, "At constant pressure, the volume of the gas is directly proportional to its temperature."

$$\frac{V}{T} = \text{constant, p, constant}$$

$$\Rightarrow \boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}} \leftarrow \text{Charle's law}$$

Gay – Lussac's law states that, "At constant volume, the pressure of the gas is directly proportional to its temperature.

$$\frac{P}{T} = \text{constant, v, constant}$$

$$\Rightarrow \boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2}} \leftarrow \text{Gay – Lussac's law}$$

General Gas law, it is a more general relation that combines the three gas laws.

$$\frac{PV}{T} = \text{constant}$$

$$\Rightarrow \boxed{\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}} \leftarrow \text{General gas law}$$

Illustrative Example

52. What volume of hydrogen gas at atmospheric pressure is required to fill a 5000 cm^3 tank under a pressure of 631 kPa?

Solution: $P_1V_1 = P_2V_2$

$$\Rightarrow V_1 = \frac{P_2V_2}{P_1} = \frac{(631)(5000)}{101} = 31100\ cm^3$$

53. A large balloon filled with air has a volume of 200L at 0°C. What will its volume be at 57°C at constant pressure?

Solution: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, $T_1 = 0 + 273 = 273K$

$$T_2 = 57 + 273 = 330k$$

$$V_2 = \frac{T_2}{T_1}V_1 = \left(\frac{300}{273}\right)(200) = 242L$$

54. An automobile tire is inflated to a pressure of 30kPa at a temperature of 20°C. After the car is driven, the temperature of the air in the tire is increase to 35°C. What will be the new pressure if the volume is fixed?

Solution: $\frac{P_1}{T_1} = \frac{P_2}{T_2}$, $T_1 = 20 + 273 = 293K$

$$T_2 = 35 + 273 = 308K$$

$$P_2 = \frac{T_2}{T_1}P_1 = \left(\frac{308}{293}\right)(30) = 31.5kPa$$

55. A tank with an initial volume of 20L is filled with oxygen at 6×10^6 Pa and 20°C. If the pressure and temperature are changed to 7×10^4 Pa and –20°C respectively. What will be the volume under these conditions?

Solution: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow V_2 = \left(\frac{P_1V_1}{T_1}\right)\left(\frac{T_2}{P_2}\right)$

$$V = \frac{(6\times10^6)(20)}{(293)}\left(\frac{253}{7\times10^4}\right)$$

$$V_2 = 1480L$$

56. An ideal gas of volume $10m^3$ is enclosed in a container at a temperature of 27°C and a pressure of 120 kPa. If its pressure is doubled and its temperature is tripled, what will be its new volume?

Solution: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$, $P_2 = 2P_1$ and $T_2 = 3T_1 \Rightarrow \frac{P_1V_1}{T_1} = \frac{2P_1V_2}{3T_1}$

$$V_1 = \frac{2}{3}V_2 \Rightarrow V_2 = \frac{3}{2}V_1 = 1.5V_1 = (1.5)(10) = 15m^3$$

The Ideal Gas Equation

Though ideal gas do not actually exist in nature, the behavior of real gas is approximated by that of ideal gas.

Ideal Gas: Is an idealized model for real gases that have sufficiently low densities. The condition of low density means that the molecules of the gas are so far apart that they do not interact except elastic collisions.

- Low density can be achieved at low pressure and high temperature. At this state the gas can be ideal.

The law that expresses the relationship between pressure, temperature, volume and number of moles of the gas is known as *ideal gas equation or equation of state*, and is given by:

$$\frac{PV}{T} = \text{constant} = nR$$

$\boxed{Pv = nRT}$ ← ideal gas equation

Where, R is the universal gas constant and has the value of 8.31J/mol.k.

Illustrative Example

57. The standard temperature and pressure (STP) is at a temperature of 0°C and pressure of 1.01×10^5Pa. Find the volume that 1000 mole of an ideal gas occupies at STP.

Solution:

$$PV = nRT \Rightarrow V = \frac{nRT}{P}, \quad T = 0 + 273 = 273k$$

$$V = \frac{(1000)(8.31)(273)}{1.01\times10^5} \simeq 22.4m^3$$

58. How many grams of oxygen will occupy a volume of $1.6m^3$ at a pressure of 200kPa and a temperature of 27°C?

Solution: $PV = nRT$, $n = \frac{m}{M}$, $M = 32g$, $T = 27 + 273 = 300k$

$$PV = \frac{m}{M}RT$$

$$m = \frac{PvM}{RT} = \frac{(200,000)(1.6)(32)}{(8.31)(300)} = 4110g$$

59. A container with an initial volume of $0.06m^3$ holds 2.5 moles of a monatomic ideal gas at a temperature of 315k. If the gas is compressed adiabatically to $0.03m^3$. Find
   a) The final pressure
   b) The final temperature

Solution: From ideal gas equation, $pv = nRT$

a) $P_1v_1 = nRT_1 \Rightarrow P_1 = \frac{nRT_1}{v_1} = \frac{(2.5)(8.31)(315)}{0.06} = 1.1\times10^5\,Pa$

then, $P_1v_1^{\gamma} = P_2v_2^{\gamma}$

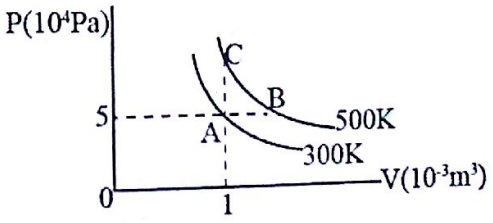
$$P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma} = 1.1\times10^5\left(\frac{0.06}{0.03}\right)^{1.67} = 3.5\times10^5\,Pa$$

b) From $pv = nRT$

$P_2v_2 = nRT_2$

$$\Rightarrow T_2 = \frac{P_2V_2}{nR} = \frac{(3.5\times10^5)(0.03)}{(2.5)(8.31)} = 505K$$

60. A p – v graph is shown for a fixed mass of a monatomic ideal gas at 300k and 500k. The gas is in a cylinder fitted with a freely movable piston. Calculate:
   a) The number of moles of the gas.
   b) The volume of gas B.
   c) The total internal energy of the gas at point A.
   d) The internal energy change along path.
      i) AC
      ii) BC
      iii) AB



Solution: a) At point A, $P_AV_A = nRT_A$

$$n = \frac{P_A V_A}{RT_A} = \frac{(5\times10^4)(1\times10^{-3})}{(8.31)(300)} = 0.02\text{mol}$$

b) The line AB is isobaric, $\frac{P_A V_A}{T_A} = \frac{P_B V_B}{T_B}, P_A = P_B$

$$\frac{V_A}{T_A} = \frac{V_B}{T_B}$$

$$V_B = T_B\left(\frac{V_A}{T_A}\right) = 500\left(\frac{1\times10^{-3}}{300}\right) = 1.7\times10^{-3}\,m^3$$

c) The internal energy at point A is,

$$u_A = \frac{3}{2}nRT_A \leftarrow \text{mono-atomic}$$

$$u_A = \frac{3}{2}(0.02)(8.31)(300) = 75.2\text{J}$$

d) The internal energy at point C is,

$$u_C = \frac{3}{2}nRT_C = \frac{3}{2}(0.02)(8.31)(500) = 125.3\text{J}$$

i) The change in internal energy along AC is,
$\Delta u_{AC} = u_C - u_A = 125.3 - 75.2 = 50.1\text{J}$

ii) The change in internal energy along BC is,
$\Delta u_{BC} = u_C - u_B$, but $u_C = u_B$, isothermal
$\Delta u_{BC} = 0$

iii) The change in internal energy along AB is,
$\Delta u_{AB} = u_B - u_A = 125.3 - 75.2 = 50.1\text{J}$

61. A vessel of volume 8L contains an ideal gas at 300k and pressure 200 kPa. If the gas leak till the pressure falls to 125kPa. Calculate the amount of gas leaked at constant temperature.

Solution: From $pv = nRT$

$$n_1 = \frac{P_1 V}{RT}, \text{ before leakage}$$

and $n_2 = \frac{P_2 V}{RT}$, after leakage

$$\Delta n = n_1 - n_2 = \frac{P_1 v}{RT} - \frac{P_2 v}{RT} = \frac{v}{RT}(P_1 - P_2)$$

$$\Delta n = \frac{8\times10^{-3}}{(8.31)(300)}(200\times10^{3} - 125\times10^{3}) = 0.24\text{mole}$$

62. A closed container of volume $0.02m^3$ contains a mixture of neon and argon gases at 27°C and 1×10^5Pa with a total mass of 28g. If molecular mass of neon and argon are 20g and 40g respectively, find the masses of the individual gases in the container.

Solution: Let m_1 be mass of neon and m_2 mass of argon

$m_1+m_2=28$ g ..... (*)

From $pv = nRT$, $n = \frac{m}{M}$

$$Pv = \left(\frac{m_1}{M_1} + \frac{m_2}{M_2}\right)RT$$

$$\frac{m_1}{M_1} + \frac{m_2}{M_2} = \frac{Pv}{RT}$$

$$\frac{m_1}{20} + \frac{m_2}{40} = \frac{(1\times10^5)(0.02)}{(8.31)(300)}$$

$2m_1 + m_2 = 32$ ..... (**)

From equation (*) and (**)

$m_1 = 4$g and $m_2 = 24$g

63. An ideal gas undergoes an isothermal process starting with a pressure of 2×10^5 Pa and a volume of 6 cm^3. Which one of the following might be the pressure and volume of the final state? (UEE, 2007)

A. 4×10^5 and 4 cm^3
B. 4×10^5 and 3 cm^3
C. 5×10^5Pa and 3 cm^3
D. 10^5 Pa and 10 cm^3

Solution: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$, $T_1 = T_2 \rightarrow$ isothermal

$\Rightarrow P_1V_1 = P_2V_2$

$(2 \times 10^5)(6 \times 10^{-6}) = 1.2$J

the combination that gives 1.2 is (4×10^5Pa, 3cm^3)

Answer: B

64. A given weight of gas has a temperature of 27°C and volume v. Keeping the pressure constant, the volume is increased to 3v by varying the temperature. What is the final temperature of the gas? (UEE, 2007)

A. 81°C B. 546°C C. 627°C D. 900°C

Solution: $\frac{P_1v_1}{T_1} = \frac{P_2v_2}{T_2}$, $P_1 = P_2$

$$\frac{v_1}{T_1} = \frac{v_2}{T_2}, \quad T_1 = 27 + 273 = 300k$$

$$\frac{v}{300} = \frac{3v}{T_2}$$

$T_2 = 900k$, $T_2 = 900 - 273 = 627°C$

$\Rightarrow T = 627°C$

Answer: C

Avogadro's Law, states that, "Equal volumes of all gass at the same temperature and pressure contain equal number of molecules. $\boxed{n_1 = n_2}$"

That is, if $P_1 = P_2$, $v_1 = v_2$ and $T_1 = T_2$ and $P_2v_2 = n_2RT_2$ and

$P_1v_1 = n_1RT_1$

Taking ratio, $\frac{P_1v_1}{P_2v_2} = \frac{n_1RT_1}{n_1RT_2}$

$$1 = \frac{n_1}{n_2} \Rightarrow \boxed{n_1 = n_2} \leftarrow \text{Avogadro's law}$$

Dalton's Law of Partial Pressure states that, "The total pressure of a mixture of gases, which do not interact chemically, is equal to the sum of the pressures that each gas would exert if it alone occupied the volume containing the mixture. $\boxed{P = P_1 + P_2}$"

Suppose we have a volume v that contains n_1 moles of a gas with partial pressure P_1 and n_2 moles of a gas with a partial pressure P_2, at constant temperature T. Then

$P_1v = n_1RT$ ..... (*)

$P_2v = n_2RT$ ..... (**)

Taking ratio, $\frac{P_1}{P_2} = \frac{n_1}{n_2}$

$$P_2 = P_1\frac{n_2}{n_1}, \text{ and } P = P_1 + P_2 = P_1 + P_1\frac{n_2}{n_1}$$

$$\Rightarrow P = \frac{n_1P_1 + n_2P_1}{n_1} = P_1\left(\frac{n_1 + n_2}{n_1}\right)$$

Similarly, $\boxed{\Rightarrow P_1 = P\left(\frac{n_1}{n_1 + n_2}\right)}$ $\boxed{\Rightarrow P_2 = P\left(\frac{n_2}{n_1 + n_2}\right)}$ } Dalton's partial pressure

Illustrative Examples

65. If air is taken to consist of 80% N_2 and 20% O_2, what are the partial pressures of these two gases at an atmospheric pressure of 101kPa?

Solution: Let P_1 and n_1 be for N_2 and P_2, n_2 for O_2

Now, $P_1 = P\left(\frac{n_1}{n_1 + n_2}\right) = 101\left(\frac{0.8}{0.8 + 0.2}\right) = 80.8\text{kPa}$

$$P_2 = P\left(\frac{n_2}{n_1 + n_2}\right) = 101\left(\frac{0.2}{0.8 + 0.2}\right) = 20.2\text{kPa}$$

66. A vessel of volume 2L contains 0.2 mole of O_2 and 0.5 mole CO_2. If the temperature of the mixture is 300k, find its pressure.

Solution: From ideal gas equation, pv = nRT

$$P_1 = \frac{n_1RT}{v} = \frac{(0.2)(8.31)(300)}{2\times10^{-3}} = 2.5\times10^5\text{Pa}$$

$$P_2 = \frac{n_2RT}{v} = \frac{(0.5)(8.31)(300)}{2\times10^{-3}} = 6.2\times10^5\text{Pa}$$

From Dalton's law, the total pressure of the mixture is,

$P = P_1 + P_2 = (2.5 \times 10^5) + (6.2 \times 10^5)$

$P = 8.7 \times 10^5$ Pa

67. Two gases A and B having the same temperature T, same pressure P and same volume v are mixed. If the mixture is at the

same temperature and occupied a volume v, the pressure of the mixture is.

A. P B. 2P 3. 4P 4. 6P

Solution: From Dalton's law

$P_T = P_1 + P_2 = P + P \Rightarrow P_T = 2P$ **Answer: B**

9.3 The Kinetic Theory of Gases

The kinetic energy of gases applies only to ideal gases and it makes the following assumptions.

1. The internal energy of the gas is made up of random kinetic energies of particles only. There is no potential energy component because there is no interaction between particles except collision, since they are far apart. No force of attraction or repulsion.
2. Collisions between particles and between particles and walls are perfectly elastic, thus there is no loss of kinetic energy during collision.
3. The volume occupied by a single particle is negligible compared to the volume of the gas as a whole.
4. The distribution of velocities of particles is random. That is, the average distribution of velocities is the same in all direction.
5. Newton's laws can be applied to all collisions.
6. The collisions are instantaneous, that is, the time spent by a molecule in a collision is very small.
7. The gas molecule behave as rigid, elastic and smooth spheres.

Illustrative Examples

68. Which statement is NOT among the assumptions made in the kinetic energy of gas? (UEE, 2004)
    A. There is no interaction b/n particles or b/n particles and the wall except during collisions.
    B. Collisions between particles and b/n particles and wall are perfectly elastic.
    C. The volume occupied by the particles is negligible compared to the volume of the gas as a whole.
    D. The distribution of velocity of particles is uniform.

    Hint: velocity distribution is random. **Answer: D**

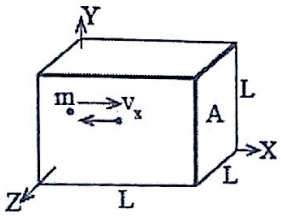
69. Which of the following statements follows from kinetic theory of gases?

A. The pressure of the gas is proportional to the average of square of molecular speed of particles.
B. The molecules undergo inelastic collisions with each other and with the walls of the container.
C. The forces between molecules of gas are strong at all times.
D. Newton's laws of motion is a consequence of kinetic theory of gases.

Hint: $P = \frac{1}{3}\rho < C^2 , \Rightarrow P \sim < C^2 >$ **Answer: A**

Pressure In terms of Microscopic Parameters

The pressure exerted by a gas is a result of the momentum transfer that occur every time a molecule bounce off the wall of a container.



Now consider a molecule of mass m moving in the x – direction and collides with the face of a cube of side L.

Change in momentum, $\Delta P = mv_x - (-mv_x) = 2\,mv_x$

Impulse, $F\Delta t = \Delta P \Rightarrow F = \frac{\Delta P}{\Delta t}$, time taken in one bounce, $\Delta t = \frac{2L}{v_x}$

$$F = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}$$

Pressure exerted on area A, by a single molecule.

$$P = \frac{F}{A} = \frac{mv_x^2}{L/L^2}, A = L^2$$

$$P = \frac{mv_x^2}{L^3} = \frac{mv_x^2}{v}, v = L^3$$

Pressure exerted on area A, by N molecules is then

$$P = N\frac{m(v_x^2)}{v} \quad \ldots\ldots\ldots\

Substituting this into equation (*), we have

$$P = \frac{1}{3} N \frac{m(v^2)}{v} av$$

Let, $(v^2)_{av} = \langle C^2 \rangle$, mean square speed

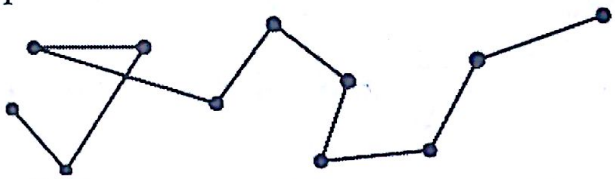
and $N\frac{m}{v} = \rho \rightarrow$ density of N molecules of ideal gas then, we get,

$$\boxed{P = \frac{1}{3}\rho \langle C^2 \rangle} \leftarrow \text{pressure in kinetic theory of gas}$$

☛ **Mean square speed $< C^2 >$:** Is the average value of the squares of the speed of particles in a gas.

Brownian Motion and Diffusion

Brownian motion: Is the apparently random movement of particles in a fluid. It was first observed by the Scottish botanist Robert Brown. He observed pollen grain suspended in water 'jiggle' around when seen under a microscope.



Mean Free Path: Is the average distance covered by a particle between impacts or collisions in Brownian motion.

Diffusion: Is the process where particles in a fluid spread from a region of high concentration to a region of low concentration.

Graham's Law of Diffusion

Graham's law of Diffusion; states that; "The rate of diffusion of a gas is inversely proportional to the square root of its density."

Consider two gasses at the same pressure P. Let C_1 and C_2 be the rms speeds of their molecules, and let ρ_1 and ρ_2 be their densities respectively. Now, According to kinetic theory of gases,

$$P = \frac{1}{3}\rho \langle C_1^2 \rangle$$

$$\Rightarrow P_1 = \frac{1}{3}\rho_1 \langle C_1^2 \rangle \ldots\ldots\ldots\

$$P_2 = \frac{1}{3}\rho_2 < C_2^2 > \quad \ldots\ldots\ldots\

$$\frac{2}{3}<K_E>=\frac{n}{N}RT,\ N=nN_A \Rightarrow \frac{n}{N}=\frac{1}{N_A}$$

$$\frac{2}{3}<k_E>=\frac{1}{N_A}RT=\frac{R}{N_A}T$$

$$\frac{2}{3}<k_E>=kT, k=\frac{R}{N_A}=\frac{8.31}{6.02\times10^{23}}=1.38\times10^{-23}\ J/k$$

$$\boxed{<K_E>=\frac{3}{2}kT}$$ ← Average kinetic energy of a single particle or molecule.

Where k is called *Boltzmann's constant*, $k = 1.38 \times 10^{-23}$J/k.

From $<K_E>=\frac{3}{2}kT, \frac{1}{2}m<C^2>=\frac{3}{2}kT$

$$m<C^2>=3kT \Rightarrow <C^2>=\frac{3kT}{m} \Rightarrow \sqrt{<C^2>}=\sqrt{\frac{3kT}{m}}$$

$$\boxed{C_{rms}=\sqrt{\frac{3kT}{m}}}$$, or using molar mass M, $m=\frac{M}{N_A}$

$$C_{rms}=\sqrt{\frac{3kT}{M/N_A}}=\sqrt{\frac{3N_AkT}{M}}$$, but $N_Ak = R$

$$\boxed{C_{rms}=\sqrt{\frac{3RT}{M}}}$$ Where, C_{rms} is called *root mean square speed.*

Room Mean Square (C_{rms}): Is the square root of the mean square speed.

That is, $\boxed{C_{rms}=\sqrt{<C^2>}}$

The average kinetic energy, that is, internal energy of N molecules of a mono atomic ideal gas is then given by:

$$\boxed{<kE>=u=\frac{3}{2}NKT=\frac{3}{2}nRT}$$ ← Average kinetic of N molecules or n moles of gas molecules.

Where, $NK = N\frac{R}{N_A} = \left(\frac{N}{N_A}\right)(R) = nR$

Conceptual Examples

70. One container is filled with Helium gas and another with Argon gas. If both containers are at the same temperature which molecules have the higher rms speed? Explain

Explanation: Molar mass of the gases are He = 4g, Ar = 40g

$C_{rms} = \sqrt{\frac{3RT}{M}}$, the smaller the molar mass, the higher the rms speed. Hence helium gas molecules move faster.

71. If the molar mass of a gas increases, what can you say about the rms speed of the gas molecule? Explain

Explanation: $C_{rms} = \sqrt{\frac{3RT}{M}}$, as M increases C_{rms} decreases.

72. If you put a helium filled balloon in a refrigerator, will its volume increase, decrease or stay the same? Explain

Explanation: From PV = nRT, as T decreases, V decreases.

73. If the rms speed of the atoms of an ideal gas is to be doubled, by what factor must the temperature be increased?

Explanation: $C_{rms} \sim \sqrt{T} \Rightarrow \frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{C_1}{2C_1} = \sqrt{\frac{T_1}{T_2}}$

$\frac{1}{4} = \frac{T_1}{T_2} \Rightarrow T_2 = 4T_1$, four times, it increases

74. Two different gases at the same temperature have equal root mean square speed. True or false? Explain.

Explanation: False, $C_{rms} = \sqrt{\frac{3RT}{M}}$, as M differs so does C_{rms}.

75. The average kinetic energy of one mole of all ideal gases at the same temperature is the same. True or False? Explain.

Explanation: True. Because, $\langle K_E \rangle = u = \frac{3}{2}nRT$

76. The average kinetic energy of 1g of all ideal gases at the same temperature is the same. True or False? Explain

Explanation: False. Because, $<K_E> = u = \frac{3}{2}nRT$. Where $n = \frac{m}{M}$, for different gases molar mass (M) is different.

77. When a container filled with a gas is placed in a fast moving train, will the temperature of the gas increase, decrease, or unchanged. Explain

Explanation: Unchanged, because the distribution of speed of molecules relative to their centre of mass remains fixed.

Illustrative Example

78. You are given the following group of particles with *ni* number of molecules and *Ci* speeds. Calculate

a) The average speed (C_{av})
b) The root mean square speed (C_{rms})
c) The most probable speed (C)

ni	C_i (m/s)
2	1
4	2
8	3
6	4
3	5

Solution: a) $C_{av} = \frac{\sum niCi}{\sum ni}$

$$= \frac{n_1C_1 + n_2C_2 + n_3C_3 + n_4C_4 + n_5C_5}{n_1 + n_2 + n_3 + n_4 + n_5}$$

$$= \frac{(2)(1) + (4)(2) + (8)(3) + (6)(4) + (3)(5)}{2 + 4 + 8 + 6 + 3} = \frac{73}{23} = 3.17 m/s$$

b) $(C^2)_{mean} = \frac{\sum niCi^2}{\sum ni} = \frac{n_1C_1^2 + n_2C_2^2 + n_3C_3^2 + n_4C_4^2 + n_5C_5^2}{n_1 + n_2 + n_3 + n_4 + n_5}$

$$= \frac{(2)(1)^2 + (4)(2)^2 + (8)(3)^2 + (6)(4)^2 + (3)(5)^2}{2 + 4 + 8 + 6 + 3}$$

$$<C^2> = \frac{281}{23}, C_{rms} = \sqrt{<C^2>} = 3.36 m/s$$

c) The most probable speed the speed possessed by many particles. Thus, C = 3 m/s.

79. If the rms speed of a helium atom is at room temperature is 1350 m/s, what is the rms speed of an oxygen molecule at this temperature? (The molar mass of O_2 is 32 and that of He is 4)

Solution: $\frac{1}{2}mC^2 = \frac{3}{2}kT$

$$C_{rms} = \sqrt{\frac{3KT}{m}}, \text{ but } m = M/N_A$$

$$C_{rms} = \sqrt{\frac{3N_A kT}{M}}, N_A k = R$$

$$C_{rms} = \sqrt{\frac{3RT}{M}},$$

$$\text{Thus, } \frac{C_1}{C_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{32}{4}} = \sqrt{8}$$

$$\Rightarrow C_2 = \frac{C_1}{\sqrt{8}} = \frac{1350}{\sqrt{8}} = 477.3 \text{m/s}$$

80. Calculate the average kinetic energy of oxygen molecule at 17°C.

Solution: $<kE> = \frac{3}{2}kT = \left(\frac{3}{2}\right)(1.38\times10^{-23})(290) = 6\times10^{-21}\text{J}$

81. At what temperature is the rms speed of H_2 equal to the rms speed of O_2 at 0°C.

Solution: $C_{rms} = \sqrt{\frac{3RT}{M}},$

$$C_1 = \sqrt{\frac{3RT_1}{M_1}}, \text{ and } C_2 = \sqrt{\frac{3RT_2}{M_2}}, \text{ but } C_1 = C_2$$

$$\Rightarrow \sqrt{\frac{3RT_1}{M_1}} = \sqrt{\frac{3RT_2}{M_2}} \Rightarrow \frac{T_1}{M_1} = \frac{T_2}{M_2}$$

$$\frac{T_1}{2} = \frac{273}{32} \Rightarrow T_1 = \frac{273}{16} = 17\text{k}$$

82. An ideal gas is at a temperature of 300k. At what temperature will the molecules have twice the root mean square speed.

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}}, C_2 = 2C_1$

$$\frac{C_1}{2C_1}=\sqrt{\frac{300}{T_2}}\Rightarrow\frac{1}{2}=\sqrt{\frac{300}{T_2}}$$

$$\frac{1}{4}=\frac{300}{T_2}\Rightarrow T_2=(4)(300)=1200K$$

83. The number of gas particle in a 100mL container is 5×10^{21} and the average kinetic energy of each gas particle is 6×10^{-21}J. Calculate the pressure of the gas.

Solution: $P=\frac{1}{3}\rho\langle C^2\rangle=\frac{1}{3}N\frac{m}{v}\langle C^2\rangle=\frac{2}{3}\frac{N}{v}\frac{1}{2}m\langle C^2\rangle$

$$=\frac{2}{3}\left(\frac{5\times10^{21}}{100\times10^{-6}}\right)(6\times10^{-21})=2\times10^5\text{Pa}$$

84. If the density of oxygen at STP is 1.43 kg/m^3. Calculate the root mean square speed of oxygen molecule.

Solution:

$$P=\frac{1}{3}\rho\langle C^2\rangle\Rightarrow C_{rms}=\sqrt{\frac{3P}{\rho}}=\sqrt{\frac{3(1.01\times10^5)}{1.43}}=460\text{m/s}$$

85. By what factor does the kinetic energy of a gas changed if its temperature is tripled?

Solution: $\langle kE\rangle=\frac{3}{2}kT, T_2=3T_1$

$$\frac{kE_1}{kE_2}=\frac{\frac{3}{2}kT_1}{\frac{3}{2}kT_2}\Rightarrow\frac{kE_1}{kE_2}=\frac{T_1}{3T_1}\Rightarrow kE_2=3kE_1$$

86. Consider gases of the following atoms or molecules, all at the same temperature 293k. which atom or molecule have the largest average kinetic energy and the fastest average speed? He, H_2, N_2, O_2

Solution: The average kinetic energy is the same for all gas molecules at the same temperature.

$$\langle kE\rangle=\frac{3}{2}kT=\frac{3}{2}(1.38\times10^{-23})(293)=6.07\times10^{-21}\text{J}$$

The rms speed of the molecules is not the same, but depends on the mass. The rms speed for each type of molecules is

$$\text{He, } C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{(3)(8.31)(293)}{4\times10^{-3}}} = 1360\text{m/s}$$

$$H_2, C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{(3)(8.31)(293)}{2\times10^{-3}}} = 1911\text{m/s}$$

$$N_2, C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{(3)(8.31)(293)}{28\times10^{-3}}} = 511\text{m/s}$$

$$O_2, C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{(3)(8.31)(293)}{32\times10^{-3}}} = 478\text{m/s}$$

Therefore, H_2 molecule is fastest due to its smallest mass.

87. Calculate the average kinetic energy, that is, internal energy of 2g of helium at 17°C.

Solution: $< kE > = u = \frac{3}{2}nRT, n = \frac{m}{M} = \frac{2}{4} = 0.5$

$$u = \left(\frac{3}{2}\right)(0.5)(8.31)(290), T = 17 + 273 = 290k = 1807\text{ J}$$

88. Suppose two molecules of different mass are kept at the same temperature. Which one of the following statement is NOT true? (UEE, 2005)

A. Both remain at rest.
B. Both move with the same speed.
C. The lighter molecule moves faster.
D. The heavier molecule moves faster.

Explanation: $C_{rms} = \sqrt{\frac{3RT}{M}}$, the lighter molecule moves faster.

Answer: D

89. An ideal gas is at a temperature of 300k. If we wish to double the root mean square speed of the molecule of the gas, to what value must we raise its temperature? (UEE, 2006)

A. 450k B. 600k C. 800k D. 1200k

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}}, C_2 = 2C_1$

$$\frac{C_1}{2C_1} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{300}{T_2}}$$

$$\frac{1}{2} = \sqrt{\frac{300}{T_2}} \Rightarrow \frac{1}{4} = \frac{300}{T_2} \Rightarrow T_2 = 1200K$$ **Answer: D**

90. Gas at pressure P is contained in a vessel. If the masses of all the molecules are doubled and their speeds halved, then what will be the resulting pressure? (UEE, 2005)

A. 2P B. $P/3$ C. $P/2$ D. 3P

Solution: $P_1 = \frac{1}{3}\rho_1 \langle C_1^2 \rangle = \frac{1}{3}\frac{m_1}{v}\langle C_1^2 \rangle$ ..... (*)

$$P_2 = \frac{1}{3}\rho_2 \langle C_2^2 \rangle = \frac{1}{3}\frac{m_2}{v}\langle C_2^2 \rangle$$ ..... (**)

taking ratio, $\frac{P_1}{P_2} = \frac{m_1 \langle C_1^2 \rangle}{m_2 \langle C_2^2 \rangle}$, but, $m_2 = 2m_1$ & $C_2 = \frac{C_1}{2}$

$$\frac{P_1}{P_2} = \frac{m_1}{2m_1} \frac{\langle C_1^2 \rangle}{\left\langle \left(\frac{1}{2}C_1\right)^2 \right\rangle} \Rightarrow \frac{P}{P_2} = 2 \Rightarrow P_2 = \frac{P}{2}$$ **Answer: C**

9.4 Entropy Second law of Thermodynamics and Efficiency

Entropy: Is a measure of amount of disorder, chaos or lack of arrangement in a system.

If Ω (Omega) is the amount of disorder in a system, then the entropy (s) of the system is given by

$$S = k \ln \Omega$$

Where; k is Boltsmann constant

If heat (Q) flows into a system at constant temperature (T), the change in entropy of the system is given by

$\boxed{\Delta S = \frac{Q}{T}}$, T is in Kelvin

- SI unit of entropy is Joule/Kelvin = J/k.

Important point to remember

Reversible process: Is a process that can be reversed slowly without transfer of energy. $\frac{Q_C}{T_C} = \frac{Q_h}{T_h}$.

Irreversible process: Is a process that cannot be reversed .Natural processes are irreversible processes. $\frac{Q_C}{T_C} > \frac{Q_h}{T_h}$

Cycle: Is an interval of time over which a system passes through a series of different states before returning to its original state.

- The change in entropy depends only on the initial and final state of the system.

Entropy change in reversible process

Assume hot and cold bodies are in thermal contact. Then, heat (Q) leaves the hot body. Thus the entropy of the hot body decrease by $\left(\frac{-Q}{T_h}\right)$.

$$\Delta S_{hot} = \frac{-Q}{T_h}, \text{ and}$$

Heat (Q) is added to the cold body, hence its entropy increase by $\left(\frac{Q}{T_C}\right)$.

$$\Delta S_{cold} = \frac{Q}{T_C}$$

$$\Delta S_{total} = \Delta S_{hot} + \Delta S_{cold} = \frac{-Q}{T_{hot}} + \frac{Q}{T_C}.$$

$\boxed{\Delta S_{total} = 0}$, $\frac{-Q}{T_C} = \frac{Q}{T_h}$ for reversible process.

- For reversible process, the change in entropy is zero. That is, in reversible process, entropy is constant.

Entropy change in irreversible process

When heat (Q) flows from a hot system to a cold system ($T_H > T_C$) the total entropy change of the system is:

$$\Delta S_{total} = \Delta S_H + \Delta S_C = \frac{-Q}{T_H} + \frac{Q}{T_C}$$

but, $T_H > T_C$ hence $\frac{Q}{T_h} < \frac{Q}{T_C}$

Thus, $\boxed{\Delta S_{total} > 0}$

- **Important points to remember**
  - For irreversible process, the change in entropy is none zero and entropy is increasing.
  - The increase in entropy of a colder body is greater than the decrease in entropy of the hotter body.
  - The entropy of the universe never decreases.

Illustrative Examples

91. A hot reservoir at the temperature of 576k transfers 1050J of heat to a cold reservoir at temperature of 305k. Calculate the change in entropy of the universe.

Solution: $\Delta S = \frac{-Q}{T_h} + \frac{Q}{T_C} = \frac{-1050}{576} + \frac{1050}{305} = 1.62 J/k$

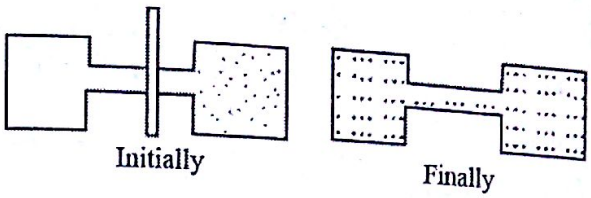
92. A 1500kg car traveling at 20 m/s crashes into concrete wall. If the air temperature is 20°C. What is the change in entropy of the universe?

Solution: $\Delta S = \frac{\frac{1}{2}mv^2}{T} = \frac{(0.5)(1500)(20)^2}{293} = 1024 J/k$

93. Suppose 0.8 mol of N_2 gas is confined to the right side of a container by a stop cock. When the stop cock is removed, the volume of the gas tripled. What is the change in entropy of the gas?

Solution:

$$\Delta S = \frac{Q}{T} = \frac{nRT\ln\left(\frac{v_f}{v_i}\right)}{T}$$



$$\Delta S = nR \ln\left(\frac{v_f}{v_i}\right) = (0.8)\ (8.31)(\ln 3) = 7.3\ J/k$$

94. The change in entropy is zero for (UEE, 2006).s
A. Reversible isobaric process
B. Reversible adiabatic process.
C. Reversible isothermal process.
D. Reversible process where no work is done.

Explanation: In reversible and adiabatic process, no energy transfer takes place and entropy is constant. **Answer: B**

95. Which of the following statement is NOT true? (UEE, 2002)
A. If a heat Q is transferred from a hot reservoir at temperature T_h to a cold reservoir at temperature T_C, the entropy of the cold reservoir increases.
B. The increase in entropy of the cold reservoir is greater than the decrease in entropy of the hot reservoir.
C. The change in entropy of the system between any two equilibrium states does not depend only on the initial and final states.
D. Metals are good heat conductors because of their large number of electrons. **Answer: C**

96. In a reversible thermodynamic process, the system (UEE, 2005)
A. Is always close to equilibrium state.
B. Might never be close to any equilibrium state.
C. Is close to equilibrium states only at the beginning and end.
D. Is close to equilibrium states throughout, except at the beginning and end. **Answer: A**

Second Law of Thermodynamics

The second law can be expressed in different statements but similar meanings as follows.

A) **Entropy Statements**

Second law of thermodynamics states that; "No process is possible in which there is an overall decrease in the entropy of the universe."

The second law of thermodynamic is also expressed as:

B) Energy Statement

"The complete conversation of energy from a hot source into work is not possible."

The very same second law of thermodynamics can also be stated as:

C) The heat flow statement

"The spontaneous transfer of energy from a cooler body to a hotter body is not possible."

Conceptual Examples

97. According to the second law of thermodynamics, heat energy can not be converted completely to mechanical energy. Is it possible to convert mechanical energy completely into thermal energy?
Explanation: Yes. It is possible.

98. Is it possible for the transfer of heat from cold to hot system? Explain
Explanation: Yes. This is possible by using external force.

Illustrative Examples

99. The second law of thermodynamics doe NOT imply that: (UEE 2003)
   A) There is not any process in which there is an overall decrease in the entropy of the universe.
   B) The complete conversation of energy from a hot source into work is not possible.
   C) Heat flows from colder to hotter body as in refrigerator without work done on the system.
   D) Perfectly reversible process are impossible.

Answer: C

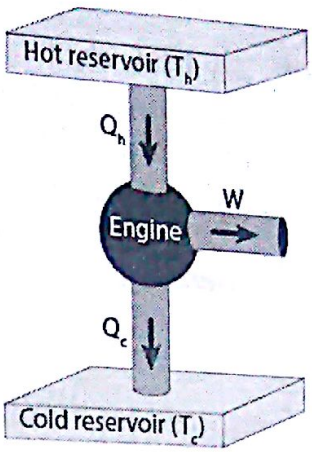
Heat Engines

Heat Engine: Is a device that converts heat into work or mechanical energy or electrical energy.

For instance, in a power plant, coal or other fuel is burned and the thermal energy produced is used to convert water to steam. This steam is directed at the blades of a turbine to rotate it and drive an electric generator. Moreover, petrol engine, diesel engine, gasoline engine or jet engine in an automobile or air plane extract thermal engine from a burning fuel and convert part of it to mechanical energy motion.

A heat engine has three essential features.

1. A high temperature region or hot reservoir that supplies heat to the engine.
2. A working substance of the engine.
3. A low temperature region or cold reservoir or sink where 'waste' heat is released.



Now, from the first law of thermodynamics, $\Delta u = \Delta Q - W$, but since the cycle repeats, and return to its initial state, the internal energy is constant, $\Delta u = 0$.

$$\Delta Q = W$$

$$\boxed{Q_H - Q_C = W}$$

Remember: The engine absorbs heat (Q_h) from a hot reservoir performs work (W), and the remainder of the heat (Q_C) is exhausted or rejected to the cold reservoir.

Efficiency of a heat engine

$$\text{Efficiency} = \frac{\text{Work done by the engine}}{\text{Total energy absrobed}}$$

$$\boxed{\eta = \frac{W}{Q_h}}$$

but, $Qh - Q_C = W$

$$\Rightarrow \eta = \frac{Q_h - Q_c}{Q_h} = \frac{Q_h}{Q_h} - \frac{Q_c}{Q_h}$$

$$\boxed{\eta = 1 - \frac{Q_C}{Q_h}}$$

To convert efficiency to percentage, we can multiply the above equation by 100.

Note: According to second law of thermodynamics, No heat engine is 100% efficient. In fact a more common value for diesel or petrol engine is between 30% and 40%.

Illustrative Examples

100. A heat engine absorbs 1300J of heat from the hot reservoir and exhausts 760J of heat to the cold reservoir. Calculate
   a) The efficiency of an engine.
   b) The work done by the engine.

Solution: a) $\eta = 1 - \frac{Q_C}{Q_h} = 1 - \frac{760}{1300} = 0.415 = 41.5\%$

b) $\eta = \frac{W}{Q_h} \Rightarrow w = (\eta)(Q_n) = (0.415)(1300) = 540J$

101. A heat engine performs 200J of work in each cycle and has an efficiency of 25%. For each cycle, how much thermal energy is:
   a) Absorbed b) Expelled

Solution: a) $\eta = \frac{Q_C}{Q_h} \Rightarrow \frac{25}{100} = \frac{200}{Q_h} \Rightarrow Q_h = 800J$

b) $Q_c = Q_h - W = 800 - 200 = 600J$

102. A particular engine has a power output of 5kw and an efficiency of 25%. If the engine expels 8000J of thermal energy in each cycle. Find:
   a) The heat absorbed in each cycle and
   b) The time for each cycle.

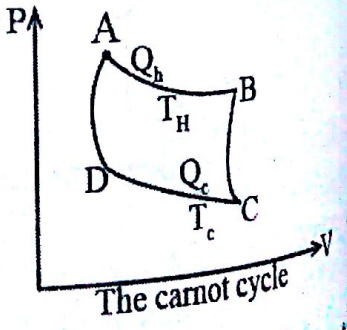
Solution: a) $\eta = 1 - \frac{Q_C}{Q_h} \Rightarrow \frac{25}{100} = 1 - \frac{8000}{Q_h} \Rightarrow Q_h = 10667J$

b) $W = Q_h - Q_c \Rightarrow Pt = Q_h - Q_c \Rightarrow t = \frac{Q_h - Q_c}{P} = 0.5\,sec$

Maximum theoretical efficiency of a heat engine

A French engineer Sadi Carnto developed an idealized heat engine with the maximum possible efficiency.

- The Carnot engine operates with an ideal gas, reversibly through a sequence of isothermal and adiabatic steps known as the *Carnot cycle*.



The carnot cycle

According to Carnot, the maximum possible efficiency of an ideal heat engine is given by

$$\eta_{max} = 1 - \frac{T_C}{T_H}$$

Where T_C and T_H are the absolute temperatures of the cold and hot reservoir, respectively.

Moreover, for a Carnot engine it is true that:

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H} \rightarrow \text{for reversible process}$$

- **Remember:** No engine, no matter how sophisticated or technologically advanced it may be, can have a greater efficiency than Carnot's engine.

Conceptual Example

103. If friction could be totally removed from a Carnot engine, would it be 100% efficient? Explain

Explanation: No. Because, even a frictionless engine has an efficiency always less than 100%. The efficiency could be 100% if $T_C = 0K$. But the absolute zero could never be reached. Hence, efficiency of a heat engine is always less than 100%.

Conceptual Example

104. A heat engine operates between temperature of 20°C and 300°C. What is the maximum efficiency?

Solution: $\eta = 1 - \frac{T_C}{T_h}$, but $T_C = 20 + 273 = 293K$,

$T_H = 300 + 273 = 573K$.

$$\eta = 1 - \frac{293}{573} = 0.49 = 49\%$$

105. The highest theoretical efficiency of a steam engine based on the Carnot cycle is 40%. If this engine gives off its gas into the atmosphere at temperature of 400K, calculate the temperature in the reservoir after combustion.

Solution: $\eta = 1 - \frac{T_C}{T_H} \Rightarrow \frac{40}{100} = \frac{1-400}{T_H} \Rightarrow T_H = 667K$

106. A reversible heat engine operates between thermal reservoirs at 900K and 200K. If 9000J are absorbed from the hot reservoir. How much work is done by the heat engine?

$$\eta_{max} = 1 - \frac{T_C}{T_H}$$

Where T_C and T_H are the absolute temperatures of the cold and hot reservoir, respectively.

Moreover, for a Carnot engine it is true that:

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Heat Death

The universe is "running down." That is, the universe constantly increases in entropy or disorder, as it does, the amount of energy available decreases. A possible fate of the universe is heat death, in which everything is at the same temperature.

Conceptual Example

109. Is it possible to cool your kitchen by leaving the refrigerator door open? Explain

Explanation: No! the heat in the kitchen is $Q_H = Q_C + W$ hence, rather than cooling, the open refrigerator warms the room up.

Illustrative Example

110. A refrigerator of COP 5 takes in 120J of energy from a cold reservoir in each cycle. Find the work required in each cycle.

Solution: $COP = \frac{Q}{W} \Rightarrow 5 = \frac{120}{W} \Rightarrow W = 24J$

111. Which of the following changes would allow your refrigerator to use less energy to run?
   A. Increasing the temperature inside the refrigerator.
   B. Increasing the temperature of the kitchen.
   C. Decreasing the temperature inside the refrigerator.
   D. Decreasing the temperature of the kitchen.

   Explanation: The closer the temperature of the hot and cold reservoirs, the more efficient the refrigerator. **Answer: A & D**

112. An ordinary refrigerator is turned on and its door is left open in a thermally insulated kitchen. What is the temperature of the room (UEE, 2005)
   A. Increase according to the first law of thermodynamics.
   B. Decrease according to the first law of thermodynamics.
   C. Remains constant according to the first law of thermodynamic.
   D. Remains constant according to the second law of thermodynamic.

   Hint: $Q_H = Q_C + W$ **Answer: A**

Additional Examples

113. The first law of thermodynamics is a restatement of the law of conservation of:
A. Momentum C. Mass
B. Charge D. Energy **Answer: D**

114. An ideal gas changes from state a to state b as shown. What is the work done by the gas in the process?
A. Zero C. Negative
B. Positive D. Infinite **Answer: D**

Explanation: Since $P \sim T$, volume is constant. Thus, $\Delta W = P\Delta V = 0$ **Answer: A**

115. For an ideal gas, the heat capacity at constant pressure is larger than that at constant volume because:
A. Work is done during expansion of the gas by the external pressure.
B. Work is done during expansion by the gas against external pressure.
C. Work is done during expansion by the gas against intermolecular forces of attraction.
D. More collision occur per unit time when volume is kept constant. **Answer: B**

116. Heating of water under atmospheric pressure is an
A. Isothermal process C. Isobaric process
B. Adiabatic process D. Isochoric process **Answer: C**

117. A thermodynamic system goes from state
a) P_1, v to $2P_1$, v b) P, v to P, 2v
What is the work done in the two cases respectively?
A. Zero, zero C. Zero, Pv
B. Pv, zero D. pv, pv

Explanation: Case a, $\Delta W = P\Delta v$ = zero and
Case b, $\Delta W = P\Delta v = P(2v - v) = pv$
Hence, the answer is zero, pv. **Answer: C**

118. In adiabatic expansion
A. $\Delta u = 0$ C. Δu = negative
B. Δu = positive D. ΔW = zero

Explanation: From the first law of thermodynamics, $\Delta u = \Delta Q + \Delta W \Rightarrow \Delta u = \Delta W$. And for an adiabatic expansion, work done is negative, hence, Δu is also negative since it decreases.
Answer: C

119. In an adiabatic process, there is no:
A. Change in temperature C. Exchange of heat
B. Change in internal energy D. Work done
Answer: C

120. In a reversible isochoric change
A. $\Delta W = 0$ B. $\Delta T = 0$ C. $\Delta Q = 0$ D. $\Delta u = 0$
Answer: A

121. An ideal gas performs the most when it expands:
A. Isothermally C. Isobarically
B. Adiabatically D. Isochorically
Explanation: In isobaric process, maximum expansion occurs.
Answer: C

122. In the equation pv^{γ} = constant. If the value of γ is one. Then the process is.
A. Isothermal C. Adiabatic
B. Isobaric D. Isochoric
Explanation: If γ is one, then $P_1v_1 = P_2v_2$. Which is an isothermal process.
Answer: A

123. The work done in an adiabatic process depends only on the change in
A. Specific heat C. Volume
B. Pressure D. Temperature
Explanation: For adiabatic process, $\Delta W = \frac{R}{\gamma - 1}\Delta T$.
Answer: D

124. In a cyclic process, the change in internal energy of a system in one complete cycle is:
A. Unknown C. Positive
B. Negative D. Zero
Answer: D

125. When an ideal gas undergoes an isothermal expansion, the pressure of the gas decreases. This is due to:
A. Decrease in change in momentum per collision.
B. Decrease in frequency of collision.
C. Both a decrease in frequency of collision and momentum.
D. Neither a decrease in frequency and momentum.
Answer: B

126. In a thermodynamical process, 400J of heat is given to a gas and 100J of work is also done on it. The change in the internal energy of the system is:
A. 300J B. 4J C. 0.25J D. 500J

Solution: $\Delta u = \Delta Q + \Delta W = 400 + 100 = 500J$ **Answer: D**

127. Free expansion of a gas is the expansion of the gas against.
A. Zero external pressure
B. Against standard pressure
C. Atmospheric pressure
D. Constant pressure
Answer: A

128. In which process does the internal energy of the system remains constant.
A. Iso choric
B. Adiabatic
C. Iso thermal
D. Iso baric
Answer: C

129. The internal energy is a unique function of state because it,
A. Does not depend upon path.
B. Depends upon the path.
C. Corresponds to an adiabatic process.
D. Corresponds to an isothermal process.
Answer: A

130. The internal energy of a gas increases in:
A. Adiabatic expansion
B. Isothermal expansion
C. Adiabatic compression
D. Isothermal compression
Explanation: For adiabatic process, the work done is given by $\Delta W = \frac{P_1 v_1 - P_2 v_2}{\gamma - 1}$ which is positive for compression, and equals to the increase in internal energy.
Answer: C

131. During adiabatic expansion, an ideal gas cools because:
A. Internal energy decreases
B. Internal energy increases
C. Internal energy unchanged
D. Nothing can be said
Answer: A

132. In an isobaric process, there is no change in
A. Pressure
B. Temperature
C. Volume
D. Heat
Answer: A

133. The gas law $\frac{Pv}{T}$ = constant is true for
A. Isothermal changes only.
B. Adiabatic changes only.
C. Both isothermal and adiabatic changes.
D. Neither isothermal nor adiabatic changes.
Answer: C

134. A given system undergoes a change in which the work done by the system equals the decrease in its internal energy. The system must have undergone an:
A. Isothermal change
C. Adiabatic change

148. Which of the following temperatures is the highest?
A. 100K B. –20°C C. –13°F D. –23°C
Solution: $T_C = Tk - 273 = 100 - 273 = -173°C$, and

$$\frac{T_C}{5} = \frac{T_F - 32}{9} \Rightarrow \frac{T_C}{5} = \frac{-13-32}{9} \Rightarrow T_C = -25°C$$

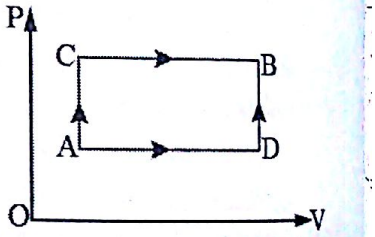
Answer: B

149. A monatomic ideal gas expands at constant pressure, with heat Q supplied. The fraction of Q which goes as work done by the gas is:
A. 1 B. $\frac{2}{3}$ C. $\frac{3}{5}$ D. $\frac{2}{5}$
Solution: $Q_P = C_P\, n\, \Delta T$ and $\Delta W = P\Delta v = nR\Delta T$

For monatomic gas, $C_P = \frac{5}{2}R$, hence, $\frac{\Delta W}{\Delta Q} = \frac{nR\Delta T}{\frac{5}{2}Rn\Delta T} = \frac{2}{5}$

Answer: D

150. When a thermodynamic system is taken from state A to B via path ACB, 420J heat is given to the system and 252J of work is done. If along the path ADB the work done is 84J, what is the heat flow into the system in this case?
A. 540J C. 588J
B. 168J D. 252J



Solution: $\Delta u = \Delta Q + \Delta W = 420 - 252 = 168J$, also

$$168 = \Delta Q - 84 \Rightarrow \Delta Q = 252J$$

(the work done is by the gas)

Answer: D

151. A gas at NTP is slowly compressed to one – fourth of its original volume. If γ is $\frac{3}{2}$, then final pressure is,
A. 4 atm B. 8 atm C. 0.75atm D. 0.25atm
Solution: The process is slow, hence isothermal.

$$P_1v_1 = P_2v_2 \Rightarrow P_2 = \frac{P_1v_1}{v_2} = \frac{P_1v_1}{\frac{1}{4}v_1} = 4P_1 = 4atm$$

Answer: A

152. A gas at NTP is suddenly compressed to one – fourth of its original volume. If γ is $\frac{3}{2}$, then final pressure is,

A. 4 atm B. 8 atm C. 0.75atm D. 0.25atm

Solution: The process is sudden, hence adiabatic.

$$P_1v_1\gamma = P_2v_2^{\gamma} \Rightarrow P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma} = P_1\left(\frac{v_1}{1/4\, v_1}\right)^{\gamma} = P_1(4)^{3/2} = 8P$$

Answer: B

153. If one mole of a monatomic gas $\left(\gamma = 5/3\right)$ is mixed with one mole of a diatomic gas $\left(\gamma = 7/5\right)$, the value of γ for the mixture is,

A. 1.40 B. 1.50 C. 1.53 D. 3.07

Solution: Let u_1 and u_2 be internal energies per mole for a monatomic and diatomic gas respectively. Then

$$u_1 = \frac{3}{2}RT \text{ and } u_2 = \frac{5}{2}RT \text{, now}$$

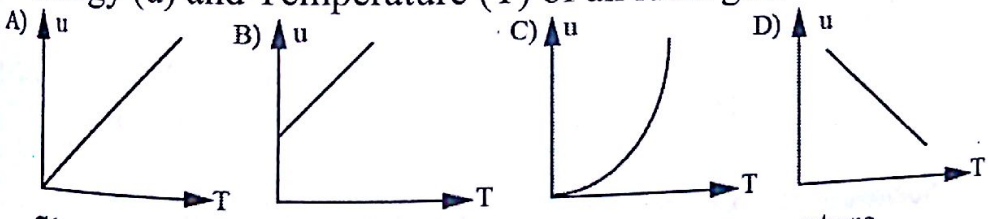
$$u = \frac{u_1 + u_2}{2} = 2RT;\ C_v = \frac{\Delta u}{\Delta T} = \frac{2R\Delta T}{\Delta T} = 2R, \text{ and}$$

$C_P = C_v + R = 2R + R = 3R$. Then,

$$\gamma = \frac{C_P}{C_v} = \frac{3R}{2R} = 1.5$$

Answer: B

154. Which graph best illustrates the relationship between internal energy (u) and Temperature (T) of an ideal gas?



Hint: internal energy directly depends on temperature.

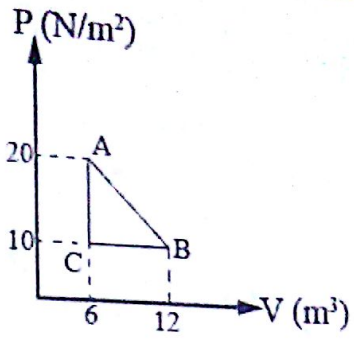
Answer: A

155. One mole of an ideal gas requires 207J heat to raise the temperature by 10k at constant pressure. If the gas is heated at constant volume to raise the temperature by the same 10k, then the heat required is:

A. 198.7J B. 215.3J C. 124J D. 24J

Solution: $\Delta Q_P = C_P n\Delta T \Rightarrow C_P \times 1 \times 10 = 207 \Rightarrow C_P = 20.7\text{J/mol.k}$

158. The p – v graph for a thermo dynamical system is shown. What is the work done by the system in the process A to B?



A. 90J C. 0J
B. 60J D. 0J

Solution: W_{AB} = Area under $\overline{AB}$

$= \frac{1}{2}(6)(10) + (16)(10) = 90$ J

Answer: A

159. The work done in the complete cycle ABCA is:
A. 90J B. 60J C. 0J D. 30J

Solution: W_{ABCA} = Area of triangle ABC $= \frac{1}{2}(6)(10) = 30$J

Answer: D

160. Two samples A and B of a gas which are initially at the same temperature and pressure compressed from volume v to $\frac{v}{2}$ (A isothermally and B adiabatically). The final pressure of A is:
A. Greater than that of B. C. Less than that of B.
B. Equal to that of B. D. Twice that of B.

Solution: For A, isothermal

$$P_1v_1 = P_2v_2 \Rightarrow P_1v_1 = P_2\frac{v_1}{2} \Rightarrow P_2 = 2P_1$$

For B, adiabatic

$$P_1v_1^{\gamma} = P_2v_2^{\gamma} \Rightarrow P_1v_1^{\gamma} = P_2\left(\frac{v_1}{2}\right)^{\gamma} \Rightarrow P_2 = 2^{\gamma}P_1$$

Since $\gamma > 1$, $2^{\gamma}P_1 > 2P_1$, hence $P_2B > P_2A$ **Answer: C**

161. 5 mol of oxygen is heated at constant volume from 10°C to 20°C. If C_P = 33.6J/mol°C and R = 8.3J/mole. The amount of heat consumed by oxygen is:
A. 420J B. 840J C. 1265J D. 1680J

Solution: $\Delta Q = C_v n\Delta T$, $C_v = C_P - R = 33.6 - 8.3 = 25.3$

$= (25.3)(5)(20 - 10) = 1265$J **Answer: C**

162. A gas $\left(\gamma = 7/5\right)$ at pressure P is adiabatically compressed so that its density becomes twice that of initial value. What will be the final pressure of the gas?

A. P B. 2P C. 2.6P D. 1.4P

Solution: $P_1 v_1^{\gamma} = P_2 v_2^{\gamma}$, but $v = \dfrac{m}{\rho}$

$$P_1\left(\frac{m}{\rho_1}\right)^{\gamma} = P_2\left(\frac{m}{\rho_2}\right)^{\gamma},\ m_1 = m_2 \text{ and } \rho_2 = 2\rho_1$$

$$\Rightarrow P_1\left(\frac{m}{\rho_1}\right)^{\gamma} = P_2\left(\frac{m}{2\rho_1}\right)^{\gamma} \Rightarrow P_2 = P_1(2)^{\gamma} = P_1(2)^{1.4} = 2.6P_1$$

Answer: C

163. In a thermodynamic process, the process is changed in such a way that, the gas gives out 30J of heat and 10J of work is done on the gas. If the initial internal energy of the gas was 40J, then the final internal energy will be:

A. 0 B. 80J C. 20J D. −20J

Solution: The decrease in internal energy is

$\Delta u = \Delta Q + \Delta W = -30 + 10 = -20J$

also, $\Delta u = u_f - u_i \Rightarrow -20 = u_f - 40 \Rightarrow u_f = 20J$ **Answer: C**

164. A 588J of heat is required to raise the temperature of 2 mol of a diatomic gas at constant pressure from 0°C to 10°C. How much heat will be required to raise the temperature from 0°C to 10°C at constant volume?

A. 252J B. 422J C. 588J D. 756J

Solution: $\Delta Q_P = C_P n\Delta T \Rightarrow C_p = \dfrac{\Delta Q_P}{n\Delta T} = \dfrac{588}{(2)(10-0)} = 29.4 J/mol°C$

and $C_v = C_P - R = 29.4 - 8.3 = 21.1$ J/mol.°C, then

$\Delta Q_v = C_v n\Delta T = (21.1)(2)(10-0) = 422J$ **Answer: B**

165. What is the amount of heat needed to raise the temperature of 2 mol of an ideal monatomic gas from 273kto 373k when no work is done?

A. 830J B. 1245J C. 2490J D. 4150J

Solution: The process is isochoric and the gas is monatomic hence,

$\Delta Q_v = C_v n\Delta T$, but $C_v = \frac{3}{2}R$, monatomic

$$\Delta Q_v = \frac{3}{2}Rn\Delta T = \left(\frac{3}{2}\right)(8.3)(2)(373-273) = 2490J$$ **Answer: C**

166. During the adiabatic expansion of 2 mole of a gas, the internal energy was found to have decreased by 100J. The work done by the gas in this process is:

A. Zero B. -100J C. 200J D. 100J

Solution: $\Delta u = \Delta Q + \Delta W$, $\Delta Q = 0$, adiabatic

$100 = \Delta W$, Δu, decreased **Answer: D**

167. An ideal gas with pressure P is compressed isothermally till the mean separation between molecules of the gas is reduced to half of the original value. The final pressure will be:

A. P B. 2P C. 8P D. 4P

Solution: Volume, $v = L^3$ and $v_1 = L_1^3$ and $L_2 = \frac{L_1}{2}$

$$v_2 = L_2^3 = \left(\frac{L_1}{2}\right)^3 = \frac{L_1^3}{8} = \frac{V_1}{8}$$

Now, for isothermal, process,

$P_1V_1 = P_2V_2$

$$P_2 = P_1\frac{v_1}{v_2} = P_1\left(\frac{v_1}{\frac{v_1}{8}}\right) = 8P_1$$ **Answer: C**

168. A gas is expanded adiabatically at an initial temperature of 300K so that its volume is doubled. The final temperature of the gas is ($\gamma = 1.4$)

A. 227.3k B. 500.3K

C. 454.76K D. −47°C

Solution: For adiabatic process, $T_1v_1^{\gamma-1} = T_2v_2^{\gamma-1}$, $v_2 = 2v_1$

$$T_2 = T_1\left(\frac{v_1}{v_2}\right)^{\gamma-1} = T_1\left(\frac{v_1}{2v_1}\right)^{\gamma-1} = 300\left(\frac{1}{2}\right)^{1.4-1} = 227.3K$$

Answer: A

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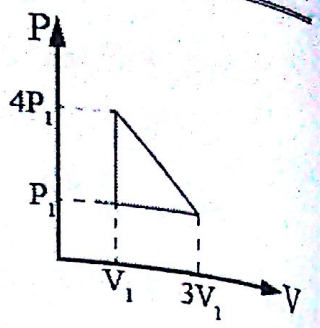
169. An ideal gas is taken through series of changes as shown. What is the net work done by the gas at the end of the cycle?



A. P_1V_1 B. $3P_1V_1$
C. $6P_1V_1$ D. $12P_1V_1$

Solution: ΔW = Area of the triangle

$$= \frac{1}{2}(3v_1 - v_1)(4P_1 - P_1) = 3P_1V_1$$

Answer: B

170. A mono atomic gas $\left(\gamma = 5/3\right)$ is suddenly compressed to $\frac{1}{8}$th of its original volume adiabatically. The pressure of the gas will change to:

A. $24/5$ B. 8 C. $40/3$ D. 32 times the initial

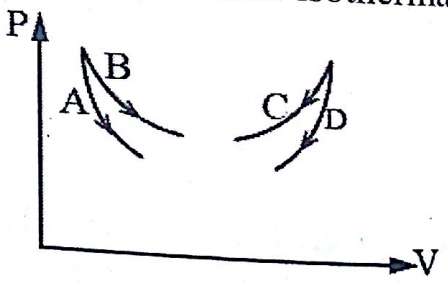
Solution: For adiabatic process, $P_1v_1^{\gamma} = P_2v_2^{\gamma}, V_2 = \frac{1}{8}V_1$

$$\Rightarrow P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma} = P_1\left(\frac{v_1}{\frac{v_1}{8}}\right)^{\gamma} = P_1(8)^{5/3} = 32P_1$$

Answer: D

171. The curves which represent adiabatic and isothermal expansion are respectively.

A. C and D
B. A and B
C. D and C
D. B and A



Answer: B

172. A polyatomic gas $\left(\gamma = \frac{4}{3}\right)$ is compressed to $\frac{1}{8}$th of its volume adiabatically. If its initial pressure is P_o, its new pressure will be:

A. $8P_o$ B. $16P_o$ C. $6P_o$ D. $2P_o$

Solution: $P_1v_1^{\gamma} = P_2v_2^{\gamma}$, adiabatic $P_ov_1^{\gamma} = P_2\left(\frac{v_1}{8}\right)^{\gamma}, v_2 = \frac{v_1}{8}$

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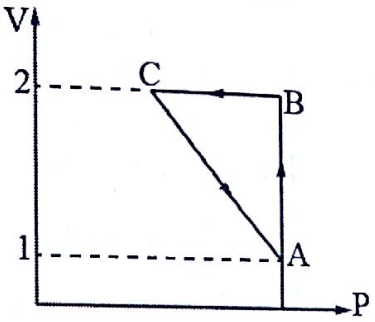
$$P_2 = P_o\left(\frac{\frac{V_1}{V_1}}{\frac{8}{}}\right)^{\gamma} = P_o(8)^{\gamma}$$

$$P_2 = P_o(8)^{4/3} = 16P_o$$

Answer: B

173. An ideal gas is taken through the cycle A→B→C→A. If the net heat supplied to the gas in the cycle is 5J, the work done by the gas in the process C→A is:



A. −5J C. −10J
B. −15J D. −20J

Solution:
For a cyclic process, $\Delta W = \Delta Q$
$\Delta W = \Delta Q_{ABCA} = W_{AB} + W_{BC} + W_{CA}$
$5 = 10 + 0 + W_{CA} \Rightarrow W_{CA} = -5J$

Answer: A

174. A gas is compressed at a constant pressure of 50 N/m^2 from a volume of $10m^3$ to a volume of $4m^3$. Energy of 100J is then added to the gas by heating. Its internal energy is:

A. Increased by 400J C. Increased by 100J
B. Increased by 200J D. Decreased by 200J

Solution: $\Delta u = \Delta Q + \Delta W$
$\Delta u = \Delta Q - P\Delta v$
$= 100 - 50\,(4 - 10) = 400J$

Answer: A

175. During the adiabatic expansion of 2 mol of a gas, the internal energy of the gas is found to decrease by 2J. The work done during the process on the gas will be equal to:

A. 1J B. −1J C. 2J D. −2J

Solution: $\Delta u = \Delta W$, adiabatic

Answer: D

176. If the amount of heat given to a system be 35J and the amount of work done by the system be −15J, then the change in the internal energy of the system is:

A. −50J B. 20J C. 30J D. 50J

Solution: $\Delta u = \Delta Q + \Delta W = 35 - (-15) = 50J$

Answer: D

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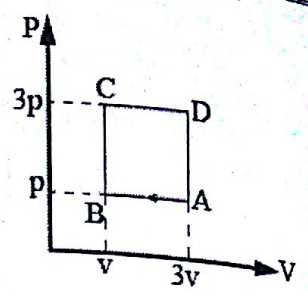
177. An ideal monatomic gas is taken around the cycle ABCDA as shown. The work done during the cycle is:

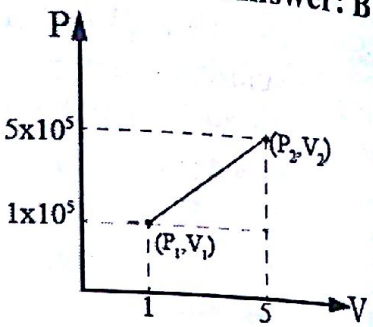


A. PV C. 2PV
B. 4PV D. Zero

Solution:
ΔW = Area of ABCD = (3P – P)(3V – V)
= (2P)(2V) = 4PV

Answer: B

178. A system changes from the state (P_1, V_1) to (P_2, V_2) as shown. The work done by the system is:



A. 7.5×10^5J C. 5×10^5J
B. 12×10^5J D. 6×10^5J

Solution: ΔW = Area under the curve

$$= \frac{1}{2}(4\times10^5)(4) + (1\times10^5)(4)$$

$$= 12\times10^5 \text{J}$$

Answer: B

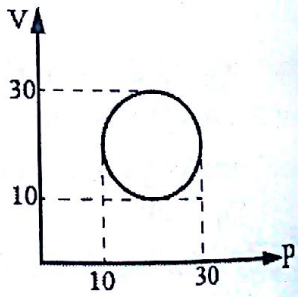
179. A fixed mass of an ideal gas absorbs 1000J of heat and as a result expands at a constant pressure of 2×10^4Pa from 0.05m^3 to 0.075m^3. What is the effect on the internal energy of the gas?

A. Increased by 1000J C. Decreased by 500J
B. Decreased by 1000J D. Increased by 500J

Solution: $\Delta u = \Delta Q + \Delta W = \Delta Q - P\Delta V$
$= 1000 - 2 \times 10^4(0.075 - 0.05) = 500$J

Answer: D

180. The heat energy absorbed by a system in going through a cyclic process is:



A. $10^7\pi$J
B. $10^4\pi$J
C. $10^2\pi$J
D. $10^{-3}\pi$J

Solution: For a cyclic process

$\Delta Q = \Delta W$ = Area = πr^2

$= \pi r^2 = \pi (10)^2 = 10^2\pi$, D = 20, $r = \frac{D}{2} = \frac{20}{2} = 10$

Answer: C

181. The internal energy of an ideal gas depends on:

A. Specific volume C. Pressure
B. Temperature D. Density

Answer: B

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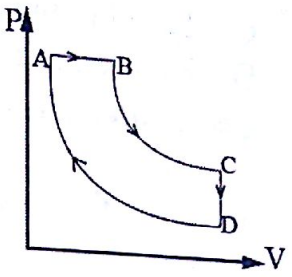
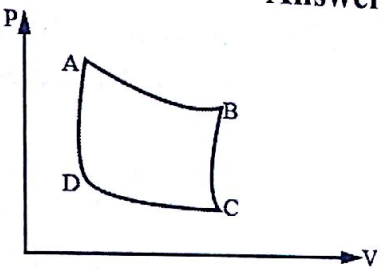
182. In the pressure – volume diagram shown, the isochoric, isothermal, isobaric and adiabatic parts respectively are:



A. BA, AD, DC, CB
B. DC, CB, BA, AD
C. AB, BC, CD, DA
D. CD, DA, AB, BC

Answer: D

183. The P–V graph of an ideal gas cycle is shown. The adiabatic process is described by:



A. AB and BC
B. AB and CD
C. BC and DA
D. BC and CD

Answer: C

184. Calculate the work done when 1 mole of mono atomic ideas gas at initial pressure of 10^5Pa is adiabatically compressed from 6L to 2L.

A. –957J B. –57J C. 95J D. 157J

Solution: $P_1v_1^{\gamma} = P_2v_2^{\gamma} \Rightarrow P_2 = P_1\left(\frac{v_1}{v_2}\right)^{\gamma} = 10\left(\frac{6}{2}\right)^{5/3} = 6.2\times10^5\,\text{Pa}$

then, $W = \frac{1}{1-\gamma}(P_2v_2 - P_1v_1) = -957\text{J}$

Answer: A

185. In an adiabatic process

A. There is no transfer of heat, hence the temperature remains constant.
B. There is a transfer of heat in such a way that the temperature remains constant.
C. There is no transfer of heat, but the temperature may vary.
D. There is free transfer of heat as well as variation in the temperature.

Answer: C

186. Air in a cylinder is suddenly compressed by a piston. If the volume after compression is half the original volume, the final pressure immediately after compression is:

A. Greater than twice the initial pressure.
B. Equal to twice the initial pressure.

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C. Equal to the initial pressure.
D. Half the initial pressure.

Explanation: The process is adiabatic. Hence $P_2 = P_1 (2)^{\gamma}$

Answer: A

187. The complete conversion of heat into useful mechanical work occurs:
A. Often B. Seldom C. Never D. Depends on the device

Answer: C

188. The complete conversion of mechanical energy into heat occurs:
A. Often B. Seldom C. Never D. Depends on friction

Answer: A

189. Two cylinders A and B contain equal amount of diatomic ideal gas at 300K. The piston of A is free to move, while that of B is fixed. When the cylinders are given the same amount of heat, the rise in temperature of the gas in A is 30k. What is the rise in temperature of the gas in B?
A. 30K B. 18K C. 50K D. 42K

Solution: $Q_A = Q_B$, A is isobaric & B is isochoric

$$C_P n\Delta T_A = C_v n\Delta T_B \Rightarrow \left(\frac{7}{2}\right)(30) = \left(\frac{5}{2}\right)(\Delta T_B) \Rightarrow T_B = 42K$$

Answer: D

190. During an adiabatic expansion of a 2 moles ideal gas, the internal energy was found to decrease by 100J. The work done by the gas in this process is:
A. Zero B. −100J C. 200J D. 100J

Solution: $\Delta u = \Delta W \Rightarrow -100 = -\Delta W$, $\Delta W = 100J$ **Answer: D**

191. If R is the universal gas constant, the amount of heat needed to raise the temperature of 2 moles of an ideal mono atomic gas from 273K to 373K when no work is done is:
A. 100R B. 10R C. 300R D. 500R

Solution: Work is zero for iso choric process.

$$\Delta Q_v = \frac{3}{2} nR\Delta T = \frac{3}{2}(2)(R)(373 - 273) = 300R$$

Answer: C

Kinetic Theory of Gases

192. If k, u and E represent kinetic energy, potential energy and total energy of one mole of an ideal gas then:
A. K = 0, E = u C. E = 0, K = u

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B. $u = 0, E = K$ D. $E = 0, k = -u$

Explanation: For an ideal gas there is no intermolecular force and hence no potential energy.

Answer: B

193. Two identical cylinders contain Helium at 2.5atm and Argon at 1 atm. If both the gases are filled in one of the cylinders, the pressure would be:

A. 3.5 atm B. 1.75atm C. 1.5 atm D. 1 atm

Solution: From Dalton's partial pressure, the total pressure is,

$P = P_1 + P_2 = 2.5 + 1 = 3.5$ atm

Answer: A

194. Two vessels having equal volume contain H_2 at 1 atm and He at 2 atm. If both samples are at the same temperature, the mean velocity of H_2 molecule is:

A. Equal to that of He C. Half that of He

B. Twice that of He D. $\sqrt{2}$ times that of He

Solution: $<KE> = \frac{3}{2}kT$

$$\frac{1}{2}M_1C_1^2 = \frac{3}{2}kT \text{ ..... for } H_2$$

$$\frac{1}{2}M_2C_2^2 = \frac{3}{2}kT \text{ ..... for He}$$

Taking ratio, $\frac{M_1C_1^2}{M_2C_2^2} = 1 \Rightarrow \frac{C_1^2}{C_2^2} = \frac{M_2}{M_1}, M_1 = M_{H2} = 2g$

$$M_2 = M_{He} = 4g$$

$$\frac{C_1}{C_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{4}{2}} = \sqrt{2}$$

$$\frac{C_1}{C_2} = \sqrt{2} \Rightarrow C_1 = \sqrt{2}\,C_2$$

Answer: D

195. The average velocity of the molecules in a gas in equilibrium is:

A. Proportional velocity to $\sqrt{T}$ C. Proportional to T^2

B. Proportional to T D. Equal to zero.

Explanation: Motion of molecules is random, and mean velocity of random motion is zero.

Answer: D

196. In the equation $Pv = RT$, v stands for volume of:

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A. Any amount of gas
B. One gram of gas
C. One gram mole of gas
D. One litre of gas

Answer: C

197. A container contains He gas. When it is heated from 300K to 600K, the average kinetic energy of the He atoms is:
A. Halved
B. Unchanged
C. Doubled
D. Become $\sqrt{2}$ times

Explanation: $<KE> = \frac{3}{2}kT$, hence

$<kE> \sim T$, and as T is doubled, $<kE>$ is doubled

Answer: C

198. A gas at a pressure P_o is contained in a vessel. If masses of all molecules are halved and their speeds doubled, the resulting pressure P would be equal to:

A. $4P_o$ B. $2P_o$ C. P_o D. $P_o/2$

Solution: $P = \frac{1}{3}\rho<C^2>, \rho = \frac{m}{v}, v_1 = v_2$

$$P_o = \frac{1}{3}\frac{m_1}{v}<C_1^2> \quad \ldots\ldots (1)$$

$$P_2 = \frac{1}{3}\frac{m_2}{v}<C_2^2>, m_2 = \frac{m_1}{2} \text{ and } C_2 = 2C_1$$

$$P_2 = \frac{1}{3}\frac{m_1}{2v}<(2C_1)^2>$$

$$P_2 = \frac{1}{3}\frac{m_1}{2v}4<2C_1^2> \quad \ldots\ldots (2)$$

taking ratio of equation (2) to (1)

$$\frac{P_2}{P_o} = 2 \Rightarrow P_2 = 2P_o$$

Answer: B

199. The temperature of an ideal gas is increased from 27°C to 927°C. The root mean square speed of its molecules becomes:
A. Twice B. Half C. Four times D. one forth

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Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{300}{1200}} = \sqrt{\frac{1}{4}} = \frac{1}{2} \Rightarrow C_2 = 2C_1$

Answer: A

200. Oxygen and hydrogen gases are at the same temperature T. The kinetic energy of an oxygen molecule will be equal to:
A. 16 times the kinetic energy of a hydrogen molecule.
B. 4 time the kinetic energy of a hydrogen molecule.
C. The kinetic energy of a hydrogen molecule.
D. One – fourth the kinetic energy of a hydrogen molecule.

Explanation: $\langle kE \rangle = \frac{5}{2} kT$, diatomic gas

$\langle kE \rangle = \frac{3}{2} kT$, mono atomic gas

At the same temperature, the molecules of H_2 and O_2 do have the same kinetic energy.

Answer: C

201. If pressure of a gas contained in a closed vessel is increased by 0.4% when heated by 1°C, the initial temperature must be:
A. 250k B. 250°C C. 25000k D. 25°C

Solution: From, $pv = nRT$

$P_1 v = nRT_1$ ..... (1)

and, $P_2 v = nRT_2$, but $P_2 = \frac{0.4}{100} P_1 + P_1 = 1.04P_1$

$T_2 = T_1 + 1$

$1.04P_1 v = nR(T_1 + 1)$ ..... (2)

taking ratio of equation (1) to (2)

$$\frac{1}{1.04} = \frac{T}{T_1 + 1} \Rightarrow T_1 + 1 = 1.04T_1, T_1 = \frac{1}{0.04} = 250k$$

Answer: A

202. Jar A is filled with a gas characterized by parameters p, v and T and jar B is filled with a gas with parameters 2p, $\frac{v}{4}$ and 2T. The ratio of the number of moles of A to B is:
A. 1: 1 B. 1: 2 C. 2: 1 D. 4: 1

Solution: $Pv = nRT$ ..... (1)

$(2P)\left(\frac{v}{4}\right) = n'R(2T)$ ..... (2)

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taking ratio, $\frac{n}{2n'} \Rightarrow 2 \Rightarrow n: n' = 4: 1$ **Answer: D**

203. A gas in container A is in thermal equilibrium with a gas of the same number of moles in container B. Then:

A. $P_A = P_B, V_A \neq V_B$ C. $P_A \neq P_B, V_A = V_B$

B. $P_A/V_A = P_B/V_B$ D. $P_AV_A = P_BV_B$

Explanation: In thermal equilibrium, temperature is same.

Hence, $P_AV_A = P_BV_B \rightarrow$ Boyle's law, or

From $Pv = nRT$

$P_AV_A = nRT_A$ .....(1)

$P_BV_B = nRT_B$ .....(2)

taking ratio, $\frac{P_AV_A}{P_BV_B} = 1 \Rightarrow P_AV_A = P_BV_B$ **Answer: D**

204. A jar has a mixture of hydrogen and oxygen gases in the ratio 1: 5. The ratio of the mean kinetic energies of hydrogen and oxygen molecules is:

A. 1: 16 B. 1: 4 C. 1: 5 D. 1: 1

Explanation: Average kinetic energy of molecules of any diatomic gas is given by $\langle kE \rangle = \frac{5}{2} kT$. And it's the same for H_2 and O_2 molecules. **Answer: D**

205. The temperature of a gas is a measure of:

A. The average kinetic energy of the gaseous molecules.
B. The average potential energy of the gaseous molecules.
C. The average distance between the molecules of the gas.
D. The size of the molecules of the gas.

Answer: A

206. A vessel has 6g of oxygen at pressure P and temperature 400k. A small hole is made in it so that oxygen leaks out. How much oxygen leaks out if the final pressure is $P/2$ and temperature 300K?

A. 5g B. 4g C. 2g D. 3g

Solution: $P_1V = n_1RT_1$, $n_1 = \frac{m_1}{M} = \frac{6}{32}, n_2 = \frac{m_2}{m} = \frac{m_2}{32}$

$P_2V = n_2RT_2$

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$$Pv = \frac{6}{32}(R)(400) \ldots\ldots\ldots\ldots\ldots (1)$$

$$\frac{P}{2}V = \frac{m_2}{32}(R)(300) \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Taking ratio, (1) to (2)

$$2 = \frac{(6)(400)}{(m_2)(300)} \Rightarrow m_2 = 4g$$

$m_2 + m_{leaked} = 6 \Rightarrow M_{leaked} = 6 - 4 = 2g$ **Answer: C**

207. At room temperature, the rms speed of the molecules of a certain diatomic gas is found to be 1920 m/s. The gas is:

A. H_2 B. F_2 C. O_2 D. Cl_2

Solution: $<kE> = \frac{3}{2}RT$

$$\frac{1}{2}MC^2 = \frac{3}{2}RT$$

$$M = \frac{3RT}{C^2} = \frac{(3 \times 8.31 \times 10^3 \times 300)}{(1920)^2} = 2g$$ **Answer: A**

208. If one mole of mono atomic gas $\gamma = 5/3$ is mixed with one mole of a diatomic gas $\gamma = 7/5$, the value of γ for the mixture is:

A. 1.4 B. 1.5 C. 1.53 D. 3.07

Solution: $(C_v)_{mix} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$

$$\frac{R}{\gamma_{mix} - 1} = \frac{n_1\left(\frac{R}{\gamma_1 - 1}\right) + n_2\left(\frac{R}{\gamma_2 - 1}\right)}{n_1 + n_2} = \frac{\frac{1}{\left(\frac{5}{3} - 1\right)} + \frac{1}{\left(\frac{7}{5} - 1\right)}}{2} = \frac{\frac{3}{2} + \frac{5}{2}}{2}$$

$$\frac{1}{\gamma_{mix} - 1} = 2 \Rightarrow \gamma_{mix} - 1 = \frac{1}{2} \Rightarrow \gamma_{mix} = \frac{3}{2} = 1.5$$

Answer: B

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209. When an ideal diatomic gas is heated at constant pressure, the fraction of heat energy that increase the initial energy of the gas is:

A. 2/5 B. 3/5 C. 3/7 D. 5/7

Solution: From first law, $\Delta u = \Delta Q - P\Delta v$

at constant pressure,

$$\Delta u = C_P nR\Delta T - P\Delta v,\ C_P = \frac{7}{2}R \text{, diatomic}$$

$$\Rightarrow \Delta u = \frac{7}{2}nR\Delta T - nR\Delta T = \frac{5}{2}nR\Delta T,\ P\Delta v = nR\Delta T$$

$$\frac{\Delta u}{\Delta Q} = \frac{\frac{5}{2}nR\Delta T}{\frac{7}{2}nR\Delta T} = \frac{5}{7}$$

Answer: D

210. The absolute temperature of a gas is increased three times. The root mean square speed of the gas molecules will increase.

A. 3 times B. 9 times C. $\frac{1}{3}$ times D. $\sqrt{3}$ times

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{T_1}{3T_1}} = \frac{1}{\sqrt{3}} \Rightarrow C_2 = \sqrt{3}C_1$ **Answer: D**

211. The average translational energy and the rms speed of molecules in a sample of O_2 gas at 300k are 6.21×10^{-21}J and 484 m/s respectively. The corresponding values at 600K are nearly.

A. 12.42×10^{-21}J, 968 m/s C. 6.21×10^{-21}J, 968 m/s

B. 8.78×10^{-21}J, 684 m/s D. 12.42×10^{-21}J, 684 m/s

Solution: $\langle kE \rangle \sim T \Rightarrow \frac{kE_2}{kE_1} = \frac{T_2}{T_1} \Rightarrow \frac{600}{300} = 2 \Rightarrow kE_2 = 2kE_1$

and, $C \sim \sqrt{T} \Rightarrow \frac{C_2}{C_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{600}{300}}\sqrt{2} \Rightarrow C_2 = \sqrt{2}C_1$ **Answer: D**

212. 2 mole of mono atomic gas $\gamma = \frac{5}{3}$ and 1 mole of a diatomic gas $\gamma = \frac{7}{5}$ are mixed. The γ value for the mixture is:

A. 5/3 B. 7/5 C. 17/11 D. 11/7

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Solution: $\frac{R}{\gamma_{mix}-1}=\frac{n_1\left(\frac{R}{\gamma_1-1}\right)+n_2\left(\frac{R}{\gamma_2-1}\right)}{n_1+n_2}$

$$\frac{1}{\gamma_{mix}-1}=\frac{\frac{2}{\left(\frac{5}{3}-1\right)}+\frac{1}{\left(\frac{7}{5}-1\right)}}{2+1}$$

$$\frac{1}{\gamma_{mix}-1}=\frac{11}{6}\Rightarrow\gamma_{mix}-1=\frac{6}{11}\Rightarrow\gamma_{mix}=\frac{17}{11}$$ **Answer: C**

213. The molecules of a given mass of a gas have rms speed of 200 m/s at 27°C and 10^5Pa. When the temperature is doubled and pressure halved, the rms speed of the molecules of the gas is,

A. 200 m/s B. 400m/s C. $200\sqrt{2}$m/s D. $100\sqrt{2}$m/s

Solution: $\frac{C_1}{C_2}=\sqrt{\frac{T_1}{T_2}}=\sqrt{\frac{T_1}{2T_1}}=\sqrt{\frac{1}{2}}\Rightarrow C_2=\sqrt{2}C_1$ **Answer: C**

214. Oxygen and helium are at the same temperature T. The mean kinetic energy of helium is E, then that of oxygen will be:

A. E B. 4E C. $\frac{5}{3}$E D. $\frac{3}{5}$E

Solution: For diatomic gas molecule $<kE> = \frac{5}{2}kT$

and, for mono atomic gas molecule $<kE> = \frac{3}{2}kT$

taking ratio, $\frac{<kE_0^2>}{E}=\frac{5}{3}\Rightarrow<kE>_0^2=\frac{5}{3}E$.

Answer: C

215. At a given temperature the rms speed of hydrogen molecule is V then than for oxygen at the same temperature will be:

A. V B. 4V C. $\frac{V}{4}$ D. $\frac{V}{16}$

Solution: $<kE>=\frac{3}{2}kT$ ..... Diatomic

$\frac{1}{2}M_1C_1^2=\frac{3}{2}KT$ ..... (1) Hydrogen

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$\frac{1}{2}M_2C_2^2 = \frac{3}{2}KT$ ..... (2) Oxygen

Taking ratio, $m_1C_1^2 = m_2C_2^2$

$$\frac{C_1}{C_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{32}{2}} = 4$$

$$\frac{C_1}{C_2} = 4 \Rightarrow \frac{v}{C_2} = 4 \Rightarrow V/4$$

Answer: C

216. A vessel contains 1 mole of O_2 gas at temperature T. The pressure of the gas is P. The same vessel when containing one mole of the gas at temperature 2T has a pressure of:

A. $P/8$ B. P C. 2P D. 8P

Solution: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$, $V_1 = V_2$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow P_2 = \frac{P}{T}(2T) = 2P$$

Answer: C

217. Which of the following molecular properties is the same for all ideal gases at a given temperature?

A. Momentum
B. Mean kinetic energy
C. rms speed
D. Mean path free

Answer: B

218. For an ideal gas, the intermolecular interaction is:

A. Attractive B. Repulsive C. very large D. Zero

Hint: molecules far apart

Answer: D

219. The temperature of an ideal gas is increased from 120K to 480K. If the rms speed at 120K is v, then at 480K, it will be:

A. 4V B. 2V C. $v/2$ D. $v/4$

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v}{C_2} = \sqrt{\frac{120}{480}} = 2 \Rightarrow C_2 = V/2$

Answer: C

220. A vessel contains 16g of oxygen at 27°C. The translational kinetic energy of oxygen is:

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A. 800J B. 1125J C. 1868J D. 3120J

Solution:

$$u = \langle kE \rangle = \frac{3}{2}nRT = \left(\frac{3}{2}\right)\left(\frac{m}{M}\right)(R)(T) = \left(\frac{3}{2}\right)\left(\frac{16}{32}\right)(8.3)(300)$$

$= 1868J$

Answer: C

221. By what percent should the pressure of a gas be increased to decrease its volume by 10% at constant temperature?

A. 8.1% B. 9.1% C. 10.1% D. 11.1%

Solution: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$, $T_1 = T_2$ and $V_2 = 0.9V_1 = V_1 - 0.1V_1$

$$P_1V_1 = P_2V_2$$

$$P_2 = P_1\left(\frac{V_1}{V_2}\right) = P_1\left(\frac{1}{0.9}\right) = \frac{10}{9}P_1 = (1.11)P_1 = P_1(1 + 0.11)$$

$$P_2 = P_1 + \frac{11}{100}P_1 \Rightarrow \Delta P = 11\%$$

Answer: D

222. If P represents the pressure exerted by an ideal gas and E represents the translational kinetic energy per volume of gas, then:

A. $P = E$ B. $P = 2E$ C. $P = 3E$ D. $P = \frac{2}{3}E$

Solution: $P = \frac{1}{3}\rho\langle C^2 \rangle = \frac{1}{3}\frac{m}{v}\langle C^2 \rangle = \frac{2}{3}\frac{m}{v}\frac{1}{2}\langle C^2 \rangle$

$$P = \frac{2}{3}\frac{1}{v}\frac{1}{2}m\langle C^2 \rangle = \frac{2}{3}E, \frac{\langle kE \rangle}{v} = E, \text{ given}$$

Answer: D

223. What is the ratio of the total energy of all the molecules of one mole of O_2 to the total energy of all the molecules of two moles of He at the same temperature.

A. 1: 2 B. 2: 1 C. 2: 3 D. 3: 2

Solution: $u_1 = \frac{3}{2}n_1RT$ ..... O_2

$u_2 = \frac{3}{2}n_2RT$ ..... He

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taking ratio, $\frac{u_1}{u_2} = \frac{n_1}{n_2} = \frac{1}{2} = 1:2$

Answer: A

224. According to the kinetic theory of gases, the pressure exerted by a gas the walls is:
A. Rate of change of momentum per second per unit area.
B. Momentum imparted per second to the wall per unit area.
C. Change of momentum imparted per unit area.
D. Change in momentum per unit volume.

Answer: B

225. At a certain temperature, the rms speed of O_2 is 400 m/s. At the same temperature, the rms speed of H_2 molecule will be:
A. 100 m/s B. 25 m/s C. 1600 m/s D. 6400m/s

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{2}{32}} = \frac{1}{4} \Rightarrow C_2 = 4C_1 = 4(400) = 1600\text{m/s}$

Answer: C

226. The density of an ideal gas is:
A. Directly proportional to both its pressure and temperature.
B. Inversely proportional to both its pressure and temperature.
C. Directly proportional to its pressure and inversely proportional to its temperature.
D. Inversely proportional to its pressure and directly proportional to its temperature.

Explanation: $PV = nRT,\ v = \frac{m}{\rho}$

$$P\frac{m}{\rho} = nRT \Rightarrow \rho = \frac{Pm}{nRT},$$

Answer: C

227. The fastest moving molecules are those of:
A. N_2 B. O_2 C. H_2 D. Cl

Explanation: $C \sim \frac{1}{\sqrt{m}}$, the smaller the mass the larger the speed.

Answer: C

228. N is the total number of molecules in an ideal gas on an average, the number of molecules moving in a particular direction is:
A. N/2 B. N/3 C. N/4 D. N/6

Explanation: The molecules move in x, y and z direction equally.

Answer: B

229. The internal energy of a monatomic ideal gas is:

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A. Partly kinetic and partly potential
B. Totally kinetic
C. Totally potential
D. Neither kinetic or potential

Answer: B

230. Boyle's law is applicable in
A. Isobaric process C. Isothermal Process
B. Isochoric process D. Adiabatic process **Answer: C**

231. The average translational kinetic energy of the molecules of a gas will be doubled if at constant.
A. Volume, Pressure is doubled.
B. Temperature, Pressure is doubled.
C. Pressure, volume is halved.
D. Temperature, volume is doubled. **Answer: A**

232. An ideal gas is found to obey VP^2 = constant. The gas is initially at temperature T and volume V. If its expands to a volume 2V, the temperature becomes:

A. T B. 2T C. $T\sqrt{2}$ D. T/2

Solution: From $PV = nRT$, $P = \frac{nRT}{v}$ and $V_1P_1^2 = V_2P_2^2$, $V_2 = 2V$

$$V\left(\frac{nRT}{v}\right)^2 = 2V\left(\frac{nRT_2}{2v}\right)^2$$

$$T^2 = \frac{T_2^2}{2} \Rightarrow T_2 = \sqrt{2}T$$

Answer: C

233. The temperature at which the rms speed of the gas molecules become twice of its value at 0°C is:
A. 819°C B. 1092°C C. 1100°C D. 1400°C

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{273}{T_2}}$

$$\frac{C_1}{2C_1} = \sqrt{\frac{273}{T_2}} \Rightarrow \frac{1}{4} = \frac{273}{T_2} \Rightarrow T_2 = 4(273) = 1092K$$

$T_2 = 1092 - 273 = 819°C$ **Answer: A**

234. A sample of O_2 is compressed to half of its original volume at constant temperature. If the rms speed of its gas molecules was originally C, their new rms speed is:
A. 4C B. $\sqrt{2}$ C C. C D. C/L

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Explanation: No change in temperature and hence no change in rms speed.

Answer: C

235. A vessel contains a mixture of different types of gases. Then:
A. On an average, the heavier molecules have higher speed.
B. The average speed of different molecules is the same.
C. The average kinetic energy of different molecules is the same.
D. On average, the heavier molecules have higher kinetic energy.

Answer: C

236. Rms speed of gas molecule is 300 m/s. If their mass is doubled and the temperature is halfed, the new rms speed is:
A. 300 m/s B. 600 m/s C. 75m/s D. 150m/s

Solution: $u = \frac{3}{2}nRT$

$$\frac{1}{2}M_1C_1^2 = \frac{3}{2}nRT_1 \ldots\ldots\ldots\ldots (1)$$

$$\frac{1}{2}M_2C_2^2 = \frac{3}{2}nRT_2 \ldots\ldots\ldots\ldots (2), M_2 = 2m_1, T_2 = \frac{1}{2}T_1$$

taking ratio, $\frac{C_1^2}{2C_2^2} = 2 \Rightarrow C_2^2 = \frac{C_1^2}{4}$

$$C_2 = \frac{C_1}{2} = \frac{300}{2} = 150 m/s$$

Answer: D

237. A box contains x molecules of a gas. How will the pressure of the gas be affected if the number of molecules is made 2x?
A. Pressure will decrease C. Pressure will be doubled
B. Pressure will remain unchanged D. Pressure will be 3 times

Explanation: $Pv = nRT \Rightarrow P \sim n$

Answer: C

238. If the kinetic energy of an ideal gas is 60J and its volume is 3L, then its pressure will be:

A. $2 \times 10^{-4} Pa$

B. $4 \times 10^{-4} Pa$

C. $\frac{4}{3} \times 10^4 Pa$

D. $\frac{2}{3} \times 10^{-4} Pa$

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Solution: $P = \frac{1}{3}\rho\langle C\rangle = \frac{1}{3}\frac{m}{v}\langle C\rangle = \frac{2}{3}\frac{1}{2}\frac{m\langle C\rangle}{v} = \frac{2}{3}\frac{\langle kE\rangle}{v}$

$$P = \frac{2}{3}\frac{(60)}{3\times10^{-3}} = \frac{4}{3}\times10^{4}\text{Pa}$$

Answer: C

239. One mole of H_2 and one mole of O_2 are at same temperature. The ratio of the rms speed of H_2 to O_2 molecules is:

A. 4:1 B. 1: 4 C. 1.1 D. 16:1

Solution: $\frac{C_1}{C_2} = \sqrt{\frac{m_2}{m_1}} \Rightarrow \frac{C_1}{C_2} = \sqrt{\frac{32}{2}} = 4 \Rightarrow C_1 : C_2 = 4:1$

Answer: A

240. A container has N molecules at temperature T. If the number of molecules is doubled but kinetic energy in the box remains as before, the temperature will become:

A. T B. 2T C. $T/2$ D. Zero

Solution: $u = k_{E_1} = C_v n_1 R T_1$ and $k_{E_2} = C_v n_2 R T_2$, but $k_{E_1} = k_{E_2}$

$C_v n_1 R T = C_v (2n_1) R T_2 \Rightarrow T_2 = T/2$

Answer: C

Second Law of Thermodynamics and Efficiency

241. If the door of an operating refrigerator is opened. The temperature of the room will:

A. Increase C. Decrease

B. Unchanged D. Decrease in summer, increase in Winter

Explanation: In a refrigerator, heat is pumped from the cold interior to the hot exterior (room). Hence temperature of the room increases since $Q_H = Q_C + W$.

Answer: A

242. At absolute zero, entropy of a substance becomes:

A. Positive C. Zero

B. Negative D. Infinity

Explanation: Entropy $= \frac{\Delta Q}{T}$, at T = 0K, entropy is ∞

Answer: D

243. What is the approximate coefficient of performance of a refrigerator working between 30°C and 0°C?

A. 0 B. 1 C. 9 D. 10

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Solution:

$$\text{Coefficient of performance } (\beta) = \frac{Q_C}{W} = \frac{T_C}{T_H - T_C} = \frac{273}{303 - 273} = 9$$

Answer: C

244. A Carnot engine takes 1260J of heat at 500K and rejects 630J of heat to the sink. The temperature of the sink is?

A. 1000k B. 250k C. 750k D. 125k

Solution: For Carnot's engine, $\frac{T_C}{T_H} = \frac{Q_C}{Q_H}$

$$\Rightarrow \frac{T_C}{500} = \frac{630}{1260} \Rightarrow T_C = 250K$$

Answer: B

245. A Carnot engine working between 300K and 600K has a work output of 800J per cycle. The amount of heat energy supplied from the source to the engine in each cycle is:

A. 800J B. 3200J C. 1600J D. 6400J

Solution: $\eta = \frac{W}{Q_H} = 1 - \frac{T_C}{T_H} \Rightarrow \frac{800}{Q_H} = 1 - \frac{300}{600} \Rightarrow Q_H = 1600J$

246. The efficiency of a Carnot engine:

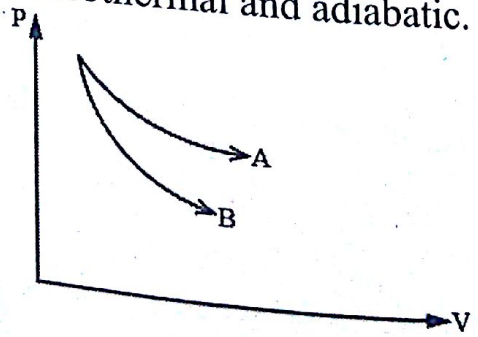
Answer: C

A. Is independent of the temperature of the source and the sink.
B. Is independent of the working substance.
C. Can be 100%.
D. Is independent of heat capacity of the engine parts.

Answer: B

Unit 9: Conceptual Questions

1. Internal energy of an ideal gas depends only on temperature. True or False? Explain
2. When the door of a running refrigerator is left open in a closed room, the room gets cold. True or False? Explain
3. Identify the curves as isothermal and adiabatic.



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4. What fraction of heat energy causes the internal energy of a gas to increase when a monatomic gas is heated at constant pressure?
5. When an ideal gas expands adiabatically, its internal energy decreases. True or False? Explain
6. In an isobaric process, the change in internal energy of n moles of an ideal gas when its temperature rises by ΔT is $nC_P\Delta T$. True or False? Explain
7. If heat is supplied to a gas in an isothermal process, the internal energy of the gas increases. True or False? Explain
8. The efficiency of an ideal engine such as Carnot can be 100%? True or False? Explain
9. The change in entropy for a reversible, adiabatic process is zero. True or False? Explain
10. The entropy change in an adiabatic process must be zero, because $Q = 0$. True or False? Explain
11. Can the coefficient of performance of a heat pump greater than unity? Explain
12. When does an automobile burn more gas. In winter or in summer? Why?
13. Is it possible to describe an irreversible process by a path in a p–v diagram?
14. If the temperature T of an ideal gas doubles, by what factor does the average kinetic energy of the atoms change?

Unit 9: Problems

First law of thermodynamics

15. A gas is compressed at a constant pressure of 3 atm from 8L to 2L. In the process, 500J of heat leaves the gas by heat. What is:
    a) The work done on the gas.
    b) The change in its internal energy.
16. When 1210J of heat is added to one mole of monatomic gas, its temperature increase from 272k to 276k, find the work done by the gas in the process.
17. What is the internal energy of a monatomic ideal gas that contains 4 moles at a temperature of 400k.
18. The temperature of a 7.5 moles of ideal gas decreases by 200k. If 3500J of heat is extracted from it, what is the work done by the gas.

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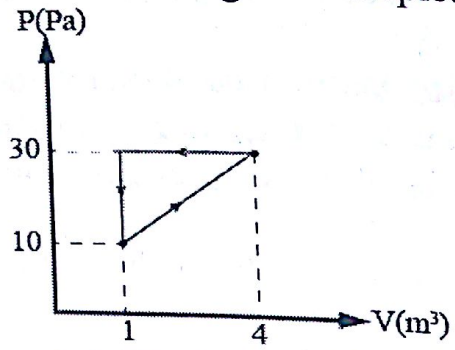
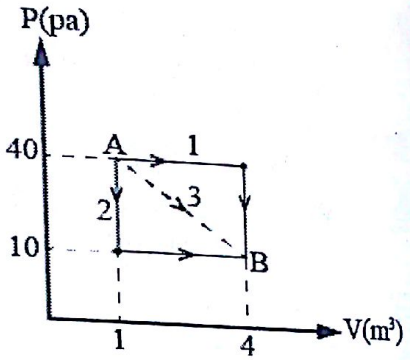
19. A system does 164J of work on its environment and in the process gains 77J of heat. Find the change in the internal energy of:
    a) The system
    b) The environment

20. A gas under goes a cyclic process as shown. Calculate the energy added to the system as heat during one complete cycle.



21. A gas expands from A to B along the three paths shown. Calculate the work done by the gas along paths.
    a) 1
    b) 2
    c) 3



22. A 3 moles of a monatomic ideal gas expand adiabatically and its temperature decreases from 370k to 290k. Calculate:
    a) The work done.
    b) The change in its internal energy.

23. A 2 kg copper of volume $0.08m^3$ is heated at constant pressure of 1.01×10^5Pa and its temperature changes from 30°C to 90°C. If volume expansion of copper is $\gamma = 51 \times 10^{-6}$/°C and its specific heat capacity is C = 387J/kg.k. Calculate:
    a) The work done.
    b) The heat transferred.
    c) The change in internal energy.

24. Find the amount of heat needed to increase the temperature of 2.5 mole of an ideal gas by 15k.
    a) At constant pressure.
    b) At constant volume.

25. At constant pressure, it takes 210J of heat to raise the temperature of an ideal monatomic gas by 17k. How much heat is required for the same temperature change at constant volume?

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Kinetic theory of gases

26. At what temperature is the rms speed of H_2 equal to the rms speed that O_2 has at 303k?
27. What is the average kinetic energy of a molecule of oxygen at a temperature of 300k?
28. A 3 mol of gas at temperature of $-120°C$ fills a 2L container. What is the gas pressure?
29. At what temperature will the molecules of an ideal gas have twice the rms sped they have at 20°C?

Efficiency of heat engine

30. What is the efficiency of an engine that exhausts 870J of heat while doing 340J of useful work?
31. An engine takes in 8000J of heat and discards 5000J each cycle. Calculate:
    a) The work output of the engine.
    b) The efficiency of the engine.
32. A Carnot engine working between 500k and 300k receives 5000J of heat from the hot reservoir.
    a) How much energy does it reject to the cold reservoir?
    b) What is the efficiency of the engine?
33. A heat engine absorbs 1200J of heat from the hot reservoir and exhausts 680J of heat to the cold reservoir. How much work is done by the engine?

Unit 9: Explanation to Questions

1. True. Ideal gas molecules do not exert intermolecular forces.
2. False, Refrigerator pumps heat from cold to hot region. In this case, from interior to exterior (room). Thus temperature of the room increases.
3. A represents isothermal and B represents adiabatic expansions. Adiabatic curve has steeper slope than isothermal.
4. $\frac{\Delta u}{\Delta Q} = \frac{nC_v\Delta T}{nC_P\Delta T} = \frac{C_v}{C_P} = \frac{3}{5}$
5. True. From first law, $\Delta u = \Delta Q - W = -W$
6. False. In any process, $\Delta u = C_v n\Delta T$
7. False. In an isothermal process, internal energy of an ideal gas remains constant.
8. False. It is impossible because of law of nature.
9. True. In this case entropy is constant.

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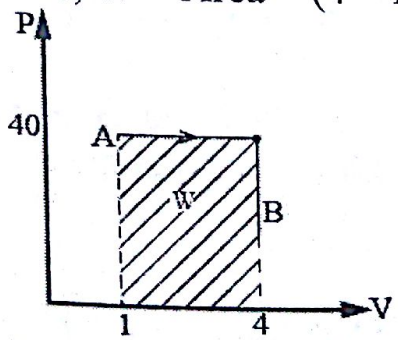
10. False. Entropy does not only depend on Q.
11. Yes. Commercially available refrigerators and air conditioners have coefficient of performance in the range 2 to 6.
12. In Winter. Because, in winter the temperature difference between the hot and cold reservoir is high.
13. No. it is not possible.
14. $\langle kE \rangle = \frac{3}{2} kT$ if T doubles then does kE.

Unit 9: Solution to Problems

15. a) $W = P\Delta v = (3.03 \times 10^5)(2 \times 10^{-3} - 8 \times 10^{-3}) = -1818J$
    b) $\Delta u = \Delta Q - P\Delta v = -500 - (-1818) = 1381J$
16. $\Delta u = \Delta Q + \Delta W \Rightarrow \Delta W = \Delta u - \Delta Q = \frac{3}{2} nR\Delta T - 1210 = \frac{3}{2}$
    $(1)(8.31)(4) - 1210 = 1160J$
17. $u = \frac{3}{2} nRT = \frac{3}{2}(4)(8.31)(400) = 19944J$
18. $\Delta u = \Delta Q + \Delta W \Rightarrow \Delta W = \Delta u - \Delta Q = \frac{3}{2}(7.5)(8.31)(200) - (-3500)$
19. a) $\Delta u = \Delta Q + \Delta W = 77 - 164 = -87J$ it decreases by 87J
    b) $\Delta u = \Delta Q + \Delta W = -77 + 164 = 87J$, it increase by 87J
20. For cyclic process, $\Delta u = 0$, and
    $\Delta W = \Delta Q = \text{Area} = \frac{1}{2}(4 - 1)(30 - 10) = 30J$
21. a) For path 1 we have, $W = \text{Area} = (4 - 1)(40) = 120J$



    b) For path 2, we have, $W = \text{Area} = (4 - 1)(10) = 30J$

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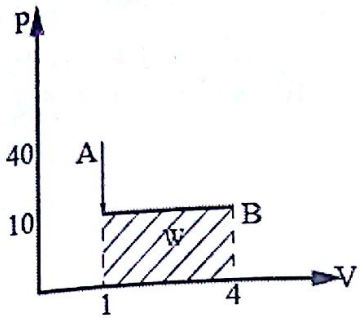

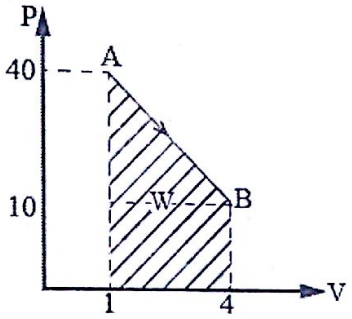

c) For path 3, we have
W = Area shaded
= (4 – 1) (40 – 10) + (4 – 1) (10)
= 75J

22. a) $\Delta W = \frac{3}{2}(n)(R)(\Delta T) = \frac{3}{2}$

$(3)(8.31)(-80) = +2992J$

b) $\Delta u = -\Delta W = -2992J$

23. a) $\Delta u = \gamma v \Delta T = (51 \times 10^{-6})(0.08)(60) = 2.45 \times 10^{-4} m^3$

b) $Q = mC\Delta T = (2)\ (387)\ (60) = 46415J$

c) $\Delta u = \Delta Q - P\Delta u = 46415 - (1 \times 10^{5})\ (2.45 \times 10^{-4})$

24. a) $\Delta Q_P = C_P n\Delta T = \frac{5}{2} nR\Delta T = \frac{5}{2}(2.5)\ (8.31)\ (15) = 799J$

b) $\Delta Q_P = C_v n\Delta T = \frac{3}{2}\ nR\Delta T = \frac{5}{2}(2.5)\ (8.31)\ (15) = 467J$

25. $\Delta Q_v = \frac{5}{2}\ nR\Delta T \Rightarrow n = \frac{2}{5}\frac{\Delta Q}{R\Delta T} = \frac{2}{5}\left(\frac{210}{8.31 \times 17}\right) = 0.6mol$

$\Delta Q_P = \frac{3}{2} nR\Delta T = \frac{3}{2}(0.6)(8.31)(17) \simeq 127J$

26. $C_{H2} = \sqrt{\frac{3RT_1}{M_1}}$ and $C_{O2} = \sqrt{\frac{3RT_2}{M_2}}$, dividing, $\frac{C_{H2}}{C_{O2}} = \sqrt{\frac{T_1 m_2}{T_2 m_1}}$

Here, $C_{H2} = C_{O2}$, $T_2 m_1 = T_1 m_2 \Rightarrow T_1 = \frac{T_2 m_1}{m_2} = \frac{(303)(2)}{32} \simeq 19k$

27. $\langle kE \rangle = \frac{3}{2} kT = \frac{3}{2}(1.38 \times 10^{-23})(300) = 6.21 \times 10^{-21} J$

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28. $Pv = nRT \Rightarrow P = \frac{nRT}{v} = \frac{(3)(8.31)(153)}{2\times10^{-3}} = 1.9\times10^{6}\,Pa$

29. $\frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}}$, $C_2 = 2C_1$ and $T_1 = 20 + 273 = 293$

$\Rightarrow \frac{1}{2} = \sqrt{\frac{293}{T_2}}$, $T_2 = 4(293) = 1172K = 899°C$

30. $\eta = \frac{W}{Q_H} = \frac{W}{W + Q_C} = \frac{340}{340 + 870} = \frac{340}{1210} = 0.28 = 28\%$

31. a) $W = Q_H - Q_C = 8000 - 5000 = 3000J$

b) $\eta = \frac{W}{Q_H} = \frac{3000}{8000} = 0.375 = 37.5\%$

32. a) $\frac{T_C}{T_H} = \frac{Q_C}{Q_H} \Rightarrow \frac{300}{500} = \frac{Q_C}{5000} \Rightarrow Q_C = 3000J$

b) $\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{300}{500} = 1 - 0.6 = 0.4 = 40\%$

33. $W = Q_H - Q_C = 1200 - 680 = 520J$

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Unit 10

Oscillation and Waves

A great many things in the world oscillate back and forth, up and down, left to right, in and out, here and there, to and fro etc. vibration of our vocal chord and vibration of strings and air inside tubes produce sounds of different kind. Parts of machines, and cars vibrate sometimes when we don't want them to. For instance, the way wind, earth quake and people act on large structures such as buildings and bridges will make them oscillate and collapse if engineers do not design these structures to absorb vibrations.

10.1 Periodic Motion

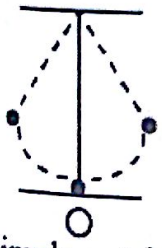
The beating of your heart, the ticking of a clock and the movement of a child on a swing are examples of periodic motion.

Periodic Motion: Is a motion that repeats itself over and over again at equal interval of time.

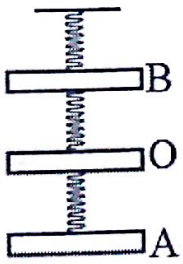
Oscillation of simple pendulum and spring – mass system are familiar examples of periodic motion.

Oscillation: Is a back and forth, up and down, here and there etc vibration of a body about its equilibrium position.

☞ **Equilibrium position** is a central unstretched or rest position of a body.



Simple pendulum



Spring mass - system

In both cases, point 'O' is the equilibrium position.

Conceptual Example

1. What is the basic cause of oscillation?

Explanation: Restoring force is the basic cause of oscillation.

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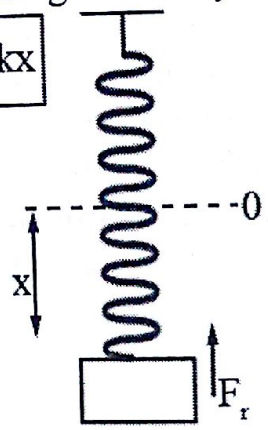
Restoring Force (F_r): Is a force that returns a displaced body to its original position.

☞ A restoring force always push or pulls a body towards the equilibrium position.

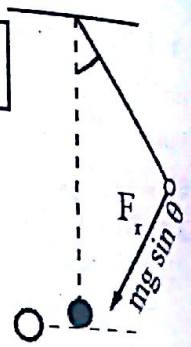
For a spring – mass system

$F_r = -kx$

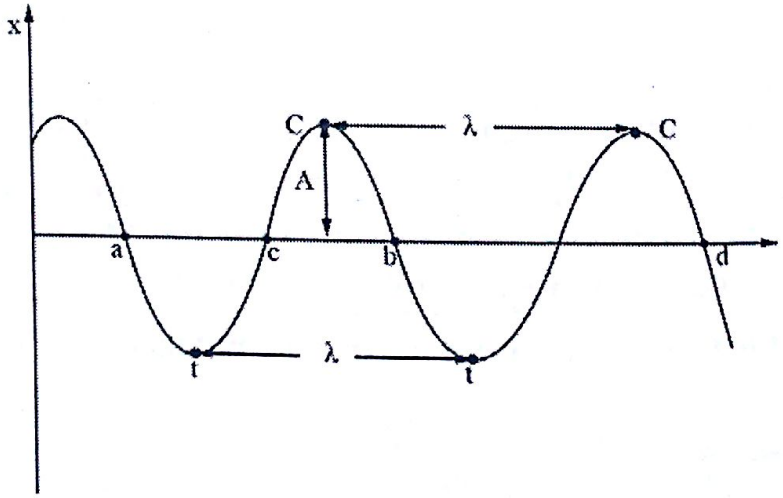


For simple pendulum

$F_r = -mg \sin\theta$



Terms frequently used in waves and oscillations. The displacement – time graph of an oscillating body can be given by a sinusoidal curve gas shown below.



Now, we have the following

Amplitude (A): Is the maximum displacement of the oscillating body from its equilibrium position. Its unit is meter.

Period (T): Is the time take for one complete cycle or oscillation.

$$T = \frac{\text{Total time taken}}{\text{Number of oscillation}}$$ Its unit is second.

Frequency (f): Is the number of oscillations per second.

$$f = \frac{\text{Number of oscillation}}{\text{Total time taken}}$$ Its unit is per second.

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The SI unit of frequency is Hertz (Hz), named for the German physicist Heinrich Hertz in honor of his studies of radio waves.

Thus, $\boxed{1\ Hz = 1\ sec^{-1}}$

☞ Period and frequency are reciprocal to each other.

$$\boxed{T = \frac{1}{f} \text{ or } f = \frac{1}{T}}$$

Wave length (λ): Is the distance covered in one complete cycle from the graph the distance from a to b, b to d, c to c and t to t are all corresponds to complete cycles as well as full wavelengths.

Angular frequency (ω): Is the angular speed of the oscillating body.

For one complete cycle, $\boxed{\omega = 2\pi f = \frac{2\pi}{T}}$

Illustrative Examples

2. A body makes 20 oscillations in 50 seconds. Calculate:
   a) The period
   b) The frequency
   c) The angular frequency

Solution: a) $T = \frac{\text{Total time taken}}{\text{Number of oscillations}} = \frac{50}{20} = 2.5\,\text{sec}$

b) $f = \frac{\text{Number of oscillations}}{\text{Total time taken}} = \frac{20}{50} = 0.4\text{Hz}$

c) $\omega = 2\pi f = (2\pi)(0.4) = 0.8\pi$ rad/sec

3. An oscillating system is used to create waves on a water. The rate of wave generation is one every 0.33 seconds. What is the period of these waves? (UEE 2005)

A. 0.3 sec B. 1 sec C. 2 sec D. 3 sec

Solution: $T = \frac{\text{Total time taken}}{\text{Number of oscillations}} = \frac{0.33}{1.0} = 0.3\,\text{sec}$

Answer: A

Simple Harmonic Motion (SHM)

It is a kind of periodic motion. Since it is the simplest of all periodic motion to analyze, we call it "*simple harmonic*". The oscillation of simple pendulum and spring – mass system can be treated as SHM.

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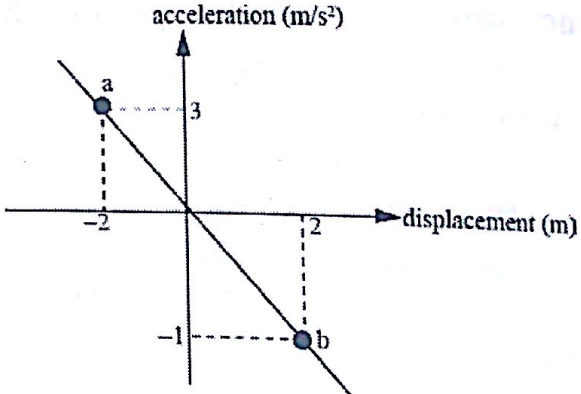
Simple Harmonic Motion: Is a periodic motion of an object about an equilibrium position such that the magnitude of its acceleration is always directly proportional but opposite in direction to its displacement.

That is, a body is said to be in SHM if and only if

- The acceleration is proportional to the displacement and,
- The acceleration is in opposite direction to the displacement.

Graphically, the acceleration Vs displacement plot of a body in SHM looks the following.



At point a) acceleration is 3 m/s^2 and the displacement is −2m, and at point b) acceleration is $-1 m/s^2$ and the displacement is 2m. Thus, the acceleration is proportional to but in opposite direction to the displacement.

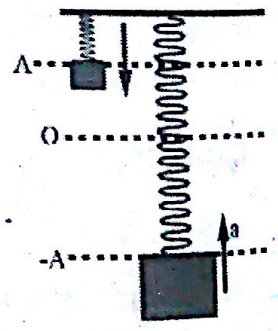
Furthermore, if you consider an oscillating spring – mass system,

Restoring force, $F = -kx$..... (*), Hooke's Law

Resultant force, $F = ma$ . . . . (**), Newton's 2nd law

Equating these two equations, we have

$$\boxed{a = -\frac{kx}{m}} \longleftarrow \text{SHM}$$



As can be seen from the relation,

- The acceleration is proportional in magnitude but opposite in direction to the displacement, $a = -\frac{k}{m}x$.

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- The acceleration is maximum when the spring is most extended or compressed from the equilibrium position. That is, a = maximum, when $x = \pm A$.
- The acceleration is zero when the mass is at the equilibrium position. That is, $a = 0$, when $x = 0$

Remember also that: The greater the displacement from the equilibrium position, the greater the restoring force and this force acts in opposite direction to the displacement creating an acceleration that always acts towards the equilibrium position.

Conceptual Example

4. Describe the behavior of acceleration and the net force in SHM.
   a) At equilibrium position.
   b) At amplitude position.
   c) At any point.

Explanation:

a) At $x = 0$, $a = -\frac{k}{m}x = 0$ and $F = ma = 0$

b) At $x = \pm A$, $a = -\frac{k}{m}A \longrightarrow$ maximum and

$F = kA = F = ma \rightarrow$ maximum

c) At any point x, $a = -\frac{k}{m}x$ and $F = -kx$

Equations of Simple Harmonic Motion

SHM is an oscillation that can be described by a sinusoidal function.

Sinusoidal: Is an equation that can be described mathematically using sine or cosine functions.

Displacement, Velocity and Acceleration in SHM

A. Displacement in SHM

The displacement of a body executing SHM is given by

$$x = A\sin\left(\frac{2\pi}{T}t\right) = A\sin\omega t$$, since $\frac{2\pi}{T} = 2\pi f = \omega$

or $$x = A\cos\left(\frac{2\pi}{T}t\right) = A\cos\omega t$$

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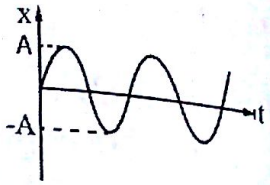
☞ **Remember:** We can use either the sine or the cosine function to describe the displacement of a body performing SHM. The only thing we need to know is the timing when the oscillation begins. That is:

- If timing starts when the system is at the equilibrium position (x = 0), use sine function.

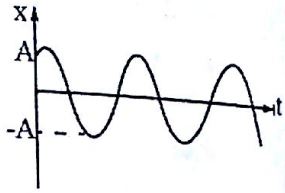


$\boxed{x = A \sin \omega t}$, thus at t = 0,

x = 0

at $t = \frac{\pi}{2}$, x = A

- If timing starts when the system is at the maximum or extreme positions (x = ±A), use cosine function.



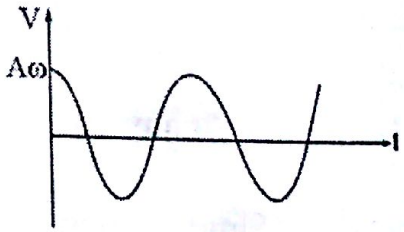
$\boxed{x = A \cos \omega t}$, thus

at t = 0, x = A

at $t = \frac{\pi}{2}$, x = 0

B. Velocity in SHM

Velocity is the rate of change of displacement.



$v = \frac{\Delta s}{\Delta t}$ and if $x = A \sin \omega t$

$v = \frac{\Delta x}{\Delta t}$, here we use the concept of derivation

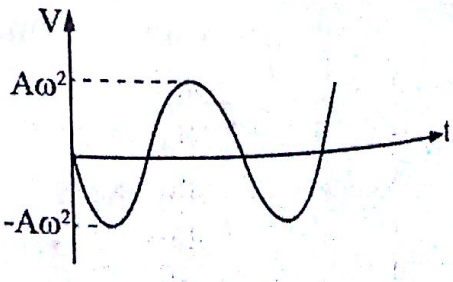
$v = \frac{dx}{dt} = \frac{d}{dt}(A \sin \omega t)$

$\boxed{v = A \omega \cos \omega t}$

C. Acceleration in SHM

Acceleration is the rate of change of velocity. Hence, when $x = A \sin \omega t$,

$v = A\omega \cos \omega t$



$a = \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d}{dt}(A\omega \cos \omega t)$

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$a = -A\omega^2 \sin \omega t$

SHM Equations Summary

In SHM, the maximum values are the following.

If

- $x = A \sin \omega t$, the maximum displacement is $x = A$
- $v = A\omega \cos \omega t$, the maximum velocity is $v = A\omega$
- $a = -A\omega^2 \sin\omega t$, the maximum acceleration is $a = -A\omega^2$

And, the displacement of a body is SHM can also be expressed as $x = A \cos(\omega t + \phi)$

Where ϕ (phi) is called the *phase angle* and it tells us at what point in the cycle the motion started at $t = 0$.

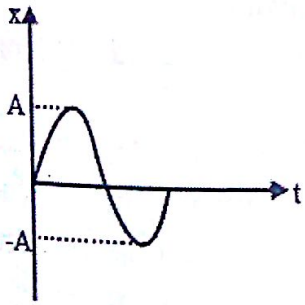
Conceptual Examples

5. In SHM, displacement and acceleration of the body can point in the same direction. True or False? Explain

   Explanation: False. In SHM acceleration and displacement are oppositely directed.

6. A mass moves in SHM with amplitude A. How far does it move in one cycle?

   Explanation: In one cycle, the oscillation pattern is,



   Hence, $x = 4A$ four times the amplitude

7. Is a bouncing ball an example of SHM? Explain

   Explanation: NO. It doesn't follow a sinusoidal function for its position as a function of time.

8. A basketball player dribbles a ball with a steady period. Is the motion of the ball SHM? Explain

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Explanation: NO. It is not because the position and velocity of the ball do not vary sinusoidally with time.

9. A mass is executing SHM. What is its speed and acceleration at
a) $x = 0$ b) $x = A$

Explanation: a) at $x = 0$ (equilibrium position)
velocity ($v = A\omega$) → maximum
acceleration ($a = -A\omega^2 x$) = 0
b) at $x = A$ (amplitude position)
velocity = 0 and
acceleration = $A\omega^2$ → maximum

☞ **Note that:** the displacement, velocity and acceleration of a body in SHM in three positions is as follows.

A. At equilibrium position ($x = 0$)
1. Velocity is maximum, $v = \omega A$
2. Acceleration is zero.
3. The net force is zero.

B. At amplitude position ($x = \pm A$)
1. Velocity is zero.
2. Acceleration is maximum, $a = A\omega^2$.
3. The net force is maximum, $F = mA\omega^2$

C. At any position x
1. Velocity, $v = \pm \omega\sqrt{A^2 - x^2}$
2. Acceleration, $a = -\omega^2 x$
3. The net force, $F = -m\omega^2 x$

Illustrative Examples

10. A pendulum has a frequency of 4 Hz and amplitude of 5 cm. Find the displacement at $t = 1.2$ sec.

Solution: $x = A \sin \omega t = A \sin 2\pi ft = (5)\sin(2\pi \times 4 \times 1.2) = -4.8\text{cm}$

11. A block suspended from a spring has a maximum displacement of 0.28m and a period of 0.4 sec. What is its displacement after 0.15 sec?

Solution: $x = A \sin \omega t = A \sin 2\pi ft = A \sin \frac{2\pi}{T} t$

$$x = (0.28)\sin\left(\frac{2\pi}{0.4}\right)(0.4) = 0.2\text{m}$$

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12. The displacement of a particle executing SHM is given by $x = (3m) \sin 0.2\pi t$. Calculate:
a) The frequency
b) The maximum velocity
c) The maximum acceleration

Solution: a) $x = 3 \sin 0.2\pi t$ and $x = A \sin 2\pi ft = A \sin \omega t$
Comparing the two equations

$$0.2\pi = 2\pi f \Rightarrow f = \frac{0.2\pi}{2\pi} \Rightarrow 0.1\text{Hz}$$

b) $V_{max} = A\omega = (3)(0.2\pi) = 0.6\pi$ m/s
c) $a_{max} = A\omega^2 = (3)(0.2\pi)^2 = 0.012\pi^2$ m/s^2 = 1.18 m/s^2

13. A particle in SHM moves according to the equation $x = 0.75 \sin 63t$ cm. Find the frequency of oscillation.

Solution: $x = 0.75 \sin 63t$, $x = A \sin 2\pi ft$
Comparing the two equations

$$2\pi f = 63 \Rightarrow f = \frac{63}{2\pi} = \frac{63}{(2)(3.14)} = 10.03 \text{ Hz}$$

14. A particle execute SHM with amplitude of 2 cm. Its velocity when passing through the rest position is 12 cm/sec. Find the period?

Solution: $x = A \sin \omega t$ and $v = \frac{dx}{dt} = A\omega \cos \omega t$

at the rest position, velocity is maximum.

$$v = A\omega \Rightarrow \Rightarrow \omega = \frac{v}{A} = \frac{12}{2} = 6 \text{ rad/sec}$$

and, $\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega} = \frac{(2)(3.14)}{6} = 1.05 \text{ sec}$

15. A body executes SHM of amplitude 1 cm and frequency 12 cycles/sec. What is the velocity when displacement is 0.5cm?

Solution: $v = \omega\sqrt{A^2 - x^2} = 2\pi f\sqrt{A^2 - x^2}$

$$v = 2(3.14)(12)\sqrt{1^2 - (0.5)^2} = 65.3\text{cm/s}$$

16. Two particles 1 and 2 start vibrating together in SHM along the same line. If their periods are 40 and 60 seconds respectively. What is their phase difference after 20 seconds?

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Solution:

$$\Delta\phi = \phi_1 - \phi_2 = \omega_1 t - \omega_2 t \left(\frac{2\pi}{T_1} - \frac{2\pi}{T_2}\right) t = 2\pi \times 20\left(\frac{1}{40} - \frac{1}{60}\right)$$

$$\Delta\phi = \frac{\pi}{3} = 60°$$

17. The velocity of a particle in SHM is 16 cm/s at a distance of 8m from the centre and 8 cm/s at a distance of 12 cm from the centre. Calculate: a) The amplitude
b) The period of motion.

Solution:

$$v = \omega\sqrt{A^2 - x^2}$$

$$16 = \frac{2\pi}{T}\sqrt{A^2 - 8^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$8 = \frac{2\pi}{T}\sqrt{A^2 - 12^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

Solving the two equations simultaneously,

a) $A = 13\text{cm}$

b) $T = 4$ sec

18. A particle that undergoes SHM has a period of 0.4 sec and an amplitude of 12mm. What is the maximum velocity of the particle?

Solution: $v = \omega A = 2\pi f A = \frac{2\pi}{T} A = \left(\frac{2\pi}{0.4}\right)(1.2) = 6\pi\text{cm/s}$

19. A body in SHM has maximum acceleration of $4\pi\text{m/s}^2$ and a maximum speed of 0.8 m/s. Find
a) The period
b) The amplitude

Solution: a. $a_{max} = \omega^2 A = \left(\frac{2\pi}{T}\right)^2 A = \frac{4\pi^2 A}{T^2}$ ..... (*)

$$v_{max} = \omega A = \frac{2\pi}{T} A \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

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$$\frac{a_{max}}{v_{max}} = \frac{\frac{4\pi^2}{T^2}A}{\frac{2\pi}{T}A} = \left(\frac{4\pi^2}{T^2}A\right)\left(\frac{T}{2\pi A}\right) = \frac{2\pi}{T}$$

$$\Rightarrow \frac{4\pi}{0.8} = \frac{2\pi}{T} \Rightarrow T = 0.4\,\text{sec}$$

b. $v_{max} = \omega A = \frac{2\pi}{T}A = 0.8 \Rightarrow \frac{2\pi}{0.4}A = 0.8 \Rightarrow A = 0.5\text{m}$

20. The period of oscillation of a particle undergoing SHM (UEE 2002)
   A. Independent of the amplitude of the motion.
   B. Directly proportional to the frequency of oscillation.
   C. Indirectly of the frequency of oscillation.
   D. Directly proportional to mass of oscillating object.

Answer: A

21. The acceleration of a SHM oscillator.
   A. Remains constant during its motion.
   B. Remains non – zero.
   C. Attains its maximum value when its amplitude is zero.
   D. Attains minimum value when its velocity is maximum.

Answer: D

22. The position of a particle moving with SHM is givne by x = A cos $(\omega t + \phi)$. If $\phi = \pi/2$, what will be the sign of the particle's position and velocity at $t = \frac{7}{8}T$ (UEE 2007)
   A. The position is positive and the velocity is negative.
   B. The position and velocity are both positive.
   C. The position and velocity are both negative.
   D. The position is negative and the velocity is positive.

Solution: $x = A\cos\left(\frac{2\pi}{T}t + \phi\right)$

$$x = A\cos\left(\frac{2\pi}{T}\left(\frac{7}{8}T\right) + \frac{\pi}{2}\right)$$

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$x = A\cos\left(\frac{9}{4}\pi\right)$, $\cos\left(\frac{9}{4}\pi\right)$ is positive and

$v = -A\omega \sin\left(\frac{9}{4}\pi\right)$, negative

Answer: A

23. A small piece of cork in a ripple tank oscillates up and down as ripples pass it. If the ripples traveling at 0.3m/s have a wavelength of 1.5πcm and the cork vibrates with an amplitude of 5mm. What is the maximum velocity of the cork?

A. 20 cm/sec B. 2 cm/sec C. 0.1m/sec D. 200 cm/sec

Solution: $v = \omega A = 2\pi fA = 2\pi\frac{v}{\lambda}A = 2\pi\frac{(0.3\times100)}{1.5\pi}\times0.5 = 20\frac{cm}{sec}$

Answer: A

24. A 0.5 kg mass attached to a spring of force constant 18 N/m vibrates in SHM with an amplitude of 5 cm. What is the maximum speed of the mass?

A. 0.2 m/sec B. 0.4m/sec C. 0.3m/sec D. 0.5m/sec

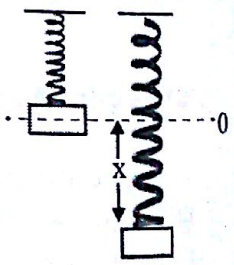
Solution: $v = \sqrt{\frac{k}{m}(A^2 - x^2)}$ velocity is maximum when $x = 0$

$$v = \sqrt{\frac{18}{0.5}(0.05)^2} = 0.3 \text{ m/sec}$$

Answer: C

Period of Spring – Mass System

Consider a body of mass m suspended at the end of a spring of force constant k. If the body is pulled down to a distance x from the equilibrium position, we have



- The force that returns the body to the equilibrium position is the restoring force given by,

$$\boxed{F = -kx} \text{ ..... Hooke's law}$$

- The body then accelerates towards the equilibrium position and hence,

$$\boxed{F = ma} \text{ .... Newton's Second Law}$$

From these two laws,

$ma = -kx$

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$$\boxed{a = \frac{-kx}{M}} \quad \ldots\ldots\ldots\

Then, $\boxed{T = 2\pi\sqrt{\frac{m}{k}}}$, where $k = k_1 + k_2$

iv) If mass m is attached to two springs connected in series, the force constant $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$.

Then, $\boxed{T = 2\pi\sqrt{\frac{m}{k}}}$, where $k = \frac{k_1 k_2}{k_1 + k_2}$

Illustrative Examples

25. When a 2kg object is suspended on a vertical spring, it oscillates with a period of 2 sec. What will be the period of oscillation if a 0.5 kg object is suspended on the same thing?

Solution: $T = 2\pi\sqrt{\frac{m}{k}}$, Hence

$$T_1 = 2\pi\sqrt{\frac{m_1}{k}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*) \text{ and}$$

$$T_2 = 2\pi\sqrt{\frac{m_2}{k}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

taking ratio.

$$\frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}} \Rightarrow T_2 = T_1\sqrt{\frac{m_2}{m_1}} = 2\left(\sqrt{\frac{0.5}{2}}\right)$$

$T_2 = 1$ sec

26. A spring stretches 4 cm when a force of 8N is applied on it. If a 2kg mass is suspended on the spring and made to vibrate, what will be the frequency of oscillation?

Solution: $F = kx \Rightarrow k = \frac{F}{x} = \frac{8}{0.04} = 200\text{N/m}$ and,

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{200}{2}} = 1.6\text{Hz}$$

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27. A 0.5kg object attached to a spring is vibrating with a period of 0.2πsec. What is,
a) The force constant of the spring.
b) The force required to stretch it by 8 cm?

Solution: a. $T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow k = \frac{4\pi^2 m}{T^2} = \frac{(4\pi^2)(0.5)}{(0.2\pi)^2} = 50\text{N/m}$

b. $F = kx = (50)(0.08) = 4\text{N}$

28. Mass m attached to a spring oscillates with a period of 2 sec. If the mass is increased by 2kg, the period increases by 1 sec. Find mass m.

Solution: $T = 2\pi\sqrt{\frac{m}{k}}$

1^{st} case, $T = 2\text{ sec} \Rightarrow 2 = 2\pi\sqrt{\frac{m}{k}}$ ..... (*)

2^{nd} case, $T = 2 + 1 = 3\text{ sec} \Rightarrow 3 = 3\pi\sqrt{\frac{m+2}{k}}$ ..... (**)

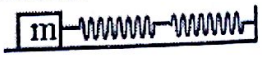
Dividing equation (**) by equation (*), we get

$$\frac{3}{2} = \sqrt{\frac{m+2}{m}} \Rightarrow \frac{9}{4} = \frac{m+2}{m} \text{ or } 9m = 4m + 8$$

$\Rightarrow 5m = 8 \Rightarrow m = 8/5 = 1.6\text{ kg}$

29. Two identical springs of k = 7500 N/m are attached to a block of mass 0.5kg. Assuming the frictionless surface. Calculate the frequency of oscillation for the two cases.

a)

b) 

Solution: a) $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{2(7500)}{0.5}} = 27.6\text{Hz}$

b) $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{(7500)(7500)}{(0.5)(7500+7500)}} = 13.8\text{Hz}$

30. A block attached to a spring of unknown force constant oscillates with a period of 2 sec. What is the period if
a) The mass is doubled.
b) The spring constant is doubled.

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Solution: a) $T_1 = 2\pi\sqrt{\frac{m}{k}}$ ..... (*)

$T_2 = 2\pi\sqrt{\frac{2m}{k}}$ ..... (**)

Taking ratio, $\frac{T_2}{T_1} = \sqrt{\frac{2m}{m}} \Rightarrow T_2 = T_1\sqrt{2} = 2\sqrt{2}$ sec

b) $T_1 = 2\pi\sqrt{\frac{m}{k}}$ ..... (*)

$T_2 = 2\pi\sqrt{\frac{m}{2k}}$ .....(**)

Taking ratio, $\frac{T_2}{T_1} = \sqrt{\frac{1}{2}} \Rightarrow T_2 = T_1\sqrt{\frac{1}{2}} = 2\sqrt{\frac{1}{2}} = \sqrt{2}$ sec

31. An object of mass m is hung from a spring and set into oscillation. The frequency is recorded as f. If the object of mass m is replaced with an object of mass 2m and set into oscillation, what will be the frequency of the motion? (UEE 2006)

A. $f/\sqrt{2}$ B. f C. $\sqrt{2}f$ D. 2f

Solution: $f_1 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ ..... (*)

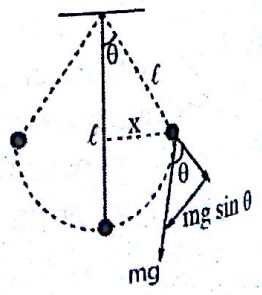
$f_2 = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$ .....(**)

Taking ratio, $\frac{f_2}{f_1} = \sqrt{\frac{1}{2}} \Rightarrow f_2 = f_1\sqrt{\frac{1}{2}} = f/\sqrt{2}$

Answer: A

Period of a Simple – Pendulum

Consider a pendulum of string length ℓ and bob mass m displaced by small angle θ from the vertical to a horizontal displacement of x from the equilibrium position. Then,



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The force that returns the bob to the equilibrium position is,

$\boxed{F = mg \sin\theta}$ ..... Restoring force

The bob accelerates to the equilibrium position when released

$\boxed{F = ma}$ ..... Newton's 2nd law

From these two laws,

$ma = -mg \sin\theta$

$a = -g \sin\theta$, but for small angle, $\sin\theta = \frac{x}{\ell}$

$$\boxed{a = -g\frac{x}{\ell}} \quad \text{..... (*)}$$

And from SHM, acceleration is,

$$\boxed{a = -x\omega^2} \quad \text{..... (**)}$$

Now, from equation (*) and (**) we have,

$$-x\omega^2 = -g\frac{x}{\ell}$$

$$\omega^2 = \frac{g}{\ell}, \text{ but } \omega = \frac{2\pi}{T}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{g}{\ell} \text{ or } \left(\frac{T}{2\pi}\right)^2 = \frac{\ell}{g}$$

$$\Rightarrow \boxed{T = 2\pi\sqrt{\ell/g}} \leftarrow \text{period of simple pendulum and}$$

since $f = \frac{1}{T}$

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}} \leftarrow \text{frequency of simple pendulum}$$

Special Cases

i) When the pendulum is infinite length

$T = 2\pi\sqrt{\frac{R}{g}} = 84.6\,\text{min}$, R radius of Earth

ii) When the pendulum has length equal to radius of Earth,

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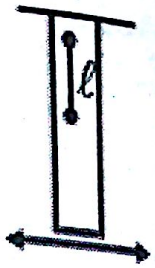
$$T = 2\pi\sqrt{\frac{R}{2g}} = 59.8\,\text{min}$$

iii) For a compound pendulum, such as a swinging meter ruler, the period with, moment of inertia $I = m\ell^2 \Rightarrow \ell = \frac{I}{m\ell}$

$$\Rightarrow T = 2\pi\sqrt{\frac{\ell}{g}}, \ell = \frac{I}{m\ell}$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{mg\ell}}} \leftarrow \text{period of compound pendulum}$$



Illustrative Example

32. A simple pendulum of 160 cm long swings to and fro on the surface of the earth. Calculate the
    a) Period of oscillation.
    b) Angular frequency of oscillation.

Solution: a. $T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1.6}{10}} = 0.8\pi$ sec

b. $\omega = 2\pi f = 2\pi\left(\frac{1}{2\pi}\sqrt{\frac{g}{\ell}}\right)$

$$\omega = \sqrt{\frac{g}{\ell}} = \sqrt{\frac{10}{1.6}} = 2.5 \text{ rad/sec}$$

33. A pendulum whose length is 1.5m oscillates 24 times per minutes in a particular location. Calculate the acceleration due to gravity.

Solution: $f = \frac{24}{60} = 0.4\text{Hz}, T = \frac{1}{f} = \frac{1}{0.4} = 2.5\,\text{sec}$ and,

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = \frac{4\pi^2\ell}{T^2} = 4\pi^2\left(\frac{1.5}{(2.5)^2}\right)$$

$g \simeq 9.5\ \text{m/s}^2$

34. How long should be a simple pendulum to have a period of 2.4 sec (use $g = 10\ \text{m/s}^2$).

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Solution: $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow 2.4 = 2\pi\sqrt{\left(\frac{\ell}{10}\right)} \Rightarrow \ell = 1.5\text{m}$

35. By what factor must the length of a simple pendulum be changed in order to double its period.

Solution: $T_1 = 2\pi\sqrt{\frac{\ell_1}{g}}$ .....(*)

$T_2 = 2\pi\sqrt{\frac{\ell_2}{g}}$ .....(**)

Taking ratio, $\frac{T_2}{T_1} = \sqrt{\frac{\ell_2}{\ell_1}}$, but $T_2 = 2T_1$

$\frac{2T_1}{T_1} = \sqrt{\frac{\ell_2}{\ell_1}} \Rightarrow \ell_2 = 4\ell_1$

Thus, the period is doubled when the length is four time its original value.

36. The period of a simple pendulum on the surface of the earth is $T = \sqrt{\frac{2}{3}}\,\text{sec}$. What will be the period of this pendulum on the surface of the moon where the acceleration due to gravity is one sixth of that on earth? $\left(g_m = \frac{1}{6}g_e\right)$.

Solution: $T_e = 2\pi\sqrt{\frac{\ell}{g_e}}$ .....(*)

$T_m = 2\pi\sqrt{\frac{\ell}{g_m}}$ .....(**)

Taking ratio, $\frac{T_m}{T_e} = \sqrt{\frac{g_e}{g_m}} \Rightarrow T_m = T_e\sqrt{\frac{g_e}{g_m}} = \left(\sqrt{\frac{2}{3}}\right)\left(\sqrt{6}\right) = \sqrt{4}$

$T_m = 2\ \text{sec}$

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37. A uniform rod of length 0.9m is pivoted at one end. What is its period?

Solution: $T = 2\pi\sqrt{\frac{I}{mg\ell}}$, its center of mass is at $\frac{L}{2}$.

The moment of inertia of a rod about its end is, $I = \frac{1}{3}mL^2$

$$T = 2\pi\sqrt{\frac{2mL^2}{3mgL^2}} = 2\pi\sqrt{\frac{2L}{3g}} = 1.6\,\text{sec}$$

☞ **Important points to remember**

Free oscillation: Is an oscillation that occurs when a body is displaced from its equilibrium position and released to oscillate without any external forces acting on it.

- A body in free oscillation has a frequency called *natural frequency* (f_o).

Forced (driven) Oscillation: Is an oscillation that is allowed to continue oscillating by external force.

- ➢ When a father pushes his son on a swing, the oscillation of the boy back and forth is a forced oscillation.
- A body in forced oscillation has a frequency called *deriving frequency* (f).

Resonance: Is oscillation of a body that occurs when the deriving frequency is the same as the natural frequency of oscillation.

- Resonance occurs when, $\boxed{f = f_o}$.

Damping force: Is a resistive force that opposes the motion of an oscillator by acting in the opposite direction to its velocity.

- Damping force is given by, $\boxed{F_b = -vb}$.

Where: b = damping coefficient which depends on the medium providing the damping.

v = the velocity of the object through the medium

Conceptual Example

38. Why do a swinging pendulum and an oscillating spring – mass system stop naturally after a while?

Explanation: Because of air resistance and other damping forces acting on these systems.

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39. When you shake up a cup of water, what happened if the deriving frequency is nearly equal to the natural frequency of the cup of water? Explain

Explanation: The amplitude of oscillation became so large and the water splashes out of the cup → resonate.

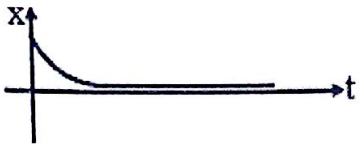
Damped oscillation: Is an oscillation that slows down and stops.

→ There are three level of damping.

1. **Under damping**: Is a damping that allows the oscillating body to move back and forth through the equilibrium position for a longer time before coming to rest.

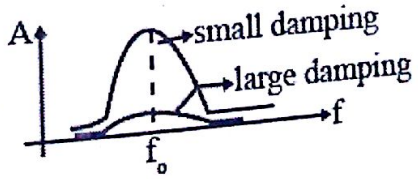
2. **Critical damping**: Is a damping that allows the oscillating body to return to its equilibrium position in the shortest possible time.
   - The system doesn't oscillates

3. **Over damping**: Is a damping that causes the oscillating body to return slowly to its equilibrium position.
   - The body doesn't oscillate



Conceptual Example

40. Explain the effect of damping on resonance and amplitude.



Explanation: Damping reduces the effect of resonance, hence, large amount of damping implies small amplitude, and
Small amount of damping implies large amplitude as shown in the diagram

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Energy in Simple Harmonic Motion

In the absence of external or damping force the mechanical energy of a body in SHM remains constant. Now, let's see the kinetic, potential and total energy of a particle executing SHM.

A) Kinetic Energy of a particle in SHM

At any instant, the displacement of a particle executing SHM may be given by:

$$x = A \sin \omega t$$

$$v = \frac{dx}{dt} = A\omega \cos \omega t, \text{ then}$$

k_E of a particle of mass m, is

$$kE = \frac{1}{2}mv^2$$

$$kE = \frac{1}{2}m(A\omega \cos \omega t)^2 = \frac{1}{2}mA^2\omega^2\cos^2\omega t$$

$$\Rightarrow kE = \frac{1}{2}mA^2\omega^2(1-\sin^2\omega t), \sin^2 \omega t + \cos^2\omega t = 1$$

$$kE = \frac{1}{2}mA^2\omega^2\left(1-\frac{x^2}{A^2}\right), x = A \sin \omega t$$

$$\boxed{kE = \frac{1}{2}m\omega^2(A^2 - x^2)}, \frac{x^2}{A^2} = \sin^2 \omega t$$

Where, $v = \omega\sqrt{A^2 - x^2}$

$$\Rightarrow \boxed{kE = \frac{1}{2}mv^2}$$

B) Potential Energy of a particle in SHM

When, $v = A\omega \cos \omega t$, then

$$a = -A\omega^2 \sin \omega t, x = A \sin \omega t$$

$$a = -x\omega^2$$

Restoring force at any instant,

$$F = -ma$$

$$F = -mx\omega^2$$

Then the work done while the particle is displaced from 0 to x.

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$$W = \int_0^x m\omega^2 x\,dx = m\omega^2 \int_0^x x\,dx$$

$$W = \frac{1}{2} m\omega^2 x^2$$

This work done is stored in the particle as P_E.

$W = P_E$

$$\boxed{P_E = \frac{1}{2} m\omega^2 x^2}$$

Where, $m\omega^2 = k$

$$\Rightarrow \boxed{P_E = \frac{1}{2} kx^2}$$

C) **Total energy, $E = k_E + P_E$**

$$E = \frac{1}{2} m\omega^2 (A^2 - x^2) + \frac{1}{2} m\omega^2 x^2$$

$$\boxed{E = \frac{1}{2} m\omega^2 A^2}$$

Since, $m\omega^2 = k$

$$\boxed{E = \frac{1}{2} kA^2}$$

☞ **Summary of energy of SHM oscillator at three positions.**

1. **At equilibrium position (x = 0)**
   - $P_E = 0$, and
   - $kE = \frac{1}{2} mv_{max}^2 = \frac{1}{2} m\omega A^2 = \frac{1}{2} kA^2$
   - Total energy = kinetic energy
2. **At amplitude position (x = A)**
   - $kE = 0$, and
   - $PE = \frac{1}{2} mv^2 A^2 = \frac{1}{2} kA^2$
   - Total energy = potential energy

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3. At any position x

- $kE = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$
- $PE = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}kx^2$
- Total energy, $\frac{1}{2}kA^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$

Conceptual Examples

41. Explain why the kE and PE of a body in SHM can never be negative?

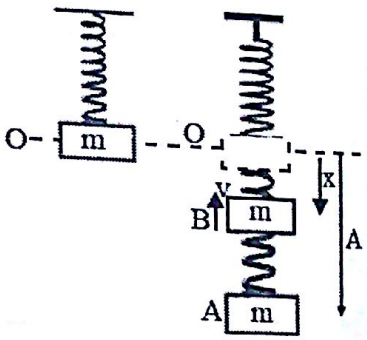
Explanation: Because, in the equation,

$kE = \frac{1}{2}mv^2$ and $PE = \frac{1}{2}kx^2$, all are non – negative

42. Consider that a mass attached to a spring is displaced to point A and then released. explain the energy of the mass at each point.

Explanation:

→ At point A, the energy is all potential $E = PE = \frac{1}{2}kA^2$



→ At point B, the body has both kE and PE

$PE = \frac{1}{2}kx^2$

$kE = \frac{1}{2}mv^2$

→ At point 0 the energy is all kinetic

$E = kE = \frac{1}{2}mv^2$

Illustrative Examples

43. A 0.5 kg body is connected to a spring of force constant 20 N/m.
   a) Calculate the total energy of the body if the amplitude is 3 cm.
   b) What is the velocity of the body at x = 2cm?
   c) Compute the kE and PE of the body at x = 2cm.

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Solution: a. $E = PE = \frac{1}{2}kA^2 = \frac{1}{2}(20)(3\times10^{-2}) = 9\times10^{-3}\,J$

b. $v = \pm\sqrt{\frac{k}{m}(A^2 - x^2)} = \pm\sqrt{\frac{20}{0.5}(0.03^2 - 0.02^2)} = \pm 0.14\,m/s$

c. $kE = \frac{1}{2}mv^2 = \frac{1}{2}(0.5)(0.14)^2 = 5\times10^{-3}\,J,$

$PE = \frac{1}{2}kx^2 = 4\times10^{-3}\,J$

44. A simple pendulum has length of 1.8m and the bob has a mass of 600g. If the pendulum is pulled to one side until it gains a height of 10 cm, calculate

a) The kinetic energy at the center of oscillation.

b) The kE and PE of the bob at a height of 8 cm.

Solution: a) At the center, velocity is maximum $\Rightarrow v = \omega A$

$$\omega = 2\pi f = 2\pi\left(\frac{1}{2\pi}\sqrt{\frac{g}{\ell}}\right) = \sqrt{\frac{g}{\ell}} = \sqrt{\frac{10}{1.8}} = 2.4 \text{ rad/sec}$$

$v = \omega A = (2.4)\,(0.1) = 0.24$ m/sec

$$\Rightarrow kE = \frac{1}{2}mv^2 = \frac{1}{2}(0.6)(0.24)^2 = 0.02J$$

b. $v = \omega\sqrt{A^2 - x^2} = 2.4\sqrt{(0.1)^2 - (0.08)^2} = 0.06m/s$

$$\Rightarrow kE = \frac{1}{2}mv^2 = \frac{1}{2}(0.6)(0.06)^2 = 0.001J$$

$$\Rightarrow PE = \frac{1}{2}mgh = (0.6)(10)(0.08) = 0.019J$$

45. A particle is vibrating in SHM with an amplitude of 4 cm. At what displacement from the equilibrium position is its energy half potential and half kinetic?

Solution: Total energy = kE = PE

$$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2, \text{ but } PE = kE$$

$$\frac{1}{2}kA^2 = \frac{1}{2}kx^2 + \frac{1}{2}kx^2$$

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$$A^2=2x^2 \Rightarrow x=\frac{A}{\sqrt{2}}=\frac{4}{\sqrt{2}}=2\sqrt{2}\text{cm}$$

46. A mass of 1 kg suspended on a spring of 16 N/m force constant is pulled down to 5 cm and then released. What is the maximum kinetic energy of the system?

Solution: $kE=\frac{1}{2}mv^2, v=\omega A=\left(\sqrt{\frac{k}{m}}\right)A$

$$kE=\frac{1}{2}kA^2=\frac{1}{2}(16)(0.05)^2=2\times10^{-2}\text{ J}$$

47. The kE of a mass of 8 kg executing SHM is 16J when it is at central position. If the amplitude of oscillation is 25cm. What is the period of oscillation?

Solution: $kE=\frac{1}{2}mv_{max}^2$, but $v_{max}=A\omega$

$$kE=\frac{1}{2}m(A\omega)^2=\frac{1}{2}mA^2\omega^2=\frac{1}{2}mA^2\left(\frac{2\pi}{T}\right)^2$$

$$kE=\frac{1}{2}mA^2\frac{4\pi^2}{T^2}\Rightarrow T=\sqrt{\frac{mA^2 4\pi^2}{2kE}}$$

$$T=\sqrt{\frac{(8)(0.25)(4)(\pi)^2}{2(16)}}=0.5\pi\,\text{sec}$$

48. The total energy of a SHM is E. What will be the kE of the particle when the displacement is half of the amplitude?

Solution: $E=\frac{1}{2}kA^2=kE+PE=kE+\frac{1}{2}kx^2$, but $x=\frac{1}{2}A$

$$kE=\frac{1}{2}kA^2-\frac{1}{8}kA^2=\frac{3}{8}kA^2$$

$$kE=\frac{3}{4}\left(\frac{1}{2}kA^2\right)=\frac{3}{4}E, \text{ since } E=\frac{1}{2}kA^2$$

49. Which one of the following is correct about damped oscillation? (UEE, 2001)

A. Runs down and stops.
B. Oscillates with resonance frequency.
C. Vibrates at a frequency higher than its natural frequency.

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D. Oscillates with increasing amplitude.

Answer: A

50. The total energy of a particle executing SHM is (UEE, 2004)
A. Zero
B. Directly proportional to the amplitude.
C. Directly proportional to the square of the amplitude.
D. Inversely proportional to the square of the amplitude.

Explanation: $E = \frac{1}{2}kA^2 \Rightarrow E \sim A^2$

Answer: C

51. Which one of the following statements is true about a spring – mass system oscillating on a frictionless surface? (UEE, 2007)
A. The kE and PE are equal at all times.
B. The kE and PE are both constant.
C. The maximum PE is achieved when the mass passes through its equilibrium position.
D. The maximum kE and maximum PE are equal.

Explanation: $kE_{max} = PE_{max} = \frac{1}{2}kA^2$

Answer: D

10.2 Wave Motion

Generally speaking, wave is a continues disturbance that moves from one place to the other carrying energy.

Basic types of Waves

There are two fundamental types of waves. These are mechanical and electromagnetic waves.

Mechanical waves: Are waves that require material to be produced and travel through. The material can be solid, liquid or gas.

Example: water wave, string wave, sound wave . . . etc 340 m/s

Electromagnetic waves: Are waves that do not require material medium for their propagation. They can travel through vacuum. They are produced due to electric and magnetic field interaction.

Example: light wave, uv light, infrared, microwave, x ray . . .

Note that: All electromagnetic waves travel at the same speed of 3×10^8 m/s is vacuum.

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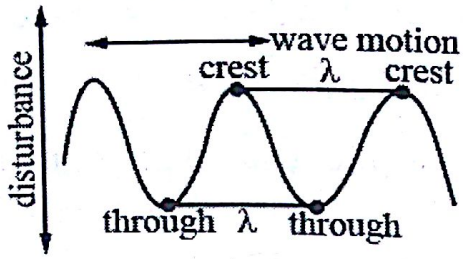
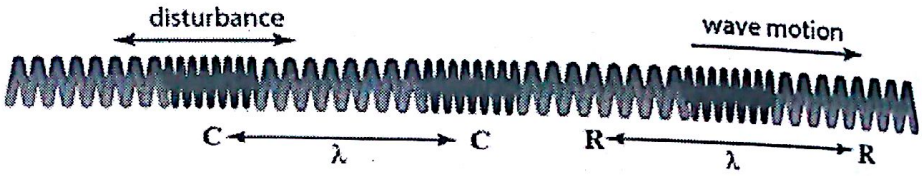
Types of Wave Motion

There are two types of wave motion. These are transverse and longitudinal.

Transverse wave: Is a wave where the direction of disturbance is perpendicular to the direction of the wave motion.



Longitudinal Wave: Is a wave where the direction of disturbance is parallel to the direction of the wave motion.



C = Compression
R = Refraction

☞ **Remember:** Waves carry energy and not particles. For instance, in a field of wheat, individual wheat stalks sway back and forth but do not change their location.

Wave Speed

The speed (v) of a wave of wavelength (λ) and period of motion (T) is given by

$$\text{from, } v = \frac{s}{t}, \text{ here } s = \lambda \text{ and } t = T$$

$$v = \frac{\lambda}{T}, \text{ but } \frac{1}{T} = f$$

$\Rightarrow \boxed{v = \lambda f}$ ← the wave equation

Note that: wave speed depends on the property of the medium through which it travels. For example, the speed of a wave through a string of mass (m), length (ℓ) and tension (T) is given by:

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$$v = \sqrt{\frac{T}{\mu}}$$

Where $\mu = \frac{m}{\ell}$, mass per length, and is called *linear density*.

Conceptual Example

52. By what factor would you have to multiply the tension in a stretched string to double the wave speed?

Explanation: from $v = \sqrt{\frac{T}{\mu}}$

$$v_1 = \sqrt{\frac{T_1}{\mu}} \text{ and } v_2 = \sqrt{\frac{T_2}{\mu}}, \text{ where } v_2 = 2v_1$$

$$\text{Taking ratio, } \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow \frac{2v_1}{v_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow T_2 = 4T_1$$

Hence, to double the speed the tension must be four times.

53. The tighter the string, and the smaller the linear density, the higher the wave speed. Is this statement true or false? Explain

Explanation: True, observe the relation $v = \sqrt{\frac{T}{\mu}}$

Illustrative Example

54. A 5m length rope with mass of 0.52kg is pulled taut with a force of 46N. What is the speed of the wave through the rope?

Solution: $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{m/\ell}} = \sqrt{\frac{T\ell}{m}} = \sqrt{\frac{(46)(5)}{0.52}} = 21\text{m/s}$

55. The wave length of a microwave is found to be 12cm. What is the frequency of the wave?

Solution: $v = \lambda f \Rightarrow f = \frac{v}{\lambda}$, since microwave is electromagnetic

$$\text{assume } v = 3 \times 10^8\text{m/s.} \Rightarrow f = \frac{3\times10^8}{0.12} = 2.5\times10^9\text{Hz}$$

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56. A wave travels with a speed of 10 m/s when the tension in the string is 8N. What should be the tension so that, the wave speed become 15 m/s in this very same string?

Solution: From the relation, $v = \sqrt{\frac{T}{\mu}}$

$$v_1 = \sqrt{\frac{T_1}{\mu}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$v_2 = \sqrt{\frac{T_2}{\mu}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

taking ratio $\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$, squaring both sides,

$$\frac{15^2}{10^2} = \frac{T_2}{8} \Rightarrow T_2 = 18N$$

57. A 4m rope is pulled tight with a tension of 90N. When one end of the rope is given a gentle wiggle, it takes 0.15 sec for the disturbance to propagate to the other end. Calculate
a) μ b) m

Solution: $v = \frac{s}{t} = \frac{\ell}{t} = \frac{4}{0.5} = 27m/s$

a) $v = \sqrt{\frac{T}{\mu}} \Rightarrow \mu = \frac{T}{v^2} = \frac{90}{(27)^2} = 0.12kg/m$

b) $\mu = \frac{m}{\ell} \Rightarrow m = \mu\ell = (0.12)(4) = 0.5\ kg$

58. A given minimum point of a sinusoidal wave is traveling 480cm along a rope in 12 sec. If the oscillator that generate the wave completes 48 vibrations in 30 sec, what is the wave length of the wave? (UEE, 2007)
A. 25m B. 15m C. 0.25m , D. 0.15m

Solution: $v = \frac{s}{t} = \frac{4.8}{12} = 0.4\ m/s$

$$f = \frac{\text{number of oscillations}}{\text{time taken}} = \frac{48}{30} = 1.6Hz$$

From wave equation,

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$$v = \lambda f \Rightarrow \lambda = \frac{v}{f} = \frac{0.4}{1.6} = 0.25m$$

59. If a long rope of uniform linear density is hung from a ceiling and waves are sent up the rope from its lower end, which one of the following is true about the speed of the ascending waves? (UEE, 2007) **Answer: C**

A. The speed decreases as the wave ascend.
B. The speed increase as the wave ascend.
C. The waves ascend with a uniform speed.
D. The speed of the waves is larger around the middle of the rope than around the beginning and end.

Explanation: $v = \sqrt{\frac{T}{\mu}}$, for a constant T and μ values, the speed is constant.

Conceptual Example **Answer: C**

60. Is the speed of transverse wave on a string the same as the speed at which a particle on the string moves? Explain

Explanation: No. it is not, for two reasons;

- If the string experience a SHM, the speed of the particle is given by $v = A\omega \sin \omega t$, not constant.
- The speed of the wave along the string is $v = \sqrt{\frac{T}{\mu}}$, which is constant.

10.3 Traveling and Standing Waves

A) Traveling Wave

A travelling wave transfer energy, or information from one place to another as it moves forward.

Traveling wave: Is a progressive wave that transfers energy from one place to another in the direction of propagation.

The displacement (y) of a traveling wave at any distance (x) and at any time (t) along its path is given by

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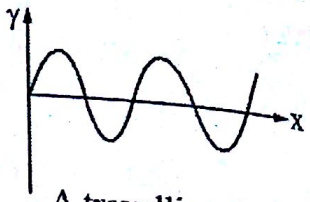
$$y = A\sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$$

or $y = A\sin(kx - \omega t)$



A travelling wave

Where, $k = \frac{2\pi}{\lambda}$ and $\omega = \frac{2\pi}{T}$

☞ Sometimes we may have $y = A\sin(kx - \omega t + \phi)$, ϕ is *phase constant.*

Illustrative Examples

61. A wave is given by, $y = 15\text{ cm}\cos(0.157x - 50.3t)$. Calculate
a) Wave length
b) Angular frequency
c) frequency
d) velocity of wave

Solution: Comparing, $y = 15\cos(0.157x - 50.3t)$

With, $y = A\cos\left(\frac{2\pi}{\lambda} - \frac{2\pi}{T}t\right)$

a) $\frac{2\pi}{\lambda} = 0.157 \Rightarrow \lambda = \frac{2\pi}{0.157} = 40\text{cm}$

b) $\omega = \frac{2\pi}{T} = 50.3\,\text{rad/sec}$

c) $\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} = \frac{50.3}{2\pi} = 8\text{Hz}$

d) $v = \lambda f = (40)(8) = 320\text{ cm/s} = 3.2\text{ m/s}$

62. Write the traveling wave equation for a wave travelling in the x direction with an amplitude of 3 cm and a frequency of 100Hz moving at a speed of 400 m/s.

Solution: $y = A\sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$

We have $A = 0.3\text{m}$, $v = 400\text{ m/s}$, $f = 100\text{ Hz}$, $T = 0.01\text{ sec}$

$$v = \lambda f \Rightarrow \lambda = \frac{v}{f} = \frac{400}{100} = 4\text{m}$$

$$\Rightarrow y = 0.3\sin\left(\frac{2\pi}{4}x - \frac{2\pi}{0.01}t\right) = 0.3\sin 2\pi\left(\frac{x}{4} - \frac{t}{0.01}\right)$$

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63. Write an expression for a wave that has a wavelength of 2.8m and propagates to the left with a speed of 13.3 m/s. The amplitude of the wave is 0.12m.

Solution: $y = A\cos\left(\frac{2\pi}{\lambda}x + \frac{2\pi}{T}t\right)$ ← to the left

$$\text{from, } v = \lambda f = \frac{\lambda}{T} \Rightarrow T = \frac{\lambda}{v} = \frac{2.8}{13.3} = 0.2\,\text{sec}$$

$$y = 0.12\text{m}\cos\left(\frac{2\pi}{2.8}x + \frac{2\pi}{0.2}t\right) = 0.12\text{m}\cos\left(\frac{\pi}{1.4}x + \frac{\pi}{0.1}t\right)$$

64. A wave is represented by $x = 0.4\cos\left(8t - \frac{1}{2}\right)$. Calculate the speed of the wave.

Solution: Comparing $x = A\cos\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)$ and

$x = 0.4\cos\left(8t - \frac{1}{2}\right)$, we have

$$\frac{2\pi}{\lambda} = \frac{1}{2} \Rightarrow \lambda = 4\pi\text{m and } \frac{2\pi}{T} = 8 \Rightarrow T = \frac{\pi}{4}\text{sec then}$$

$$v = \frac{\lambda}{T} = \frac{4\pi}{(\pi/4)} = 16\text{m/s}$$

65. A sinusoidal wave traveling in the positive x – direction has all amplitude of 12 cm, a wave length of 20 cm, and frequency of 4 Hz. The vertical displacement of the medium at t = 0 and x = 0 is also 12 cm. Calculate:

a. The angular wave number, period, angular frequency and speed of the wave.

b. Phase constant (ϕ) and write general expression.

Solution: a. We apply the traveling wave model.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{20\text{cm}}\text{rad} = 0.314\,\text{rad/cm}$$

$$T = \frac{1}{f} = \frac{1}{4\text{s}^{-1}} = 0.25\text{s}$$

$$\omega = 2\pi f = 2\pi\,(4\text{s}^{-1}) = 8\pi\,\text{rad/s}$$

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$v = f\lambda = (4s^{-1})(20\ cm) = 80\ cm/s$

b. Wave equation model:

$y = A \sin(kx - \omega t + \phi)$ when $x = 0$, and $t = 0$, the value of $y = 12\ cm \Rightarrow 12\ cm = 12\ cm \sin(0 - 0 + \phi) \Rightarrow \sin\phi = \frac{12}{12} = 1 \Rightarrow \phi = \frac{\pi}{2}$ rad.

Hence, the wave function of the form:

$$y = A \sin\left(kx - \omega t + \frac{\pi}{2}\right) = A \cos(kx - \omega t)$$

$\Rightarrow y = (12\ cm) \cos(0.314x - 8\pi t)$

$\therefore y = (12\ cm) \cos(0.314x - 8\pi t)$

Principle of Super Position

When two waves of the same type meet and overlap, they are said to super imposed or *interfere*.

> **Principle of super position;** states that; "If two or more waves pass through a point, then the resulting wave at that point has a displacement which is the sum of the displacement of the mixing up waves."

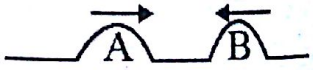
☞ When two waves meet at a point, they interfere with each other either constructively or destructively depending on their phase difference.

> **Phase difference**: Is the angular difference in timing between two waves.

☞ For example, a half cycle difference is a phase difference of π and a quarter cycle difference is a phase difference of $\pi/2$.

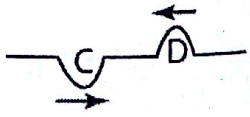
At a specific point, two waves may be in phase or out of phase.

In phase: Two waves are said to be in phase when corresponding points of each wave reach maximum or minimum displacement at the same time.



wave A & B are in phase

Out of phase: Two waves are said to be out of phase when corresponding points of each wave behave oppositely at the same time. That is, by the time one reaches maximum the other reach minimum.



Wave C and D are out of phase

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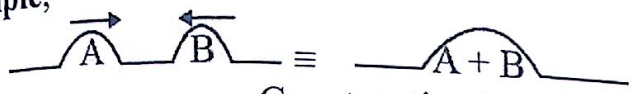
Constructive interference: Is the mixing up of two waves that are in phase.

In constructive interference, crests are aligned with crests and troughs with troughs.

☞ Constructive interference results in a wave of bigger amplitude than the mixing individual waves.

Example,



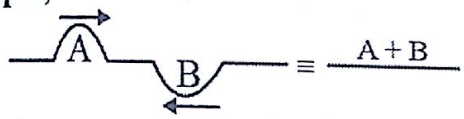
Constructive interference

Destructive Interference: Is the mixing up of two waves that are out of phase.

In destructive interference, crests are aligned with troughs.

☞ Destructive interference results in a wave of smaller amplitude than the mixing individual waves, and the waves may even cancel each other out.

Example,



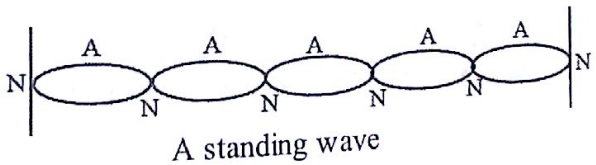
Destructive Interference

Standing Wave

Sometimes, a wave motion appears to stay stand still. When this happens we call it a *standing* or *stationary wave*. Because, the wave doesn't appear to be moving.

Standing Wave: Is set up when two waves of equal frequency wavelength, amplitude and speed moving in opposite directions mix up.

When a fixed string is disturbed, the incident and reflected wave trains going in opposite directions make constructive and destructive interference at different points along the string. Constructive interference causes a larger displacement (A) and destructive interference produces undisplaced point (N) in the string. When this happens, a standing wave is said to set up.



A standing wave

Nodes (N): Are points of zero displacement.

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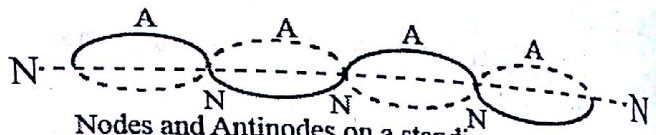
- The distance between successive nodes is half wavelength $\left(\lambda/2\right)$.

Antinodes (A): Are points of maximum displacement.

- The distance between adjacent antinodes is $\lambda/2$, and
- The distance between a node and an antinodes is $\lambda/4$.

In general,

Nodes (N): Are points where two super imposed waves are out of phase and the amplitude of oscillation is zero.



Nodes and Antinodes on a standing wave

- Nodes occur when the distance along the string is equal to $n\lambda/2$, $n = 0, 1, 2, \ldots$

Antinodes (A): Are points where two super imposed waves are in phase and the amplitude of oscillation is maximum.

- Antinodes occur when the distance along the string is equal to $\left(n+\frac{1}{2}\right)\lambda/2$, $n = 0, 1, 2, \ldots$

The Mathematics of Standing Waves

Two travelling waves moving in opposite directions can be expressed as.

$y_1 = A \sin(kx - \omega t)$ ← wave traveling to the right (incident)

$y_2 = A \sin(kx + \omega t)$ ← wave traveling to the left (reflected)

Now, according to the superposition principles, the resultant wave is the sum of this two waves.

$$y = y_1 + y_2 = A \sin(kx - \omega t) + A \sin(kx + \omega t)$$

From a trigonometric identity,

$$\sin(\theta \pm \beta) = \sin\theta \cos\beta \pm \sin\beta \cos\theta$$

So we have,

$$y = [A \sin kx \cos \omega t - A \sin \omega t \cos kx] + [A \sin kx \cos \omega t + A \sin \omega t \cos kx]$$

$$\boxed{y = 2A \sin(kx) \cos(\omega t)}$$ ← standing wave equation

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Where, $k = \frac{2\pi}{\lambda}$ and $\omega = \frac{2\pi}{T}$

- At $x = 0, \lambda/2, 3\lambda/2, \rightarrow$ Nodes
- At $x = \lambda/4, 3\lambda/4, 5\lambda/4, \rightarrow$ Antinodes

Conceptual Example

66. Like a traveling wave, a standing wave also transfers energy from one place to the other. True or False?
Explanation: False. Standing waves do not transport energy.

Illustrative Example

67. The vibration of a string of length 60 cm fixed at both ends is represented by, $y = 4\sin\left(\frac{\pi x}{15}\right)\cos(96\pi t)$

Where, x and y are in cm and t in seconds.
Calculate,
a) The maximum displacement at x = 5 cm.
b) The nodes location along the string.
c) The equation of the interfering waves.

Solution: a) The amplitude of the wave is

$$A = 4\sin\left(\frac{\pi x}{15}\right), \text{ then at } x = 5$$

$$A = 4\sin\left(\frac{5\pi}{15}\right) = 4\sin 60^\circ = 4\times\frac{\sqrt{3}}{2} = 3.5 \text{ cm}$$

b) The position of zero displacement or node is

$$4\sin\frac{\pi x}{15} = 0 \Rightarrow \frac{\pi x}{15} = n\pi, \text{ where } n = 0, 1, 2, \ldots$$

$$\Rightarrow x = 15n, = 0, 15, 30, 45, 60 \text{ cm}$$

c) Using the relation,
$2\sin\theta\cos\beta = \sin(\theta+\beta) + \sin(\theta-\beta)$

$y = 4\sin\left(\frac{\pi x}{15}\right)\cos(96\pi t)$, can be written as,

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$$y = 2\left[\sin\left(\frac{\pi x}{15} + 96\pi t\right) + \sin\left(\frac{\pi x}{15} - 96\pi t\right)\right]$$

$y = y_1 + y_2$

$$\Rightarrow y_1 = 2\sin\left(\frac{\pi x}{15} - 96\pi t\right) \text{ and}$$

$$y_2 = 2\sin\left(\frac{\pi x}{15} + 96\pi t\right)$$

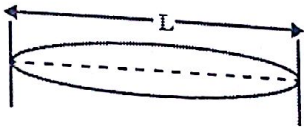
10.4 Standing waves on strings and organ pipes

Musical instruments such as piano, organ, flute, trumpet etc use standing waves on strings and organ pipes to generate sounds of different frequencies.

A) Wavelength and Frequency on fixed string

Consider a string of length L fixed at both ends and when it is plucked, the wave travels at constant speed but with different frequencies.

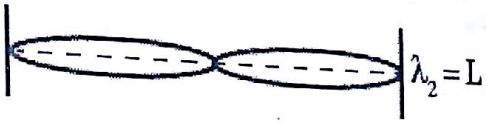
Fundamental or 1^{st} harmonic



$$\frac{\lambda_1}{2} = L \Rightarrow \lambda_1 = 2L$$

$$v = \lambda f,\ f_1 = \frac{v}{\lambda_1} \Rightarrow \boxed{f_1 = \frac{v}{2L}}$$

2^{nd} harmonic or 1^{st} overtone



$$\lambda_2 = L$$

$$f_2 = \frac{v}{\lambda_2} \Rightarrow \boxed{f_2 = \frac{v}{2L}}$$

3^{rd} harmonic or 2^{nd} overtone

$$\frac{3}{2}\lambda_3 = L \Rightarrow \lambda_3 = \frac{2}{3}L$$

$$f_3 = \frac{v}{\lambda_3} = \frac{v}{\frac{2}{3}L} \Rightarrow \boxed{f_3 = \frac{3v}{2L}}$$

In general, the n^{th} harmonic frequency is given by

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$$\boxed{f_n = \frac{nv}{2L}} \quad n = 1, 2, 3, \ldots$$

Or, $\boxed{f_n = nf_1}$, since $f_n = \frac{nv}{2L} = n\left(\frac{v}{2L}\right) = nf_1$

Remember: the fundamental frequency is the first harmonic, the second harmonic is the first over tone, the third harmonic is the second over tone and so on.

And, Since the speed of a wave on a string is given by the relation, $v = \sqrt{T/\mu}$, harmonic can be calculated using $f_n = \frac{nv}{2L}$

$$\boxed{f_n = \frac{n}{2L}\sqrt{T/\mu}}, \text{ where } \mu = m/\ell$$

Illustrative Examples

68. A steel piano wire 50 cm long has a mass of 5g and is under a tension of 400N. Calculate
    a) The fundamental frequency.
    b) The 10^{th} harmonic.

Solution: a) $f_n = \frac{n}{2L}\sqrt{T/\mu}$

$$f_1 = \frac{1}{2L}\sqrt{T/m/\ell} = \frac{1}{(2)(0.5)}\sqrt{\frac{400}{(0.005)/(0.5)}} = 200\text{Hz}$$

b) $f_n = nf_1$

$\Rightarrow f_{10} = 10f_1 = (10)(200) = 2000\text{Hz}$

69. The third overtone produced by a vibrating string of 2.5m length is 600Hz. Calculate
    a) The fundamental frequency.
    b) The speed of the wave.

Solution: a) The third overtone is the fourth harmonic.

$\Rightarrow f_n = nf_1$

$f_4 = 4f_1 = 600\text{Hz}$

$f_1 = 150\text{Hz}$

b) $f_1 = \frac{v}{2L} \Rightarrow f = (f_1)(2L)$

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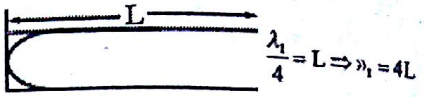
$v = (150)\ (2)\ (2.5)$

$v = 750$ m/s

B. Wave length and frequency in a pipe closed at one end.

Consider a pipe of length L where the wave speed is v.

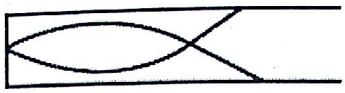


Fundamental or 1st harmonic

$$v = \lambda f \Rightarrow f_1 = \frac{v}{\lambda_1} \Rightarrow \boxed{f_1 = \frac{v}{4L}}$$

3rd harmonic or 1st over tone

$$f_3 = \frac{v}{\lambda_3} = \frac{v}{\frac{4}{3}L} \Rightarrow \boxed{f_3 = \frac{3v}{4L}}$$



5th harmonic or 2nd overtone

$$f_5 = \frac{v}{\lambda_5} \Rightarrow \boxed{f_5 = \frac{5v}{4L}}$$

In general, the n^{th} harmonic for a pipe closed at one end is given by

$$\boxed{f_n = \frac{nv}{4L}} \quad n = 1, 3, 5, 7, \ldots\ldots\ldots\ldots$$

☞ **Remember:** only the odd harmonic are allowed for a closed pipe. No even harmonies such as f_2, f_4, . . . etc. Hence the first overtone is the third harmonic, the second overtone is the fifth harmonic and so on.

C. Wavelength and frequency in an open pipe

Consider a pipe of length L open at both ends and the wave speed is v.

Fundamental or 1st harmonic

$$v = \lambda f \Rightarrow f_1 = \frac{v}{\lambda_1} \Rightarrow \boxed{f_1 = \frac{v}{2L}}$$

2nd harmonic or 1st overtone

$$f_2 = \frac{v}{\lambda_2} \Rightarrow \boxed{f_2 = \frac{v}{L}}$$

3rd harmonic or 2nd overtone

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